

Skin in the Game: Electrodermal Arousal, Expectations, and Bubbles in Experimental Asset Markets

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Abstract

We report new data on forecasts, emotional arousal, and risk preferences in experimental asset markets with a constant fundamental value (CFV). Large, sustained bubbles which crash near the terminal value are common, even with higher stakes and experienced participants. Lagged returns and forecasts best predict next-period returns. Electrodermal activity (EDA) measuring arousal adds little. However, EDA is associated with earnings: High earners are most aroused before a crash, while low earners react afterward. Forecasts fail rationality tests yet fit adaptive models. Risk preferences show no association with returns, suggesting bubbles reflect beliefs, not risk tolerance.

JEL classification: G12, G31

Keywords: Asset price bubbles; experimental asset markets; expectations; electrodermal activity; risk preferences

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1 Introduction

Behavioral finance has shown many ways in which investor and market behavior can deviate from rationality principles such as market efficiency. Financial bubbles are one such class of deviations that are often large and impactful. In a bubble, prices climb well past what fundamentals can justify — sometimes for years — before crashing (Camerer, 1989). The 2000 dot-com bust and the 2008 housing collapse are the bubbles most familiar to a contemporary reader, but hardly the only ones (Barberis, 2013).

These bubbles are evident in hindsight. But it is notoriously difficult to tell, in general, if an asset is currently in a bubble (Fama, 2014; Greenwood, Shleifer, and You, 2019).

Despite many theories and data¹ there is no consensus explanation for why bubbles form and crash. This lack of consensus is partly because bubbles in naturally occurring markets are difficult to detect and study when fundamental values are not clearly known (Garber, 1990) (exceptions include Lee, Shleifer, and Thaler 1991; Xiong and Yu 2011). In field data, price fluctuations that look like bubbles and crashes could be caused by many factors, such as changes in information and beliefs about others, discount rates and risk preferences, as well as institutional details (such as agency relations, short-selling constraints, and uncertainty about new technology).

Experiments are therefore a useful tool to study bubbles because they allow explicit control of all the determinants of the market, including an asset’s fundamental value. Many of the factors that may cause bubbles in natural asset markets can be controlled in experiments, except for endogenous beliefs and risk preferences (which can be measured). So if bubbles are common in our experiments, that suggests a possible psychological basis in both the lab and the field, which our measures can help us understand.

This paper reports experiments using a 30-period design in which the fundamental value of the asset is constant, derived only from dividends and interest rates that were carefully controlled. Bubbles are evident if prices rise well above the fundamental value.

The design combined several measures which have not been combined together in any previous experimental designs.

- Emotional arousal: In each session, electrodermal activity (EDA) was recorded from 3–5 subjects in an on-campus lab.
- Belief forecasts: In each period, just before the trading price is announced, subjects give incentivized forecasts for the current price and three future prices. We analyze whether these forecasts are rational and what models fit them.

¹See for example Santos and Woodford (1997); Abreu and Brunnermeier (2003); Scheinkman and Xiong (2003); Barberis et al. (2018); Simsek (2021).

- Risk preference: Participants’ risk preferences are measured from a simple binary choice over financial risks (unrelated to the main trading task) every period. We use these data to test theories that connect returns to time-variation in risk premia.

Together, these novel measures allow us to distinguish factors that are typically confounded in naturally occurring markets. First, belief forecasts measure what traders expect prices to do. Second, risk elicitations measure how willing they are to bear financial risk. Third, psychophysiological recordings measure both momentary and slower-moving arousal as markets rise and fall. Because these variables are collected repeatedly (and from all traders) within the same controlled asset market sessions, we can ask whether bubbles and crashes are better explained by beliefs, risk preferences, physiological states, or standard market observables such as lagged returns and order flow. (We also analyze post-experiment self-reports about trading strategies in Appendix Section C.3.)

There are two levels of results: aggregate market phenomena, and individual-level behavior.

At the aggregate market level, bubbles regularly form and typically crash to the fundamental value at the end, even with High-Stakes and trader experience. Bubbles vary across sessions and the average peak price is around three to four times the fundamental value. Returns are highly predictable (using either regression or machine learning), either from lagged returns or from traders’ forecasts. Short-term arousal, measured by EDA, has little predictive power. However, in an exploratory split, higher 5-period moving averages of arousal are associated with higher 5-period average returns, though this association does not survive the inclusion of controls.

Contrary to theories that time-varying risk premia are associated with returns, we find that trial-by-trial risk preferences are not at all correlated with returns or phases of the bubble.

Trader experience does not eliminate bubbles, but sessions with experienced traders peak and crash earlier than those of first-time participants.

The individual level of analysis is more interesting: The highest-earning traders have more stable EDA profiles across periods and their arousal peaks several periods before the price peak. Lower-earning traders have more volatile EDA profiles, and their arousal is lowest near the price peak. This pattern is consistent with some other evidence linking emotional variation to trading performance (Lo and Repin, 2002; Coates and Herbert, 2008; Smith et al., 2014). Arousal is also correlated with “de-risking” by selling assets.

Forecasts show that traders are repeatedly surprised by the prices that they themselves create (even in the High-Stakes and Experienced conditions). Forecasts mostly fail simple rationality tests because they use too little available information: Forecast errors are biased,

autocorrelated, and correlated with forecast levels. However, forecasts are accurately predicted by simple models involving adaptation and extrapolation. The best-fitting models are typically different from those that fit artificial data-generating processes (e.g., [Afrouzi et al., 2023a](#)).

2 Brief Literature Review

Several different experimental market designs have been used to study bubbles. By far the most popular is the declining fundamental value design introduced in 1988 ([Smith, Suchanek, and Williams, 1988](#)) (henceforth, SSW). In SSW, participants can buy or sell a tradable asset with a known time horizon of T total periods (typically $T = 15$). Holders of the asset get a stochastic dividend with expected value $\mathbb{E}(D)$ in each discrete period. There is no redemption value after the final period T . As a result, the rational price path of the asset is a declining step function $P_t^* = (16 - t)\mathbb{E}(D)$ for period t .

Despite this simple fundamental value structure, empirical prices often start below the fundamental, rise to a bubble price well above P_t^* , and then eventually crash toward the rational price of $\mathbb{E}(D)$ in the last market period. Subsequent work confirmed the robustness of these bubbles to scale: they emerged even in markets with participant counts exceeding 200 ([Williams, 2008](#)), and were observed consistently across computerized classroom environments as well ([Williams and Walker, 1993](#)).

The SSW bubbles are sometimes fragile to small changes, such as how the assets are described ([Kirchler, Huber, and Stöckl, 2012](#)), whether the fundamental value is shown in a chart or a table ([Huber and Kirchler, 2012](#)), and the existence of markets for future-period prices. [Noussair and Tucker \(2006\)](#) make the complementary point that adding futures markets substitutes for experience in disciplining prices. A review of SSW experiments from 1988 to 2013 shows many such effects ([Palan, 2013](#); [Bovelet, 2026](#)).

While the SSW design has certainly proved useful scientifically, the declining fundamental value is not a model for natural long-lived equities that fluctuate in value, such as stocks or real estate. Instead, SSW assets are similar to finite resources such as gold mines or oil fields, or consumer durables that depreciate, such as cars and furniture.

Furthermore, since SSW prices almost always start far below the initial fundamental value, the subsequent process of upward equilibration toward the fundamental may be what jumpstarts beliefs that prices will increase and creates a bubble, in the form of overshooting ([Smith, 2010](#); [Kirchler, Huber, and Stöckl, 2012](#)).² This pattern of equilibration is different

²These rising price paths can lead to speculative buying. [Lei, Noussair, and Plott \(2001\)](#) eliminate speculation by running sessions in which participants were assigned to the role of buyer or

from that in a constant fundamental value (CFV) design where prices begin at the CFV and immediately create bubbles (rather than converging from below).

Another important, popular paradigm, called “learning-to-forecast” (LTF) is based on the economic model of [Brock and Hommes \(1998\)](#). Pioneering experiments by [Hommes et al. \(2005, 2008\)](#) followed a similar paradigm inspired by macroeconomic models ([Marimon, Spear, and Sunder, 1993](#); [Marimon and Sunder, 1993, 1994](#)).

In LTF asset markets, participants repeatedly forecast the future price of a risky asset. The realized market price is then generated mechanically by a pricing rule with positive feedback (derived from optimal portfolio allocation), which creates higher prices from higher forecasts.³ The asset pays stochastic dividends, or traders can invest in a risk-free asset, so that there is a constant fundamental as in our design, and a known terminal period. Participants’ payoffs depend only on forecast accuracy. It’s akin to a market in which participants are financial advisors, who are incentivized to provide an accurate forecast to a client, who implements portfolio allocation mechanically from the advisor forecast.

Large bubbles and crashes reliably emerge in this paradigm, often because prices get close to an experimentally-imposed upper bound. A lot is known about how heterogeneous behavior ([Hommes, 2011](#); [Bao et al., 2012](#); [Bao, Hommes, and Pei, 2021](#)), interest rate changes ([Bao and Zong, 2019](#)), and learning selection ([Anufriev and Hommes, 2012](#)) work in this paradigm. One limit of LTF for our purposes is that prices are derived from average forecasts, so there is no scope to observe a “beliefs-portfolio” gap (Section D.2) and limited scope for forecast irrationality. Another limit is that in LTF, in practice, the upper price bound often leads to crashes, which likely generates a different underlying mechanism than in our CFV experiments.

A good design for our purposes is, as in LTF, an asset market in which stocks pay dividends but discounting is induced by an opportunity cost because experimental currency earns interest, and assets are redeemed at a known value after period T ([Bostian, Goeree, and Holt, 2005](#)). This design creates a CFV so that a bubble is easily recognized as prices being persistently above this level.⁴ This CFV design produces reliable bubbles and has

seller for the duration of the session. This eliminated speculation from the market because it was now impossible to buy in anticipation of selling at a higher price later on. They still observed bubbles, thus indicating that factors other than speculation also contribute to bubbles.

³Participants understand this qualitative relationship, but do not know the exact price-setting formula.

⁴Several other designs have induced a CFV, and generally produce mispricing. [Camerer and Weigelt \(1993\)](#) and [Ball and Holt \(1998\)](#) used a design in which discounting was induced by a probability of assets expiring in each period. [Harper, Holt, and Isaacson \(2026\)](#) used a CFV design in which the terminal payment was drawn from a known distribution, and only some “insiders” knew that payment in advance. All those designs created bubbles. [Noussair, Robin, and Ruffieux \(2021\)](#)’s design created a constant fundamental by paying expected dividends of zero. [Caginalp, Porter, and](#)

replicated successfully (Smith et al., 2014; Holt, Porzio, and Song, 2017).

Our design also measures emotional arousal. Emotions, such as exuberance or panic, have been shown to correlate with market volatility and price reversals in actual markets among professional traders (Lo and Repin, 2002; Fenton-O’Creevy et al., 2012; Weitzel et al., 2020).⁵ Breaban and Noussair (2018) found that positive emotional states (inferred from faces) correlated with purchases and overpricing, and that negative states correlated with selling, low prices, and price decreases.⁶ It has also been shown that the simple awareness of one’s heart rate (interoception) among participants is correlated with better market performance and a longer professional career (Kandasamy et al., 2016; Bossaerts, 2021). Bossaerts et al. (2024) measured heart rate and found that anticipatory emotional responses were correlated with higher earnings than reactive (non-anticipatory) responses. Other papers linked asset bubble formation to EEG activity and eye-tracking (Toma and Miyakoshi, 2021; Toma et al., 2023). Finally, a study using fMRI reported that insula activity in successful traders rose before bubble peaks and was associated with profitable selling (Smith et al., 2014). These findings motivated us to measure emotional arousal to determine whether emotions are associated with individual behavior and session-level prices in the CFV domain.

We also elicited price forecasts (following Camerer and Weigelt 1993, Haruvy, Lahav, and Noussair 2007, and others). We will compute rationality properties of these forecasts, and assess what models fit them best. Our results can then be compared to evidence from the growing amount of subjective belief data in finance and macro-finance⁷ and we also explore their association with trader behavior (Corgnet et al., 2022).

2.1 Experience Effects

Early in the history of experimental economics, participants were often recruited to participate in a second experimental session (or even a third or more) which was identical to the first session in all design details, except for the participants’ experience. This practice emerged

Smith (2001) paid terminal accumulated dividend payments rather than paying sequentially. Those two papers showed smaller bubbles than SSW.

⁵Other physiological states also matter; Berkman et al. (2017) shows that circadian mismatch and sleep deprivation can significantly amplify volatility and bubble formation in global asset markets.

⁶Inducing emotional excitement using short film clips magnified bubbles in the SSW design (Andrade, Odean, and Lin, 2015), but that finding failed to replicate (Huber et al., 2024).

⁷Many analysts have found overreaction of beliefs about financial and macroeconomic variables (De Bondt and Thaler, 1990; Amromin and Sharpe, 2014a; Greenwood and Shleifer, 2014a; Genaioli, Ma, and Shleifer, 2016; Bordalo et al., 2019a, 2020a; Barrero, 2022). A few studies show underreaction (Abarbanell and Bernard, 1992; Bouchaud et al., 2019b; Ma et al., 2020). Afrouzi et al. (2023a) find overreaction in experiments with artificial time series, echoing an older literature including Schmalensee 1976a; Lovell 1986.

because in complex markets, it was often the case that experienced subjects appeared to make less noisy choices more consistent with theory.⁸

Indeed, there is a lot of evidence from the SSW design that experience reduces bubbles (beginning with [Smith, Suchanek, and Williams 1988](#) and then [King et al. 1993](#); [Van Boening, Williams, and LaMaster 1993](#); [Haruvy, Lahav, and Noussair 2007](#)). Some studies found that experience had less impact ([Sunder, 1995](#); [Kopányi-Peuker and Weber, 2021](#)). [Dufwenberg, Lindqvist, and Moore \(2005\)](#) and [Porter and Smith \(2010\)](#) demonstrate that mixtures of new and experienced participants can sometimes reduce bubble formation, depending on the mixture ([Hussam, Porter, and Smith, 2008](#)).

Our experiments contribute new data because no CFV designs have studied the effect of previous-session experience while also measuring forecasts. The combination is important because, if experience reliably reduces bubbles, (i) that is important to know for generalizability, and (ii) it raises the question of whether experience also improves forecasts. And note that in the SSW design there is an obvious and salient declining price path that experience generally moves markets toward. That is less true for the CFV design. For example, it could be that experienced traders become even more aggressive by buying early to create bubbles, planning to sell before a crash.

Measuring time-variation in risk attitudes is also novel. More details about time-varying risk premia and other design features are in [Section A.1](#).

3 Experimental Methods

Our experiment created a series of sessions in which 15–20 participants traded a single risky asset with one another using oTree software.⁹ In each session, 3–5 participants took part in a lab at Caltech, wearing Empatica wristband sensors to measure arousal. The other 10–17 subjects participated online, recruited from Prolific. More experimental details are in [Section A.2.3](#).

Participants traded a risky asset (stock) for an experimental currency (CASH) in a call market used in [Smith et al. \(2014\)](#); [Holt, Porzio, and Song \(2017\)](#). At the end of every trading period t , the stock paid a dividend D_t , which was equally likely to be 0.4 or 1 unit of CASH (so $\mathbb{E}(D_t) = 0.7$). The dividend D_t was the same for all traders in period t , and independent and identically distributed (iid) across periods. CASH earned a fixed 5% interest.

⁸See, e.g., [Kagel and Levin \(1986\)](#); [Camerer \(1987\)](#).

⁹The software was developed at Virginia Tech using open-source oTree ([Chen, Schonger, and Wickens, 2016](#)) and was hosted on the cloud computing platform *heroku.com*.

Each participant submitted buy or sell orders for stocks, stated in units of CASH. After all orders were submitted, a uniform market-clearing price was calculated where supply met demand and every trade was executed at that single price (see Section A.2.1).

The equilibrium price P^* is the value at which an investor is indifferent between investing P^* units in CASH and earning an r percent return, or buying a share at price P and earning a dividend D_t plus an expected capital gain from reselling. Equating these two returns (and assuming risk-neutrality) gives an “Euler equation”

$$rP_t = \mathbb{E}(P_{t+1}) - P_t + \mathbb{E}(D_t) \quad (1)$$

In the steady-state solution, $P_{t-1} = P_t$, so $rP_t = 0.7$ and $P_t = 0.7/r = 14 \equiv P^*$. Intuitively, if prices are below 14, participants should bid up the price because the expected 0.7 dividend yield ($0.7/P$) is greater than their return from interest. If prices are above 14, participants should sell and invest the proceeds in cash, earning more than the dividend yield. After period 30, stock shares were redeemed for 14 units of CASH, the equilibrium price. Because the terminal redemption price is commonly known to all participants, the solution $P^* = 14$ can also be computed by backward induction.

There is a small technical aside to the discussion above: Ignoring the finite time horizon behaviorally, there can be growing bubbles, so that $P_t = P^* + B_t$, if the bubble premium $B_t > 0$, grows at an expected 5%.¹⁰

In our design (and in theory), these kinds of ‘quasi-rational’ bubbles should not exist because the asset’s finite life should kill expected bubble growth at some point (Blanchard and Watson, 1982; Tirole, 1985). If traders are forward-looking, once bubble growth is expected to be too low, the bubble should crash.¹¹

A session was 30 trading periods, each lasting approximately 80 seconds. Each of those periods consisted of four pages: an Order Submission Page, a Forecast Page, a Round Results Page, and a Risk Elicitation Page (Figure 1).

On the first page, the Order Submission Page, participants were instructed to place bid and ask limit orders in a grid. Participants had 20 seconds to submit their orders and could simultaneously input up to three bids and three asks during any given period. They could

¹⁰Suppose $\mathbb{E}(B_{t+1}) = B_t(1 + r) + \epsilon_t$ where $\mathbb{E}(\epsilon_t) = 0$. Then prices $P_t = P^* + B_t$ could include a growing-bubble component of that type, $B_t > 0$, as long as there is an expected return of r on the bubble component. Indeed, Figure 3 shows that the average price return declines slowly across periods in most sessions, but it is not far from the predicted bubble growth rate 5% for the first 15–20 periods. Note that since such bubbles must grow at rate r , this idea predicts that varying r across designs could cause varied bubble growth.

¹¹However, there are many examples in related literature, such as repeated prisoners’ dilemma and centipede games, which illustrate the empirical limits of backward induction in long-lived games (McKelvey and Palfrey, 1992).

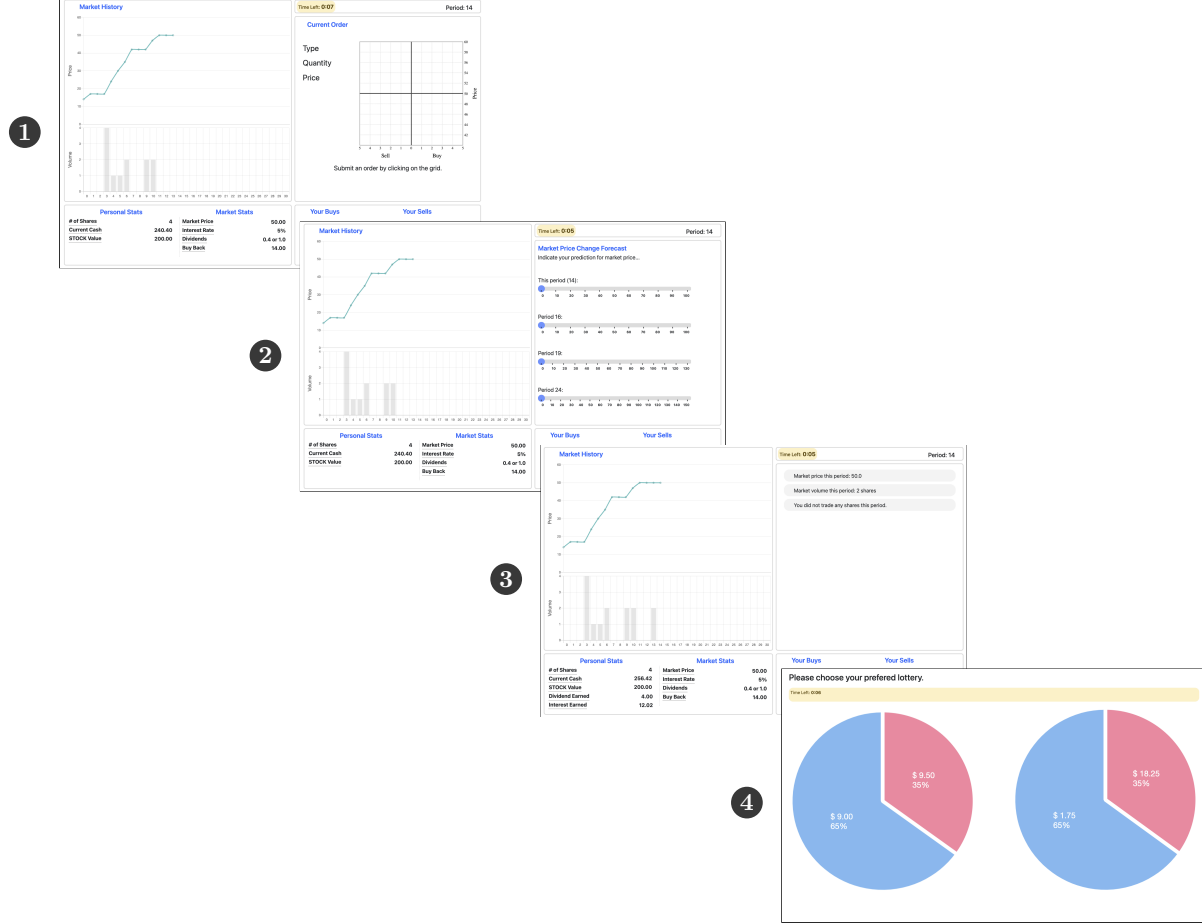


Figure 1: **Experimental Task Flow**

Each of the 30 trading periods presented four sequential pages, shown top-left to bottom-right: (1) Order Submission (20 seconds), where participants place limit orders; (2) Forecast (30 seconds), where they predict current and future prices; (3) Round Results (10 seconds), showing the realized price and their own trades; and (4) Risk Elicitation (20 seconds), an adaptive lottery task estimating risk aversion.

Alt text: Four overlapping screenshots of the trading interface arranged diagonally from top left to bottom right: the order submission grid, the forecast entry page, the round results page, and the risk elicitation page with pie-chart lotteries.

not place bids above their own ask order prices. Short sales and margin purchases were not allowed, although some studies have included those features.¹²

¹²Some experimental asset-market studies have varied these trading constraints directly. For example, [King et al. \(1993\)](#) examine the robustness of bubbles under treatments including short selling and margin purchases; [Haruvy and Noussair \(2006\)](#) study the effect of relaxing short-selling constraints in experimental spot asset markets; [Ackert et al. \(2006\)](#) study margin, short selling, and lotteries in experimental asset markets. See also [Hauser and Huber \(2012\)](#) for related evidence on short-selling and borrowing constraints in laboratory markets; [Coppock, Harper, and Holt \(2021\)](#) allowed margin purchases and varied credit constraints to assess their effect on bubbles.

After participants submitted orders, they saw a Forecast Page. Participants then had 30 seconds to make forecasts of future market prices at four different time horizons: 0, 2, 5, and 10 periods ahead. Forecasts were truncated when there were few periods left in the session. If, for example, there were only nine periods left to play, participants were not asked to predict the price ten periods ahead. We analyze whether participants traded consistently with price forecasts, whether forecasts predict prices, whether forecasts satisfy rationality tests, and what kinds of models fit forecasts.

After the Forecast Page, the price was calculated and participants saw the price and a summary of their trades, shares, and CASH.

Participants were then shown a Risk Elicitation Page, which consisted of a sequence of four pairs of monetary lotteries. Each pair of lotteries consisted of a riskier lottery and a less risky lottery, and participants had five seconds to make each selection. For each choice a participant made, the probabilities of the lotteries were updated according to the Dynamically Optimized Sequential Experimentation (DOSE) protocol (Chapman et al., 2024) to provide estimates of the participant’s constant-relative-risk-aversion parameter for that period ($\rho_{i,t}$).¹³

3.1 Data Collection

We measured physiological responses for in-lab participants using a combination of Empatica E4 devices and Empatica EmbracePlus devices.¹⁴ Most studies that have measured biological responses and their relationship to finance have employed large, complicated devices to measure various biological responses. For instance, Lo and Repin (2002) required six separate biosensors and over five minutes of set-up time. These complicated setups usually meant that only one subject could be monitored at a time. Our use of Empatica E4 wristbands circumvents this limitation, allowing us to gather data on multiple participants per session simultaneously.¹⁵ We did not record the biophysiology of the online participants.

We used these psychophysiological data to predict stock price and bubble crash dynamics. In addition, we conducted exploratory analyses to examine how psychophysiology responds to specific events, such as the revelation of forecast errors.

The experiment was approved by Caltech’s IRB. Participants were given a brief demo-

¹³An example of how the DOSE method updates probabilities and parameter estimates can be found in Figure A.1.

¹⁴These devices have been shown to provide high-quality data in several experiments (McCarthy et al., 2016; Schuurmans et al., 2020; Milstein and Gordon, 2020; Tashjian et al., 2022).

¹⁵Of primary interest for data analysis is the Round Results page, where participants learn the updated market price and presumably update their forecasts about the market. It is at this time that a subject is most likely to be surprised and emotionally aroused.

graphics survey before being shown the written and video instructions (full instructions are available in Section E.1). Subjects were then entered into the trading task. After the session ended, participants completed a post-experiment survey that included a free-response question asking participants to describe their trading strategy. Full details of these processes can be found in Section E.3.3.

We collected data from 19 Standard sessions, with an average of 19 participants per session ($N = 362$). Table A.4 reports participant statistics. We also recorded biometric data from in-lab participants (3–5 per session). Of those 90 in-lab participants, 72 yielded usable biometric data. In-lab participants were younger, made fewer errors on the comprehension quiz, and earned higher forecast bonuses (Tables A.1 and A.2; Figure A.2); within a session, they also earned higher market bonuses than online participants (Figure A.3), with other differences detailed in Section A.3.1.

These 19 Standard sessions are a benchmark for two other treatments.

To investigate, we ran four High-Stakes sessions that were identical to the Standard treatment except for the performance bonus: instead of one USD for every 200 units of CASH, participants earned five USD for every 200 units of CASH — a fivefold increase in stakes.

The two additional treatment protocols were identical to the Standard sessions except for small changes; and all participants were from Prolific (so no EDA was measured). In 4 High-Stakes sessions ($N = 59$) the performance bonuses were 5 times as large as in the Standard treatment. Paying higher stakes is motivated by a long-standing curiosity about the effects of extrinsic incentives, and a hypothesis that higher stakes can increase attention and cognitive effort, potentially moving choices closer to rational-model predictions and reducing behavioral noise (though evidence for this is mixed¹⁶).

The second treatment was 5 Experienced sessions ($N = 93$) using Prolific participants who had previously traded in one or more of the Standard sessions.¹⁷ Results are described in Section B.1.4. In all sessions, participants received a base payment of \$20 and a bonus payment based on their performance in the various experimental tasks (see Section A.2.2).

¹⁶Early summaries of the effects of higher stakes in economics experiments are (Smith and Walker, 1993; Camerer and Hogarth, 1999). Our reading of the evidence is that paying a substantial sum (small multiples of minimum wage) rather than nothing reduces noise, but paying much larger sums (5–10x as much) has little additional effect. A meta-analysis by Cala et al. (2026) shows mostly modest effects.

¹⁷Because some participants returned for more than one additional session, there are $N = 72$ distinct participants who create the $N = 93$ total.

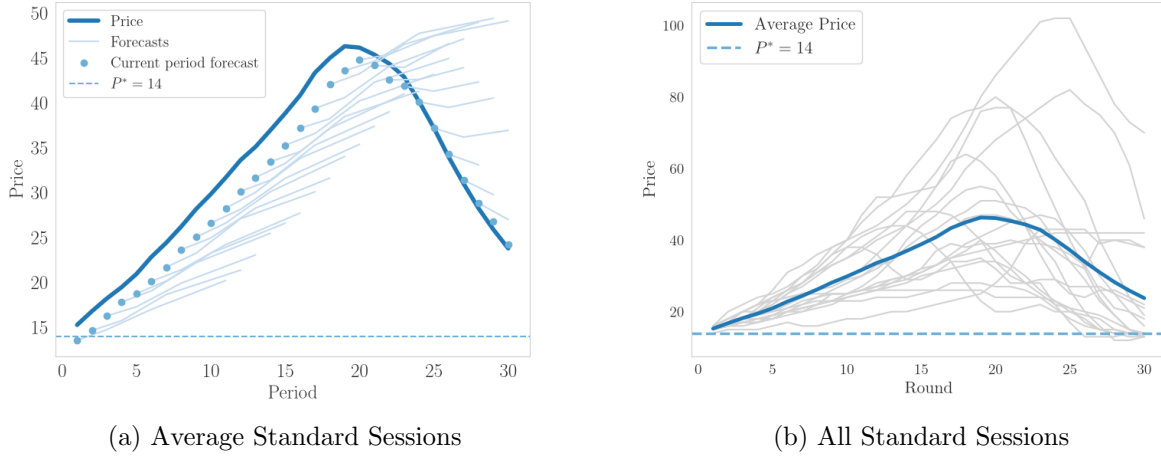


Figure 2: **Prices and Forecasts in Standard Sessions**

(a) Average prices and forecasts across 19 Standard sessions. The thick line shows the average price path. Each circular point on the graph represents the average current period forecast (just before the current price is announced) averaged across participants. The thin lines extending from the circular point are the “tongues” of average future forecasts in period $t + h$ for each of the 2-, 5-, and 10-period ahead prices. All session graphs are in Figure B.2. (b) All Standard market price paths are shown here as light gray lines. The average Standard price over time is shown as the bold line. Fundamental value is shown as a dashed line at price = 14.

Alt text: Two line charts of price against period 1 to 30. Panel (a): the average price rises from about 15 to a peak near 46 in period 19 and falls to about 24 by period 30; forecast tongues lie below the price during the rise and above it after the peak. Panel (b): 19 light gray session price paths with peaks between roughly 30 and 100, the bold average path, and a dashed fundamental value line at 14.

4 Price and Forecast Data

This section presents insights about the aggregate market behavior. We first test how well stock returns can be predicted, one period ahead, using market measures and the biometric, forecast, and risk-aversion data. Figure 2a displays the average price and forecast path of the 19 experimental sessions. This Figure shows averaged forecast “tongues” which plot forecast paths at the four horizons. Figure 2b shows all the market paths of prices.

Most sessions formed substantial bubbles, peaking on average in period 18.8 at a price of 52.0.

Bubble magnitudes and duration have conventionally been measured by a series of session-level statistics, such as relative absolute deviation (RAD) and amplitude (maximum price deviation) (Stöckl, Huber, and Kirchler, 2010). These are reported for every session in Table A.3, for comparability with previous studies.

Bubbles proved robust to two treatment variations, described fully in Sections B.1.3 and B.1.4. In four High-Stakes sessions, where the performance bonus was five times larger, and in five Experienced sessions of returning Standard participants, bubbles still formed and crashed. High-Stakes sessions had somewhat lower peaks (35.8 compared to 52.0) and Experienced sessions peaked significantly earlier (period 12.2 compared to 18.8, $p = 0.038$), consistent with evidence that experience reduces but does not eliminate bubbles (Huber et al., 2024). These small samples limit power, so the comparisons are exploratory. Participants in the High-Stakes sessions were also better forecasters, with forecast errors about half as large as in Standard sessions (Table D.2). Experienced participants forecast better only at the current horizon. That higher stakes sharpened forecasts without raising price levels matches the incentive effects Camerer and Hogarth (1999) describe for judgment tasks, where added effort chiefly reduces response variance, and is consistent with the modest average incentive effects Cala et al. (2026) document.

5 Predicting Market Phenomena

Our analysis proceeds in two stages. First, we report results of a pre-registered regression to understand which features are most correlated with asset returns. Second, we fit a pre-registered machine learning model to understand the amount of predictive power our dataset has for returns prediction.

5.1 Predicting Returns: Regression

We find that our measure of arousal, average tonic EDA activity (EDA^{ton}), is not predictive, while numerous traditional measures remain informative, suggesting that our current measure of individual arousal does not improve predictive accuracy above and beyond established price and trading metrics.

We use ordinary least squares (OLS) regression to examine the relationships between market returns, biometric indicators, and trading behavior variables across the 19 experimental sessions (Table 2). Our analysis follows our pre-registered plan with one tiny deviation.¹⁸

The final specification (column (5)) explains approximately 70% of the variance in next-period returns ($R^2 = 0.704$). Not surprisingly—given the graphs we have seen—lagged return is highly predictive on its own R_{t-1} ($\beta = 0.623$, $p < 0.001$). While the three included forecast variables (f_t^t , f_{t-2}^t , f_{t-5}^t) maintain marginal significance in the final model ($p < 0.1$ for all

¹⁸The deviation is that we eliminated f_{t-10}^t because of multicollinearity, and because including that variable required us to drop too many periods of data. A faithful execution of the pre-registered model (the Standard, backward-looking column) can be found in Table B.1.

Table 1: **Summary of Regression Notation**

Symbol	Name	Description
f_{t-h}^t	Backward-looking	For period t , made at $t-h$
f_t^{t+h}	Forward-looking	For period $t+h$, made at t
ρ_t	Risk aversion	DOSE risk-aversion parameter
$30 - t$	Periods remaining	Periods left in the session
Ω_t	Order imbalance	Sentiment from bid-ask order imbalance
Υ_t	Asset allocation	CASH-to-stock value ratio
V_{t-1}	Lagged volume	Volume, previous period
EDA_t^{ton}	Tonic EDA	Tonic electrodermal activity
$EDA_t^{\text{ton}}\text{MA}_5$	EDA moving average	5-period backward mean, $\frac{1}{5} \sum_{i=0}^4 EDA_{t-i}^{\text{ton}}$
R_{t-1}	Lagged return	Log return of previous period, $\log(P_{t-1}/P_{t-2})$

Notes: Forecasts, risk aversion, asset allocation, and tonic EDA are averaged across participants within each session-period; order imbalance, volume, and returns are period-level market aggregates. Forecasts use horizons $h \in \{0, 2, 5, 10\}$; f_t^t is the nowcast, formed just before the current price is revealed. Periods 1–5 are excluded because the longer-horizon forecasts they require are not yet available. CASH is the experimental currency.

three), they become even more significant when lagged returns are excluded (column (4)). It seems that both recent returns and near-term forecasts contain meaningful information about future price movements.

By contrast, $EDA_t^{\text{ton}}\text{MA}_5$ enters positively but is not statistically significant in the full specification. A 5-period moving average of tonic EDA shows marginal significance in a bivariate specification (Table 2, column (2)), but disappears after the inclusion of additional controls (Table 2, columns (4) and (5)).

We suspected multicollinearity among the forecast variables, and our analysis in B.3.3 bears that out. We found a high degree of correlation among those variables (Figure B.8) and the first principal component explained over 91% of their variance (Figure B.7). We then reran these models replacing the three forecast variables with their first principal component (see Table B.3) with no major changes in results.

Although asset allocation and risk-preference measures do not significantly predict returns in our full specification (a period-by-period comparison of the DOSE risk parameters against returns, Figure C.3, reaches the same conclusion), non-lagged return variables still contain substantial predictive information. In particular, when lagged returns are excluded, the model retains strong predictive power for future returns (adjusted $R^2 = 0.574$; Table 2, column (4)), driven largely by the forecast variables. We interpret this as evidence that participant forecasts are highly correlated with recent return dynamics and capture much of the same information.

Table 2: **Predicting Next-Period Log Returns**

	(1)	(2)	(3)	(4)	(5)
Intercept	-0.621*** (0.115)	-5.032* (2.660)	-0.266 (2.805)	-5.097 (4.889)	-0.331 (2.972)
R_{t-1}	0.827*** (0.034)		0.623*** (0.051)		0.623*** (0.051)
$\text{EDA}_t^{\text{ton}}\text{MA}_5$		9.783* (5.420)		-0.430 (2.791)	0.129 (1.570)
ρ_t			0.520 (0.739)	-0.831 (0.694)	0.531 (0.769)
V_{t-1}			-0.089 (0.064)	-0.029 (0.075)	-0.088 (0.067)
$30 - t$			0.179* (0.092)	0.560*** (0.154)	0.179* (0.092)
Ω_t			0.010 (0.025)	0.034 (0.032)	0.010 (0.025)
Υ_t			0.001 (0.007)	0.007 (0.011)	0.001 (0.007)
f_t^t			0.138* (0.071)	0.603*** (0.067)	0.138* (0.072)
f_{t-2}^t			-0.130* (0.071)	-0.466*** (0.107)	-0.131* (0.070)
f_{t-5}^t			-0.089* (0.044)	-0.242*** (0.070)	-0.088* (0.042)
N	456	456	456	456	456
R^2	0.668	0.012	0.704	0.582	0.704
Adjusted R^2	0.668	0.010	0.698	0.574	0.697

Notes: The dependent variable is the future log return R_t (multiplied by 100). Standard errors (in parentheses) are heteroskedasticity- and cluster-robust (clustered by session). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We repeated the OLS regression for our High-Stakes and Experienced sessions.¹⁹ These analyses do not include the EDA measure, because it was not collected from these online-only participants. Lagged returns are again highly predictive of future returns (Table B.4), with model R^2 s around 0.60. Furthermore, current-period forecasts were found to be more predictive for the High-Stakes sessions ($f_{t,\text{HS}}^t = 0.332$, $p = 0.041$) than for the Standard sessions ($f_{t,\text{Std}}^t = 0.138$, $p = 0.072$). This difference in predictive power is consistent with our earlier finding that High-Stakes participants were better forecasters.²⁰

¹⁹Table B.1 reports this pre-registered return regression for all three session types side by side.

²⁰High-Stakes forecast errors are significantly lower than in Standard sessions at every horizon (Table D.2).

5.2 Predicting Returns: Machine Learning

Using a machine learning approach focusing on Support Vector Regression (SVR),²¹ we find that while traditional market variables predict returns with substantial accuracy, the inclusion of EDA^{ton} does not meaningfully improve out-of-sample performance. This provides evidence that the null result for EDA^{ton} in our simple OLS does not reflect model misspecification but rather a true lack of predictive content, further supporting the idea that physiological arousal, as measured by EDA^{ton} , does not add incremental predictive power beyond standard metrics.

We focus on SVR, a relatively simple model that tends to generalize well when data are complex and limited; results for several other well-established ML models appear in Section B.3.6, and a LASSO (least absolute shrinkage and selection operator) regression in Table B.7.

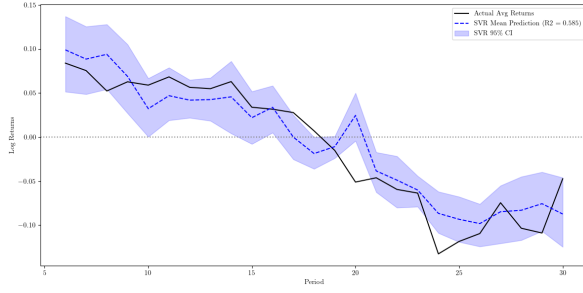
We train on period-by-period data, splitting the 19 sessions into 15 for training and 4 held out for out-of-sample testing. Within the training set, we use 10-fold cross-validation — 13 training and 2 validation sessions per fold, repeated 10 times — to train the model later evaluated on the held-out sessions; within each fold, hyperparameters are tuned by a separate 5-fold grid search. Keeping all of a session’s periods within the same fold guards against the “data leakage” that can arise when machine learning is applied to time-series data, as is standard in the machine-learning asset-pricing literature (Gu, Kelly, and Xiu, 2020).

We tuned two hyperparameters by grid search: the regularization parameter $c \in \{0.1, 1, 10\}$, which trades off training-error fit against model complexity (smaller c generalizes more, larger c fits more tightly), and the kernel — either linear, assuming a linear feature–target relationship, or the radial basis function (RBF), which maps features into a higher-dimensional space to capture non-linear patterns.

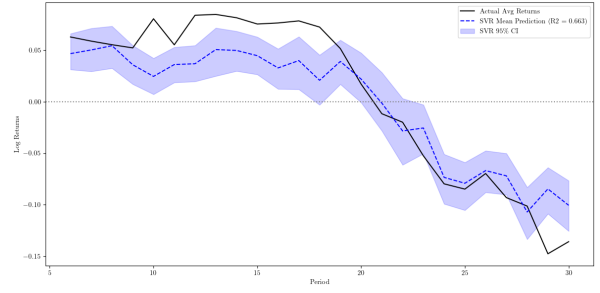
The SVR features mirror those in our regression model but exclude lagged returns, isolating our variables’ predictive content from return autocorrelation. They comprise the average DOSE risk parameters (ρ, μ) , forecasted price levels at several horizons, summary statistics of EDA^{ton} , lagged volume, and the number of periods remaining; Table B.2 (Section B.3.2) provides full details.

²¹SVR is a kernel-based machine learning algorithm that is effective for capturing both linear and non-linear relationships in data. SVR aims to find a function that deviates from the actual observed targets by a value no greater than a specified margin while being as flat as possible. See Section B.3.4.

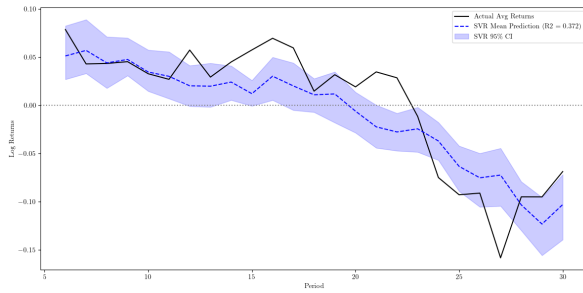
²²Bootstrapping is a statistical resampling technique that estimates variability and uncertainty by repeatedly taking random samples from the original dataset with replacement to create multiple simulated datasets. After 1000 iterations, the 95 percent confidence intervals were derived from the



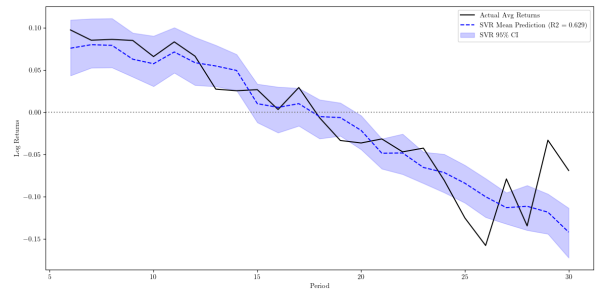
(a) Holdout: Sessions 1–4



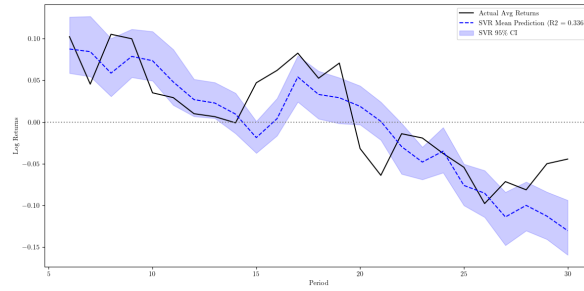
(b) Holdout: Sessions 5–8



(c) Holdout: Sessions 9–12



(d) Holdout: Sessions 13–16



(e) Holdout: Sessions 17–19

Figure 3: Mean Out-of-Sample SVR Return Predictions

Shaded regions show 95% confidence intervals from 1,000 bootstrap resamples of the training data, evaluated on a fixed held-out test set. The majority of realized returns lie within the bootstrap confidence bands across all held-out session groups, suggesting the model captures common market dynamics and provides well-calibrated out-of-sample uncertainty. For figure clarity, we choose not to plot the confidence interval of our held-out samples; for interested readers, this can be seen in Figure B.10. Confidence intervals (CIs) for model performance metrics (MAE and R^2) were estimated using bootstrapping²² with 1000 iterations.

Alt text: Five panels, one per held-out session group, each plotting realized log returns and mean SVR predictions over periods with shaded 95 percent bootstrap bands; most realized returns fall inside the bands.

Our analysis reveals that traditional measures exhibit moderate predictive power for asset returns (mean absolute error [MAE] mean = 0.046, R^2 mean = 0.544 for the non-EDA model; Table B.5, Panel B).²³ However, the inclusion of EDA^{ton} data does not improve model performance (Figure B.9). Details of feature weights can be found in Table 3 (and, under a random train/test split, Table B.6). That a few variables carry nearly all the weight parallels Gu, Kelly, and Xiu (2020), who find that machine-learning gains in asset pricing concentrate in a handful of dominant predictors.

Table 3: **SVR Feature Importance Weights**

Feature	Mean Importance	Range	
		Low	High
f_t^t	0.348	0.308	0.365
f_{t-2}^t	0.221	0.187	0.281
$30 - t$	0.130	0.116	0.148
f_{t-5}^t	0.126	0.076	0.156
$\sigma(\text{EDA}_t^{\text{ton}})$	0.053	0.028	0.076
$\text{skew}(\text{EDA}_t^{\text{ton}})$	0.053	0.025	0.070
$\text{EDA}_t^{\text{ton}}$	0.028	0.010	0.044
ρ	0.015	0.004	0.031
V_{t-1}	0.013	0.005	0.030
μ	0.012	0.001	0.025

Notes: Mean importance weights are reported across held-out session validation groups. Ranges indicate the minimum and maximum importance weight observed across held-out groups.

6 Exploratory Associations of Arousal

Earlier we observed that the $(\text{EDA}^{\text{ton}})$ measure of arousal contributes little to predicting one-period-ahead returns (Tables 2, 3). This unfortunate null result is likely due to noise in biological measures, the small size of EDA samples (only $N = 72$), and the mismatch between aggregate price changes created by all subjects and EDA measures from only a fifth of those subjects.

2.5th and 97.5th percentiles of the bootstrapped distributions, providing robust estimates of model uncertainty and generalizability.

²³To assess robustness to market selection, we repeated the w/o EDA^{ton} evaluation across 100 different random holdouts of 4 sessions. Across these splits, test performance for the SVR model averaged MAE = 0.045 (SD = 0.006) and $R^2 = 0.559$ (SD = 0.111). The wide spread in R^2 reflects genuine differences in how predictable different sessions are, not instability in the model itself; R^2 remained positive across nearly all splits.

However, there are three findings in which EDA seems to be associated with behavior and performance. They are exploratory (they weren’t pre-registered) and therefore simply suggest avenues for future research.

The first association is that above-median values of 5-period EDA are correlated with above-median 5-period forward returns (see Figure C.2 and accompanying discussion).

The second exploratory association is about “de-risking”, which is the practitioner term for reducing portfolio risk. In our simple experimental setting, the only form of de-risking is selling shares of the risky asset. A logistic regression on all subject–period observations ($N = 2,280$) shows that higher peak arousal (EDA^{peak}) significantly increases the odds of executing a sell order (see Table C.3). A one–standard deviation increase in peak arousal raises the probability of selling by approximately 9%, evaluated at the sample mean. The association appears for executed rather than attempted sales, so we read it as arousal accompanying the completion of de-risking rather than as evidence that arousal initiates the intention to sell.

We also observe differences in de-risking behavior between the high- and low-earning terciles. By looking at average share positions of the risky asset, we observe that high earners tend to accumulate shares early in the session and begin to offload them prior to the market peak, while low earners tend to gain shares during the post-peak crash (Figure C.4). This is a pattern of behavior that supports the anticipatory physiological response of the high earners, and the reactionary response of the low earners. High earners are also more accurate forecasters at every horizon (Figure C.5; Table C.5), consistent with this anticipatory profile. Participants’ own free-response strategy descriptions echo this divide: high earners more often described a buy-early, sell-before-the-peak approach (Figures C.7 and C.8; common strategy terms appear in Figure C.6).

The third association is between individual differences in the path of EDA^{ton} across market periods, and trading performance. To see this, we first relabel periods in each session so that they are zero-centered around the period in which the price was highest. Periods τ are defined as $\tau = t - t_p$, the number of periods before or after the price peak period t_p . This relabeling can find patterns that are consistent with respect to the timing of price peaks, but not consistent across periods and across sessions. For example, two periods 18 in different sessions might be zero-centered as $\tau = +1$ and $\tau = -2$ if their respective price-peak periods are in periods 17 and 20.

Our metric $\overline{EDA}_\tau^{\text{ton}}$ is the average across individual measures EDA^{ton} at each relative period τ . This statistic lets us ask whether physiological responses are anticipatory—peaking before $\tau = 0$ —or reactionary—peaking after $\tau = 0$.

We sorted participants into terciles by market-bonus performance (Table C.1). Figure C.1 shows that $\overline{EDA}_\tau^{\text{ton}}$ is less volatile across periods within a session among the highest earners

than among the lowest earners (Levene test $p = 0.003$).²⁴

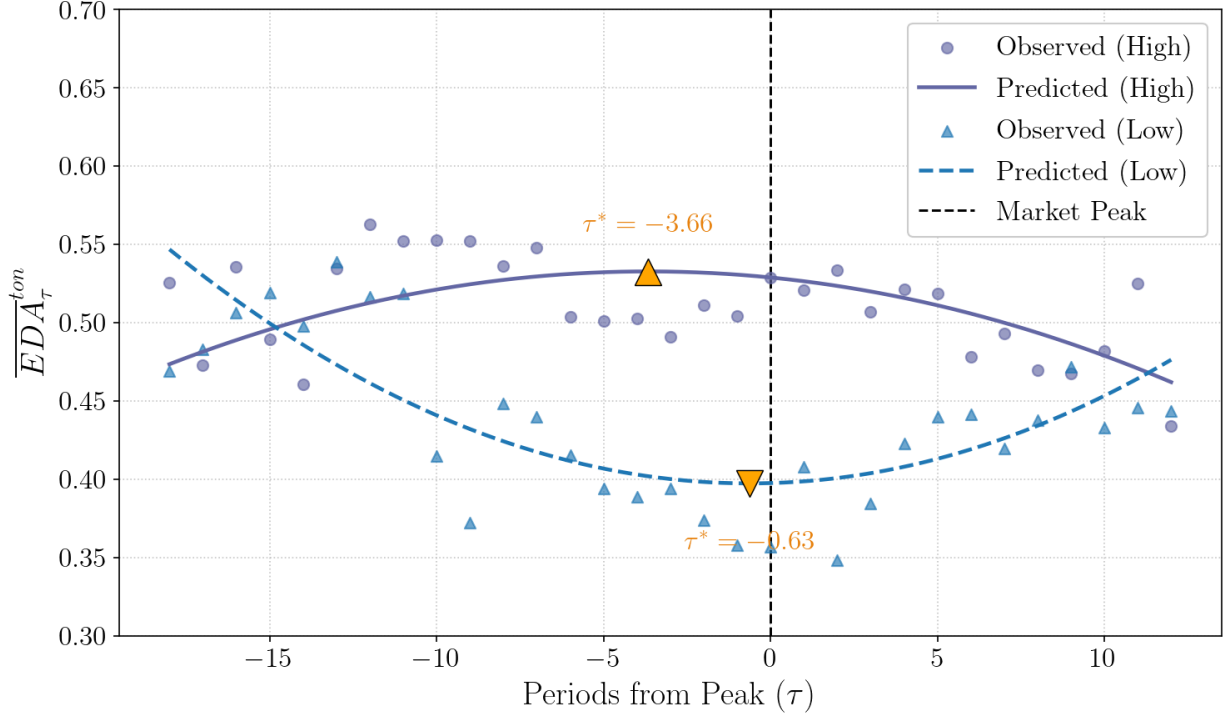


Figure 4: **Tonic EDA by Earnings Tercile Around the Market Peak**

OLS quadratic regressions of mean tonic EDA ($\overline{EDA}_\tau^{\text{ton}}$) on squared distance from the market peak (τ^2) with robust standard errors (Equation 2), estimated separately for High- and Low-earning terciles. Points are group means; curves are fitted predictions from $\overline{EDA}_\tau^{\text{ton}} = \alpha + \beta\tau + \gamma\tau^2$. Fitted curves were optimized and those optimal points are indicated by large orange triangular markers. For $\overline{EDA}_\tau^{\text{ton}}$ with 90% confidence intervals, see Figure C.1a. Detailed values can be seen in Table C.2.

Alt text: Scatter plot of mean tonic EDA against periods from the market peak, from minus 18 to plus 12, with fitted quadratic curves. The high-earner curve is an inverted U peaking about 3.7 periods before the market peak at roughly 0.53. The low-earner curve is a U with its minimum just before the peak at roughly 0.40.

To compare the two earnings groups' arousal paths across periods, we estimate separate OLS polynomial regressions of $\overline{EDA}_\tau^{\text{ton}}$ on τ

$$\overline{EDA}_\tau^{\text{ton}} = \alpha + \beta\tau + \gamma\tau^2 + \epsilon \quad (2)$$

Both regressions explain a substantial share of the variance in group-mean EDA across τ ($R^2 = 0.545$ and 0.388 for the high- and low-earning groups, respectively). The high- and low-earning relations are highly significantly different ($p < 10^{-7}$; see Tables C.2 and C.4).

²⁴An above- versus below-median split shows the same basic pattern (Figure C.1b).

The vertices of the two groups are the estimated values $\hat{\tau}$ at which $\overline{\text{EDA}}_{\tau}^{\text{ton}}$ is maximal or minimal. Those estimates are $\hat{\tau}_{\text{high}}^* = -3.66$ and $\hat{\tau}_{\text{lo}}^* = -0.63$. This indicates the high-earning group’s EDA is more anticipatory of the peak than the low-earning group’s EDA.

Our result about arousal and earnings is consistent with [Smith et al. \(2014\)](#)’s fMRI evidence. They found that high earners’ brain activity in insula cortex rises before the crash, and low earners’ insula activity dips before the crash. The insula is a region involved in translating bodily emotions into conscious feelings and is reliably associated with unpleasant sensory episodes, including financial uncertainty ([Mohr, Biele, and Heekeren, 2010](#); [Morris, Gell, and van Reekum, 2019](#)), as well as mood ([Rust et al., 2025](#)).

Our EDA findings are also consistent with the practitioner intuition that successful trading requires emotional control rather than heightened reactivity. This aligns with field evidence that emotionally reactive day-traders earn poorer returns ([Lo and Repin, 2002](#)) and that greater autonomic regulation, measured by heart-rate variability, distinguishes expert traders from novices ([Fenton-O’Creevy et al., 2012](#)). Relatedly, [Coates and Herbert \(2008\)](#) found on a London trading floor that traders’ cortisol rose with market volatility and with the variance of their own profits, linking physiological stress to the uncertainty of the trading environment.

7 Forecast Rationality and Modeling

This section examines the quality and structure of individual price forecasts, and how they are associated with trading.

The *quality* of forecasts is whether they adhere to rationality principles. The overarching principle is that good forecasts should use all observable information that can reduce forecast error; forecasts that do not do so are called “irrational” in this specific sense. As we will see, forecasts of many individuals fail these rationality tests (as other studies have shown), even for Experienced participants.

The *structure* of forecasts is which specific models, such as adaptive or extrapolative expectations, describe the forecasts most accurately.

Rationality and structure of forecasts are distinct properties. When forecast errors are autocorrelated, for example — participants sluggishly adjust forecasts — they fail rationality tests. But precisely because of those rationality failures, forecasts might be very well-described by an adaptive expectations model which reflects such autocorrelation.

7.1 Evaluating Forecast Rationality

The hypothesis of rational expectations is that agents form forecasts using all available information. In our experimental asset markets, this means participants’ price forecasts at time t should efficiently incorporate all the information they can observe and remember, including past prices, recent returns, and previous forecast errors. Forecast errors should not be systematically related to any of those variables.

Following previous studies using economic outlook surveys (Zarnowitz, 1985; Kohlhas and Walther, 2021) and experiments (Lovell, 1986), we test for three features of forecast rationality: the *unbiasedness* of forecast errors, the absence of *autocorrelation* in forecast errors, and independence of *forecast errors and forecasts*.

For each period t and forecasting horizon h , we compute mean and median forecasts across participants and across sessions. Let I_t denote the set of participants active at period t , with sample size $N_t = |I_t|$. Participant i ’s forecast at period t for the price in period $t + h$ is denoted $f_{i,t}^{t+h}$. The expression $f_{i,t-h}^t$ denotes a forecast for period t that was formed h periods earlier, i.e., a *memory* of a forecast made previously about the *current* period and the future.²⁵ The across-participant mean forecast is

$$\bar{f}_t^{t+h} = \frac{1}{N_t} \sum_{i \in I_t} f_{i,t}^{t+h}, \quad (3)$$

and the median is similar. Forecast errors are defined as the difference between realized prices and forecasts:

$$E_{i,t}^h = P_{t+h} - f_{i,t}^{t+h}, \quad (4)$$

Note that by this definition, positive errors indicate underestimation and negative errors indicate overestimation. Summary statistics are in Table D.3.

7.2 Forecast Rationality Tests

Figure 5 graphs the mean forecast error across trading periods for each horizon h . As Figure 2a, shown earlier, foreshadowed, participants start out by under-forecasting prices (positive error), then over-forecast as the crash occurs (negative error). Figure 5 shows that this pattern is consistent across all horizons h , and is bigger for longer horizons. However, we find evidence that participants do trade based on the expected price paths as indicated by their forecasts (see Section D.2).

²⁵The experiment was run in multiple sessions. For expositional simplicity, we suppress the session index s ; all aggregate quantities such as means and medians are defined at the session level.

We test to see if participants' forecasts show bias. If participant forecasts are unbiased, the distribution of forecast errors at each horizon h should center around zero (Figure D.2). The unbiasedness hypothesis is rejected for a growing share of participants as the horizon lengthens: the fraction whose forecast errors are statistically unbiased (at $p < .05$) falls from about 62% at $h = 0$ to roughly 43% at $h = 10$ (Table D.1).

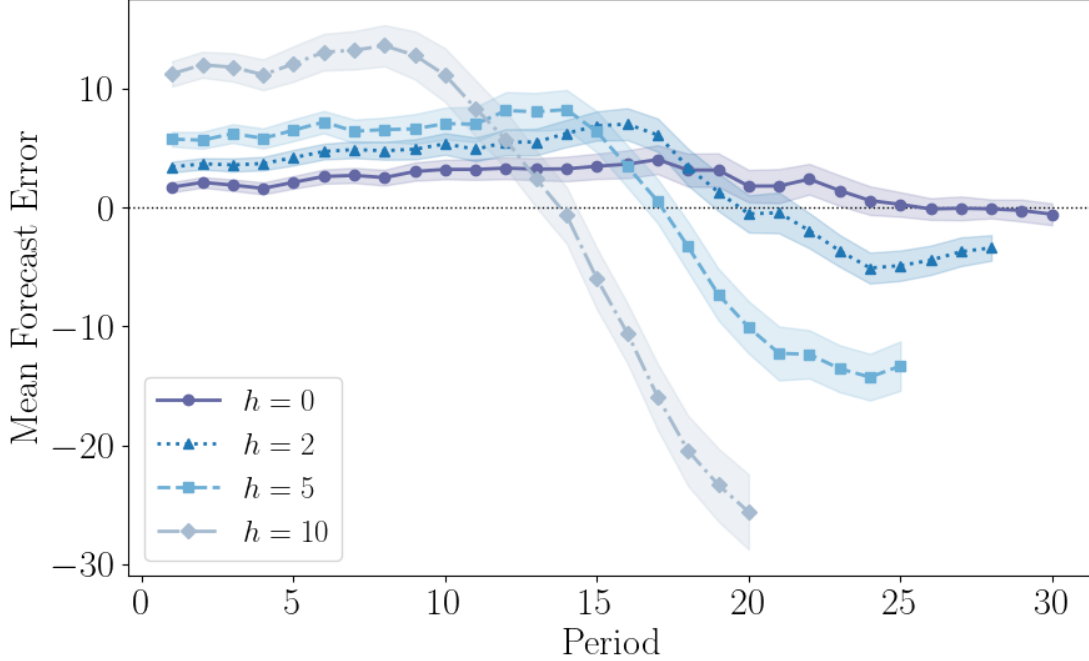


Figure 5: **Mean Forecast Error by Period and Horizon**

Mean forecast error across trading periods for each horizon h , with 95% confidence intervals shaded. As the horizon increases, errors grow in magnitude because projecting further ahead is more difficult. Across periods, as the bubble peaks and crashes, the errors go from positive to increasingly negative. The negative forecast errors show that participants are slow to revise post-peak price expectations downward. Note that the same pattern holds for the median forecast (see Figure D.3).

Alt text: Line chart of mean forecast error by period for horizons 0, 2, 5, and 10 with shaded 95 percent bands. Errors are positive early, larger for longer horizons, and turn sharply negative after period 15, reaching about minus 25 for the 10-period horizon.

Next, we test how independently participants make forecasts by checking for autocorrelation among forecast errors. Since previous forecast errors $E_{i,t}^h = P_{t+h} - f_{i,t}^{t+h}$ are observable in period $t + h$, those errors should not be autocorrelated with current errors. Autocorrelation is measured by β_1 in the OLS regression

$$E_{i,t}^0 = \beta_0 + \beta_1 E_{i,t-1}^0 + \epsilon_{i,t}, \quad (5)$$

Estimated values around $\beta_1 = 0$ indicate rationality. Estimates $\beta_1 > 0$ and $\beta_1 < 0$ indicate momentum and overreaction, respectively. We restrict this test to the current period forecast error ($h = 0$).

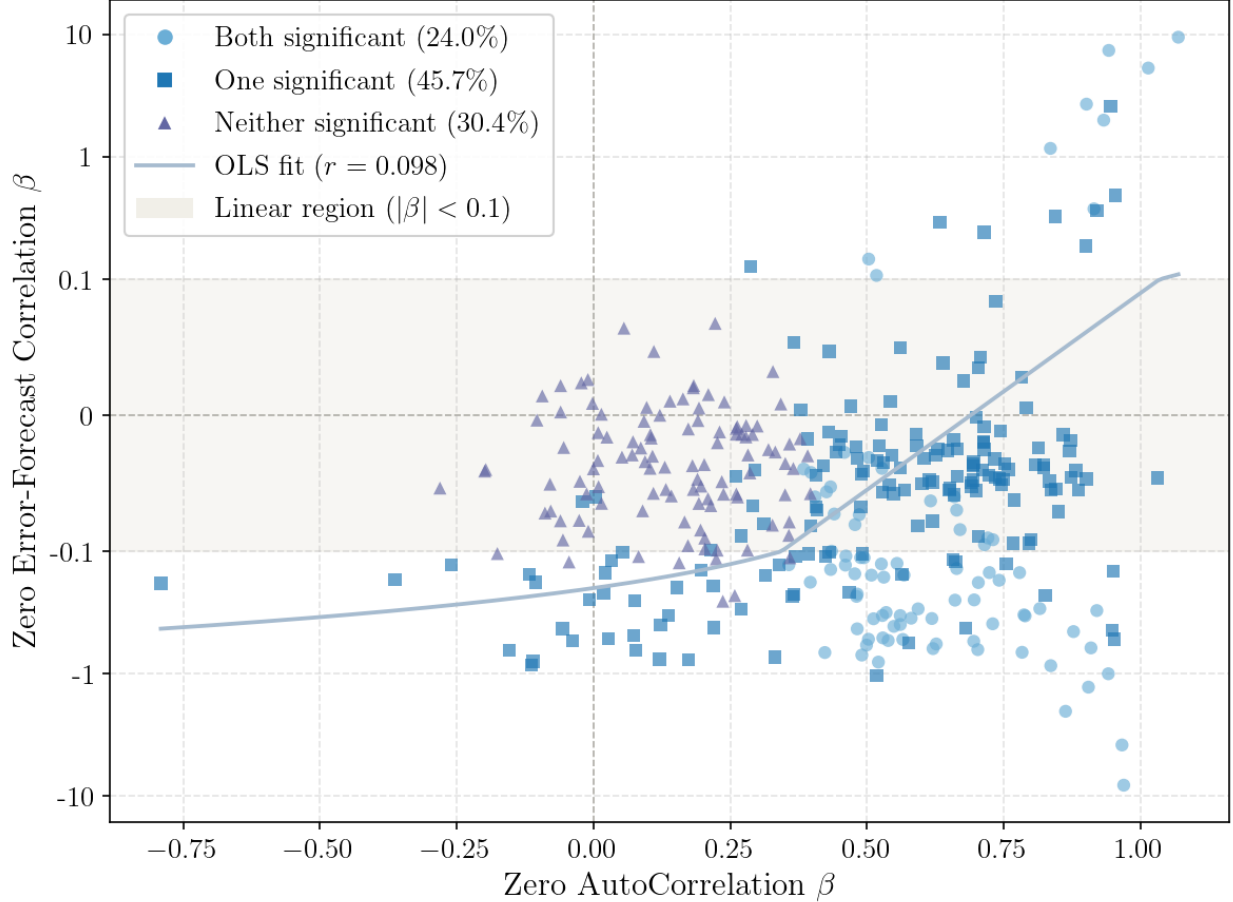


Figure 6: **Forecast-Rationality Coefficients in Standard Sessions**

Scatter plot of the forecast-rationality β s from the autocorrelation and error-forecast correlation tests (Equations 5 and 6). The figure shows that non-significance is distributed around the sample population instead of clustered around a few participants. The vertical axis is presented as a **symlog** scale, which is a logarithmic scale adapted to handle negative numbers and is linear in the light gray band between -0.1 and 0.1. This is why the trend line appears kinked. Joint significance counts can be found in Table D.4. Analogous plots for the High-Stakes and Experienced sessions are shown in panels (a) and (b) of Figure D.5.

Alt text: Scatter plot of individual error-forecast correlation coefficients on a symlog vertical axis against autocorrelation coefficients, with markers distinguishing participants for whom both, one, or neither test is significant (24, 46, and 30 percent). Points spread across the whole sample with a slightly upward-sloping fitted line.

The third rationality hypothesis is that there should be no correlation between forecast

errors and the forecasts themselves. This hypothesis is tested by estimating β_1 in

$$E_{i,t}^h = \beta_0 + \beta_1 f_{i,t}^{t+h} + \epsilon_{i,t}^h, \quad (6)$$

The intuition here needs to be spelled out. Consider a weather forecaster’s temperature forecasts across the year, in a typical region that has seasonal differences. Suppose the forecasts are much less variable (smoother) than the actual temperatures but do reflect seasonal trends. In summer with high temperatures their forecasts will be high and their errors will mostly be positive; and in winter with low temperatures their forecasts will be low and their errors will be negative. Temperature and forecast error will then be positively correlated across the year. The opposite pattern could also occur: If forecasts overreact to seasonal differences, the correlations will be negative.

Across the three tests, adherence to rational expectations weakens as the forecast horizon lengthens, most sharply for the correlation between forecast errors and forecasts.

Figure 6 plots individual-level coefficients from autocorrelation estimates in Equation 5 (on the x -axis) and Equation 6 estimates of error-forecast correlations (on the y -axis). Note that the x -axis is linear. But the y -axis uses an informative “symlog” scale which is linear for $\beta_1 \in [-.1, +.1]$ (in gray) and logarithmic outside of that range (which creates the kink at the y -axis $\beta = -.1$).

Each symbol is two estimates from one person in the 19 Standard sessions. The blue colors indicate that one coefficient is significantly different from zero (at $p < .05$) and orange colors indicate both are significant.

The reader can see that autocorrelations ($\beta_1 > 0$ on the x -axis) are overwhelmingly positive, and significantly so for about $r > .40$. Most of the error-forecast correlations (on the y -axis) are negative, indicating overreaction. Those with $\beta_1 < -.10$ are usually significant. Finally, note that the correlation between the two coefficients is only $r=.098$. So there is apparent heterogeneity in the kinds of rationality deviations these participants are making.

Taken together, these tests paint a consistent picture: individual forecasts depart systematically from rational expectations, and the departures grow with the forecast horizon. Forecast errors are biased, positively autocorrelated (momentum), and negatively correlated with the forecasts themselves (overreaction), so that forecasts leave predictable information unused precisely over the multi-period horizons most relevant to a bubble’s rise and crash. These departures are not eliminated by experience: participants who had traded in earlier sessions continue to fail the same tests, and the expectation rules we estimate for them remain firmly backward-looking — if anything, experienced traders extrapolate recent price

trends more strongly than first-time participants ($\hat{\phi} = .389$ versus .199). Prior experience thus does little to move beliefs toward rationality; the expectations driving these sessions appear adaptive and extrapolative rather than information-efficient, a characterization we make precise with the competing forecasting models estimated below.

7.3 Estimating Competing Models of Forecasts

The last section showed that forecasts often failed rationality tests. These failures of rationality immediately suggest that some alternative models which reproduce these forecast errors can probably describe forecasting more accurately. In this section, we’ll describe and estimate several forecasting models to see which ones fit best.

Note that economists (and other social scientists) have been interested in how expectations are formed for decades. Section D.1 describes this background literature.

Almost all of those previous data, however, have been generated by field surveys from households and experts about macroeconomic variables, or in experimental domains where the data generating process is controlled by the experimenter but only partially explained to subjects who have to learn it.

Our data are different. As in Williams (1987) and other experimental bubble studies that elicit traders’ price forecasts (Haruvy, Lahav, and Noussair, 2007; Cheung, Hedegaard, and Palan, 2014; Akiyama, Hanaki, and Ishikawa, 2017; Breaban and Noussair, 2018; Galí, Giusti, and Noussair, 2021; Corgnet et al., 2022), our participants are forecasting *their own future collective behavior*. This is a domain in which rationality of expectations, in the sense of Lucas and Prescott (1971), is most likely to hold.²⁶

Furthermore, under any sensible learning dynamic the forecast rationality of subjects in Experienced sessions should be better than that of subjects in Standard sessions, a hypothesis we examine by comparing forecast rationality across treatments (Figure D.5 and Table D.4).

Next, we assess how well common expectation models account for participants’ forecasts. We test four models: adaptive expectations (Cagan, 1956a; Nerlove, 1958), extrapolation (Barberis et al., 2015a; Hirshleifer, Li, and Yu, 2015), overreaction (Afrouzi et al., 2023a), and double-adaptive expectations (Holt, Porzio, and Song, 2017). This is a mixture of models with old pedigrees and some of recent interest, especially in analysis of financial market and macroeconomic expectations.

²⁶Lucas and Prescott (1971) object to adaptive expectations because “price forecasts and actual prices will have different probability distributions, and this difference will be persistent, costly to forecasters, and readily correctible.” (p 660). They continue: “To avoid this difficulty, we shall, in this paper, go to the opposite extreme, assuming that the actual and anticipated prices have the same probability distribution, or that price expectations are rational”. p 660

7.3.1 Adaptive expectations

$$f_{i,t}^t = \delta P_{t-1} + (1 - \delta) f_{i,t-1}^{t-1} + \epsilon_{i,t} \quad (7)$$

A higher δ indicates a stronger reliance on the latest information relative to past expectations. (Note that $f_{i,t}^t$ is equivalent to $f_{i,t}^{t+h}$ in previous sections, suppressing the $h = 0$ term since we only study $h = 0$ forecasting in this section.)

7.3.2 Extrapolation

$$f_{i,t}^t = P_{t-1} + \phi_i (P_{t-1} - P_{t-2}) + \epsilon_{i,t} \quad (8)$$

The parameter $\phi_i > 0$ measures the fraction of previously-observed price change which is added to the last price. Higher values of ϕ_i reflect more extrapolation.

7.3.3 Overreaction

Following [Afrouzi et al. \(2023a\)](#), we estimate the overreaction model as:

$$P_t - f_{i,t}^t = \alpha_i + \beta_i (f_{i,t}^t - f_{i,t-1}^t) + \epsilon_{i,t} \quad (9)$$

The left side of the equation is just the forecast error in period t . The parenthetical term on the right denotes the forecast revision between periods $t - 1$ and t . Parameter values $\beta_i < 0, \beta_i > 0$ indicate overreaction and underreaction.²⁷

7.3.4 Double-adaptive expectations (DAE)

The double-adaptive expectations model ([Holt, Porzio, and Song, 2017](#)) accounts for forecast errors in both price levels and price trends:

$$f_{i,t}^t = f_{i,t-1}^t + \eta (P_{t-1} - f_{i,t-1}^{t-1}) + \gamma [(P_{t-1} - P_{t-2}) - (f_{i,t-1}^t - f_{i,t-1}^{t-1})] + \epsilon_{i,t} \quad (10)$$

The coefficient η captures adjustments based on recent forecast errors in price levels, while γ captures adjustments based on errors in price trends.

²⁷Because the strict one-period-ahead forecast is unavailable in our experimental design, we approximate it using linear interpolation between the current-period and two-period-ahead forecasts. This approach preserves monotonicity without imposing additional structural assumptions on belief dynamics.

7.3.5 Forecasting model results

The three simplest models show, in Table 4: a high degree of adaptation to recent prices ($\hat{\delta} = .917$); modest positive extrapolation ($\hat{\phi} = .199$); and equal amounts of under- and overreaction ($\hat{\beta} = .009$). The R^2 s of these models range from .168 to .501.²⁸

The double-adaptive model fits forecast data substantially better (mean $R^2 = .679$). The first parameter η is exactly equivalent to δ in the simpler Adaptive model (i.e., if $\gamma = 0$ in the double-adaptive model the two parameters would be equal). The across-subject distributions of those two parameters are also quite close (the η mean is .648). The distinct feature of the double-adaptive model is that price changes which outstrip forecast changes pump up current forecasts (that’s what the parameter γ , estimated as $\hat{\gamma} = .572$, does). Figure D.4 plots histograms of the individual-level estimates of each model parameter.

Our participants show almost no overreaction. This differs from experiments in which people forecast artificial time series, where overreaction is the main departure from rationality (Afrouzi et al., 2023a). Forecasting their own collective behavior instead produces adaptive and extrapolative forecasts, so the best-fitting model depends on what is being forecast.

Table 4: **Summary Statistics of Parameter Estimates and Fit by Forecasting Model**

Model	Mean	SD	Mean R^2	N
Adaptive Expectations (δ)	0.917	0.479	0.501	354
Extrapolation (ϕ)	0.199	1.019	0.301	353
Overreaction (β)	0.009	0.612	0.168	359
DAE: Level Error (η)	0.648	0.361	0.679	358
DAE: Trend Error (γ)	0.572	0.471		

Notes: Mean and SD denote the mean and standard deviation of the OLS model parameter estimate across participants for each model, respectively. Mean R^2 reports the average OLS goodness-of-fit across participants, and N indicates the number of participants with non-missing estimates. The slight variation in sample size N across models is due to differences in model specifications (e.g., lag structures), which prevented estimation for a small subset of participants with insufficient observations. For the Double-Adaptive Expectations model, results are reported separately for the level forecast error and trend forecast error parameters.

Figure 7 illustrates how closely the actual forecasts match those fitted by the best-fitting models.

The estimated parameters differ across the Standard, High-Stakes, and Experienced conditions, and several of the pairwise comparisons are statistically significant (Table D.7). We

²⁸There are some small differences in predictive fit of forecasting models comparing Standard, High-Stakes, and Experienced sessions (compare Table 4 and Table D.5) but mostly their estimated coefficients and R^2 s are close together. We do not test for differences across session types because the design is not adequately powered to detect them.

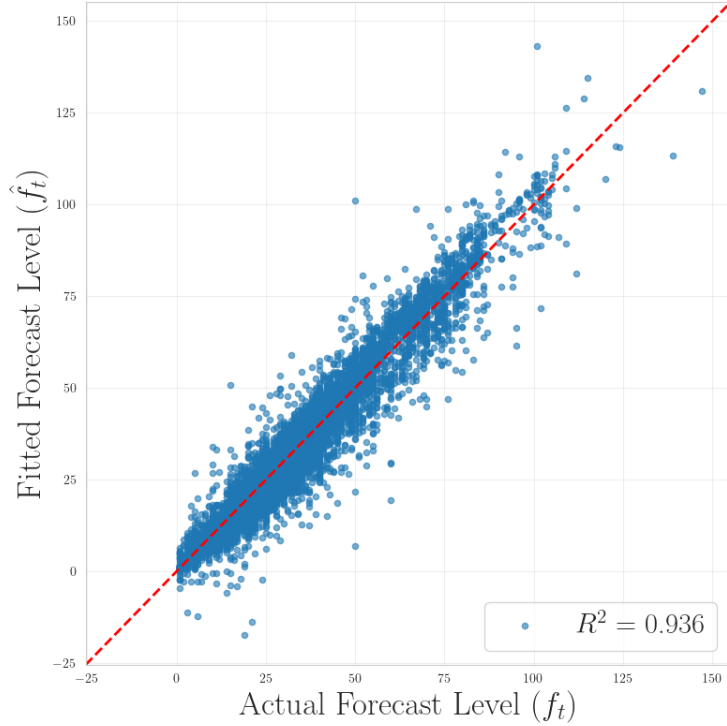


Figure 7: **Double-Adaptive Model: Actual vs. Fitted Forecasts**

We check robustness and compare realized forecast levels to the model’s fitted predictions under different representations: Figure D.6a forecasts expressed as a percentage of lagged price; Figure D.6b high- vs. low-earners; and Figure D.6c pre- vs. post-crash periods. The dashed line indicates the 45-degree line where fitted values equal observed values.

Alt text: Scatter plot of fitted versus actual forecast levels from 0 to 150 with a dashed 45-degree line; points cluster tightly along the line, with R squared equal to 0.936.

caution against reading much into these differences: they are not consistent across the five models, and no parameter shifts in a way that clearly maps to how experience or higher stakes should reshape expectation formation. We treat them as unsystematic variation rather than evidence for a specific mechanism.

Detailed parameter estimates for the High-Stakes and Experienced sessions are reported in Table D.5.

As a more demanding joint test, we regress both experimental bonus measures on all five expectation-formation parameters simultaneously (Table D.6). For each bonus type (Market & Forecast), we ran a LASSO regression against the five forecast model parameters and a battery of controls. Most controls came from survey questions and included self-reported risk preferences and demographic data. Other controls like average DOSE ρ and total number of orders placed came from the participants’ choices during the experiment. Participant location (in-lab or Prolific) was included as an indicator variable that took the value of

one if the participant was located in the lab (zero for Prolific). Each LASSO selection was followed up by a post-selection OLS regression with robust standard errors on the selected regressors. The five forecast model parameters are considered key variables in these models and so are present in the post-selection regressions regardless of whether they were selected by the LASSO. Any regressor present in Table D.6 that LASSO shrank to zero is indicated by a dagger ($\dagger$).

In predicting the forecast bonus, the adaptive and extrapolation parameters were significant. For the market bonus, the extrapolation (ϕ) and overreaction (β) parameters were significant and were selected by the LASSO, while the adaptive, level-error, and trend-error parameters were shrunk to zero and retained only by design. The forecast parameters show more significance when predicting bonuses in the absence of controls (Table D.8), but do not seem to be robust. We repeated this analysis, this time restricting the data to only the in-lab participants with biometric data. In these regressions, we included the participants' average tonic EDA activity as well as the standard deviations. Neither of these measures was selected by the LASSO regression. In fact, nearly all of the forecast parameters were shrunk to zero by the LASSO.

These joint regressions suggest that heterogeneity in forecasting behavior is correlated with the variation in participant payoffs. The adaptive model serves as the most robust predictor of the forecast bonus, while the overreaction model consistently explains the market bonus. The weaker evidence for extrapolation after controlling for session differences suggests that unobserved session-level factors may drive much of that variation.

8 Conclusion

This paper reports new evidence about asset price bubbles in experimental CFV markets. We combine trading data with measures of emotional arousal from electrodermal activity (EDA), elicited risk preferences, price forecasts, and post-experiment textual responses. No prior study has examined all these measures together. We also do so at a substantial scale (19 sessions, $N = 362$ subjects). We also compare baseline Standard sessions with smaller samples of High-Stakes ($N = 59$) and Experienced ($N = 72$) participants.

Table 5 summarizes the main results, and refers to the Tables and Figures which show those results.

At the session level, bubbles form reliably and typically, though not always, crash close to the fundamental value by the final period, even under higher stakes and among experienced traders. Next-period returns are strongly predicted by market variables, above all lagged returns and near-term forecasts. Tonic physiological arousal adds little to this period-

Table 5: **Summary of Main Empirical Results**

Result	Reference
<i>Bubble formation and return prediction</i>	
Reliable bubbles form in all three treatments	Figures 2–B.5
Next-period returns are strongly predicted by market variables (lagged returns, near-term forecasts).	Table 2
Tonic EDA arousal adds little next-period predictive power.	Tables 2, 3
<i>Physiological arousal effects</i>	
Tonic arousal is less volatile among high earners, whose arousal rises and peaks before the crash, while low earners’ arousal rises after it.	Figures C.1 and 4
<i>Forecasts and expectation formation</i>	
Forecast errors are biased, autocorrelated, and predictable from forecasts.	Figures 5, 6
Various historical models explain forecasts; double-adaptive is the best	Table 4, Figure 7

by-period prediction. Over longer horizons, arousal is more informative: When smoothed session-level arousal is high, forward returns tend to be higher, and when it is low the reverse holds.

Though aggregated arousal is uninformative about price changes after controlling for other observables, it is informative about individual traders’ behavior. Elevated arousal raises the chance of de-risking (actually selling) within a period. More importantly, EDA is less volatile within a session among high earners, whose arousal is mostly stable but rises and peaks before the crash. Low earners’ arousal is more volatile, and is low at the crash then keeps climbing after the crash (Figure 4).

That anticipatory pattern we see in high earners echoes the anterior-insula activity seen using fMRI by [Smith et al. \(2014\)](#) and the tie between emotional reactivity and weaker returns using psychophysiology (similar to ours) in [Lo and Repin \(2002\)](#). The fact that we can recover it with inexpensive EDA wristbands, rather than fMRI or more elaborate measurement, makes these markers potentially easy to scale. Their convergence with earlier evidence is reassuring and suggests a promising path to study further.

Forecasts often depart from a rational benchmark, especially as the horizon h increases. Errors are biased, autocorrelated, and predictable from the forecasts themselves. Those differences are also associated with performance: higher earners forecast more accurately and cut their holdings earlier than the price peak. Various models of expectation formation fit these forecasts well, with the double-adaptive model fitting best. Some forecast parameters (extrapolation and overreaction) are associated with earnings, though not robustly.(Table 4,

Figure 7, Table D.6).

The forecasting irrationalities are notable because the prices they are forecasting are created by the traders themselves. This is clear evidence against rationality of expectations in the sense of Muth (1961a)

It is notable that the best-fitting adaptive models are typically different from those that fit artificial data-generating processes such as AR(1) in experiments (e.g., Afrouzi et al., 2023a). One hunch is that the double adaptive model which fits well here might also fit in those domains, because the second term in the double adaptive specification is small in magnitude.

One thing we do not find is any role for risk premia: Despite eliciting risk preferences in every period (using a Bayesian-optimized adaptive design), contemporaneous risk had no relationship with prices. The bubbles here are driven by beliefs and behavior, not by any changes in risk.

We are enthusiastic about the experimental bubble research program. Because so much has been learned since 1988, it is tempting to conclude that there is not that much left to do. We disagree.

An advantage of lab experiments is that while we have to wait patiently for sporadic historical bubbles, experimental bubbles can be created over and over. Until relatively recently in financial economics, historical bubbles were thought to be exotic episodes—such as tulip bulbs and the South Sea Company—few and far between, for which there was no clear recipe. Experiments like ours show that in a lifelike stock-and-bonds design, bubbles can be created reliably. Nothing except resources and time prevents an experimenter with good ideas and tenacity from creating very large datasets of hundreds of experimental sessions, or more. There is surely a lot to learn from such a large data set—about trader skill vs luck, experience across sessions and memory (cf. Bordalo, Gennaioli, and Shleifer, 2020; Malmendier, 2021; Wachter and Kahana, 2024), and bubble sensitivity to design features such as interest rate changes. In addition, experiments are well-suited to study those asset markets (particularly emerging markets for stocks) which include a mixture of experienced professionals and inexperienced retail traders (Gao et al., 2023). In experiments that mixture of trader sophistication can be partly controlled.

Another research agenda follows from the empirical return predictability that we observed. Because short-horizon returns seem to be forecastable from observable market and behavioral variables, it should be possible to design algorithmic traders that use real-time trading data to exploit human vulnerabilities. This could be tested in hybrid algorithmic-human experiments.²⁹ This question is not new. See Farjam and Kirchkamp (2018); Miller

²⁹There is some evidence about such hybrid designs.

(2008).)

One possibility is that such traders deflate hybrid-market bubbles, perhaps timing and therefore accelerating the eventual crash. Or it could be that sophisticated algorithmic traders instead inflate bubbles to even higher prices, hoping to sell before a crash (aka “pump-and-dump”, as it is known in practice). We also have no evidence about how human traders adapt to such algorithmic traders in markets that create bubbles. It is a good topic for an energetic research program.

Finally, how much we learn from experimental data depends, of course, on whether such evidence generalizes to patterns in natural asset markets and theories about them. In this regard, note that our experiments (and many others) reproduce two important features of natural markets: (1) Negative returns are larger in magnitude than positive returns, leading to negative skewness³⁰. And (2) there is a beliefs-portfolio gap as in natural markets (see details in Section D.2). Nothing in our design or instructions mechanically creates these two patterns; they come from choices of participants. The simplest explanation is that, to some extent, there are approximately parallel mechanisms at work in the lab and field. If so, this is good news for the continuing promise of highly-controlled lab experiments to generate hypotheses about field data and new theory.

³⁰See [Mandelbrot \(1963\)](#); [Fama \(1965\)](#); [Beedles \(1979\)](#); [Kon \(1984\)](#).

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Appendix

for

Skin in the Game:

**Electrodermal Arousal, Expectations,
and Bubbles in Experimental Asset Markets**

Appendix A. Background, Design, and Sample

A.1 Extended Literature Review

A.1.1 Time-Varying Risk Premia

Time-varying risk premia have long been posited as a key mechanism underlying asset price bubbles. The concept followed soon after the onset of the efficient market hypothesis, when Shiller found that asset prices fluctuate more than their fundamental values. Further work by [Fama \(1984\)](#) and [Fama and Bliss \(1987\)](#) uncovered systematic, time-varying patterns in excess returns across the term structure, stylized facts that any dynamic model must explain. Follow-up yield-curve regressions by [Campbell and Shiller \(1991\)](#) confirmed that these movements could not be reconciled with the traditional rational expectations hypothesis, pointing instead to sizable and predictable fluctuations in term premia. Taken together, this evidence firmly established that risk compensation was state-dependent rather than constant over time and that risk premia of financial assets appeared to fluctuate over time.

A central unresolved debate concerns what drives time variation in expected returns: changes in risk preferences (or risk-bearing capacity) versus changes in beliefs about future payoffs ([Cochrane, 2011](#)). Under the first view, predictable excess returns arise because the representative (or marginal) investor becomes more risk averse in downturns (or faces tighter intermediary constraints), raising required compensation for holding risky assets; in these models, prices fluctuate largely through discount-rate variation ([Campbell and Cochrane, 1999](#); [Cochrane, 2011](#)). Under the second view, predictable returns may instead be driven by belief dynamics—learning, disagreement, extrapolative expectations, or shifts in perceived probabilities of rare disasters—so that fluctuations in expected returns reflect changing expectations about fundamentals and risks rather than changes in preferences ([Hong and Stein, 2007](#); [Gabaix, 2012](#); [Greenwood and Shleifer, 2014a](#)).

Subsequent work in macro-finance has sharpened the empirical distinction between these competing mechanisms by linking each to different observable proxies. A prominent approach measures time-varying risk aversion (or the price of risk) using option-implied quantities such as the variance risk premium, which isolates investors’ willingness to pay for protection against future volatility and has strong predictive power for aggregate equity returns ([Bollerslev, Tauchen, and Zhou, 2009](#)). A different literature measures belief dynamics using survey-based expectations and disagreement, showing that dispersion in forecasts captures slow-moving shifts in market beliefs and helps explain predictable variation in expected returns, particularly in fixed income markets ([Kim and Orphanides, 2007](#)). This measurement-based distinction reflects a broader state-of-the-art view: time-varying risk premia are generated by both preference-based and belief-based channels, but the literature increasingly

operationalizes them separately to evaluate their relative quantitative importance.

Modern empirical work increasingly attempts to separately identify these channels, recovering evidence of both time-varying risk aversion and beliefs in household and institutional portfolio choice settings.

Extended to experimental markets, as participants update their perceptions of risk—whether due to changing economic conditions, policy shifts, or psychological factors—their required compensation for bearing risk may also fluctuate. These changes in required compensation (risk premia), which can arise either from belief dynamics about the amount of risk or from shifts in risk tolerance or risk-bearing capacity, can generate asset price fluctuations independent of fundamentals, potentially leading to mispricing and bubbles.

Empirical investigations, such as those by [Cohn et al. \(2015\)](#) and [Guiso, Sapienza, and Zingales \(2018\)](#), provide supporting evidence that individual risk assessments can vary substantially over time in response to evolving market uncertainties. Collectively, this body of work reinforces the view that time-varying risk premia are a rational driver of bubble phenomena. [Brandt and Wang \(2003\)](#) also found time-varying risk aversion (TVRA) in stock and bond markets in response to inflation.

Real-world high-stakes environments, such as financial system rescues, executive compensation, and asset price bubbles, often require insights that extend beyond field data. Empirical validation of whether High-Stakes influence behavior can take months or years for suitable real-world data to emerge, which is why lab experiments are ideal incubators for simulating various behaviors, including those that occur when higher stakes are at play. By varying payoff schemes and framing within the lab, researchers can efficiently test theories about High-Stakes decision-making without waiting months or years for suitable field data to become available.

Empirical evidence on the influence of large incentives outside the laboratory is mixed. Using brokerage account records, [Sui and Wang \(2023\)](#) shows that investors exhibit more substantial biases and perform worse in higher-stakes real accounts than in lower-stakes simulated accounts, contrary to theoretical predictions that High-Stakes should improve performance. The study finds that across tasks conducted in both the United States and India, very high reward levels often lead to trading performance declines. This is in line with [Yu \(2015\)](#), who argue that, contrary to the predictions of efficiency-wage models, extremely high monetary incentives can impair rather than enhance productivity (i.e., “choking under pressure”), mainly due to high psychological stress.

Higher incentives can also influence how beliefs are formed, not merely the decisions themselves. [Ambuehl \(2017\)](#) shows that when information acquisition is costly, larger incentives lead participants to select and interpret information in a manner that favors taking

action. In asset market experiments, findings are mixed. [Cui, Feltovich, and Zhang \(2022\)](#) demonstrates that both relative-performance bonuses and loss-framed payment schemes independently drive the price of a risky asset above its fundamental value. Other studies find different results. [Enke et al. \(2023\)](#) argues that, even when stakes are raised from none to standard to very high, there is only a modest impact on reducing cognitive biases across four tasks—testing base-rate neglect, anchoring, and failure of contingent thinking.

Given the mixed and limited findings on the impact of higher incentives on asset bubble formation, we conduct a secondary analysis in [Section B.1.3](#) to investigate the effect of elevated stakes on bubble formation within our experimental framework.

A.2 Experimental Design and Procedure

A.2.1 Price Formation

For every session s and period t , the market price is determined as the price that maximizes volume. A bid submitted by participant i in period t of session s is a tuple consisting of price and quantity, (p, q) . Participants may submit up to three bids per period. The set of all bids submitted in a given session and period is represented by $B_{s,t}$. Likewise, the set of all asks is denoted by $A_{s,t}$, with similar notation for individual asks, (p, q) .

Let $P_{s,t}$ be the set of all prices submitted as bids or asks,

$$P_{s,t} = \{p \mid (p, q) \in B_{s,t} \cup A_{s,t}\}, \quad (11)$$

We define the cumulative buy quantity Q^B available at a given price p as the total quantity of all bids at or above price p ; formally:

$$Q^B(p) = \sum_{(p', q) \in B_{s,t}} q \cdot 1_{p' \geq p} \quad (12)$$

This represents the total quantity of shares that all participants are willing to purchase at a given price.

We define the cumulative sell quantity Q^A available at a given price p as the total quantity of all asks at or below price p ; formally:

$$Q^A(p) = \sum_{(p', q) \in A_{s,t}} q \cdot 1_{p' \leq p} \quad (13)$$

For a given price p , volume is determined by

$$V(p) = \min(Q^B(p), Q^A(p)). \quad (14)$$

The market price is then

$$p_{s,t}^* = \arg \max_{p \in P_{s,t}} V(p). \quad (15)$$

All prices are rounded down to the nearest integer value, a consideration made to simplify the user interface.

If there is no intersection between the bid and ask, the mid-point between the highest bid and the lowest ask is reported as the market price for that period.

A.2.2 Payment Details

- A base payment of \$20 and an average bonus of approximately \$6.66
- Bonus payment is calculated based on: (i) participants' trading results at a rate of 200 experimental CASH units to \$1, (ii) forecasting ability (i.e., each price forecast that is within 2.5 units of the actual price is rewarded with five CASH), (iii) a randomly chosen lottery that participants select from the experiment, and (iv) the quiz result (participants receive \$0.25 for each of the five comprehension questions answered correctly).

A.2.3 Experimental Procedure

1. Participants signed a consent form agreeing to participate in the experiment.
2. Participants filled in a short socio-demographic questionnaire where they provided information on their age, gender, study background, work experience, and level of English.
3. Subjects watched a video with the experimental instructions on how to trade in the software. Participants were provided with written instructions and had to respond to five comprehension questions on the experimental software. If participants could not answer at least one question correctly, they were unable to participate. This was done to ensure that we only allowed participants who understood the instructions of the experiment.
4. Participants then logged in to the market to perform the actual trading experiment. In-lab participants had their biometric data measured using either an Empatica E4

SCR device or a newer (acquired April 2024) Empatica EmbracePlus device.

5. Participants were then asked to respond to a post-experimental survey where they were asked to qualitatively outline their strategy.

A.2.4 Biometric Measure Details

- Electrodermal activity (EDA), also known as skin conductance data, refers to the measurement of the electrical conductance of the skin, which varies with its moisture level. This physiological parameter is directly influenced by the autonomic nervous system, particularly the sympathetic nervous system, and is often used as an indicator of psychological or physiological arousal. When a person experiences emotional stress or excitement, the sweat glands in the skin become more active, resulting in increased skin conductance. EDA data are collected using sensors attached to the skin, which detect these changes in conductance. The data are released in the form of continuous signals that reflect the variations in skin conductance over time. Analyzing these signals can provide insights into a person’s emotional state, stress levels, and overall autonomic activity. The Empatica E4 devices that we used sampled EDA at a rate of 4 Hz.

The EDA signal is preprocessed at a participant level by first winsorizing it to 99 percent and normalizing it to fall between 0 and 1 (inclusive) after (Cordes, Nolte, and Schneider, 2023). This step manages outliers and allows us to compare all participants. Further preprocessing and feature extraction were done using the `NeuroKit2` package available for Python (Makowski et al., 2021). The EDA signal was then decomposed into phasic (rapid, transient changes) and tonic (slow, baseline level) components by using a built-in `NeuroKit2` routine that performs these steps internally. We used the `NeuroKit2` default cutoff frequency settings of 0.05 Hz using a 5th-order Butterworth filter, thereby allowing only frequencies higher than 0.05 Hz to pass through.³¹

- Heart Rate (HR) data refer to the measurement of the number of heartbeats per minute, which serves as an essential indicator of cardiovascular health and autonomic nervous system activity. It is influenced by factors such as physical activity, emotional states, and overall health. When a person experiences stress, excitement, or physical exertion, the heart rate typically increases as part of the body’s autonomic response. EmbracePlus uses a total of nine LEDs and four photodiodes to record the photoplethysmography (PPG) at the wrist. Hence, it records a peripheral and mechanical

³¹All results are robust to frequency cutoffs of 0.01 Hz as well.

signal of cardiac activity, so it is more precise to refer to its measure of cardiac activity rate as pulse rate rather than heart rate. All wrist-worn biometric devices that use the PPG really are measuring pulse rate rather than heart rate, and FDA-cleared devices that use a similar technology usually refer to ‘pulse rate’ measurements. PPG is a non-invasive optical technique used to measure heart rate by detecting changes in blood volume in the microvascular bed of the skin. Since the PPG is a measure of blood circulation (which depends on the heart’s pumping quality), it reflects the mechanical activity of the heart rather than the electrical activity of the heart.

- Blood Volume Pulse (BVP) data refer to the measurement of the change in blood volume within the microvascular bed of tissue. It is typically detected through PPG sensors placed on the skin. BVP is a measure of the cardiovascular system’s activity, specifically the pulsatile component associated with each heartbeat. By illuminating the skin and detecting the intensity of light that either passes through or is reflected by the blood vessels, BVP sensors can capture variations in blood flow that correspond to cardiac cycles. These fluctuations allow researchers to infer heart rate and heart rate variability, both of which are valuable indicators of autonomic nervous system function, particularly the balance between sympathetic and parasympathetic activity. The BVP signal is obtained from our Empatica devices by a proprietary algorithm (of Empatica) that combines light signals of multiple specific wavelengths to illuminate tissue, and the reflected (or transmitted) light is then detected. The variations in that light—caused primarily by changes in arterial blood volume—yield the PPG signal. It has a fixed sampling rate of 64 Hz (64 times per second).
- Temperature (TEMP) refers to the measurement of a participant’s body temperature, measured in °C, at the wrist. Fluctuations in skin temperature can provide insights into an individual’s physiological state. For example, changes in skin temperature might correlate with stress levels or cognitive effort that may occur during trading.

A.2.5 DOSE Preference Measures

We use preference measures generated by DOSE, or Dynamically Optimized Sequential Experimentation ([Chapman et al., 2024](#)). DOSE is an adaptive Bayesian procedure for estimating individual-level economic preference parameters. The procedure begins with a model of preferences and a prior distribution over the parameters of that model. After each participant response, the prior is updated, and the posterior distribution is used to select the next question that is expected to be most informative.

We chose to calculate the DOSE parameters using the CRRA utility function:

$$u(x, \rho_i) = \frac{x^{1-\rho_i}}{1-\rho_i}$$

where risk aversion increases with ρ_i , and $x \in \mathbb{R}^+$ is the monetary payout presented in the DOSE task.

DOSE also explicitly models probabilistic choice. For any choice between options o_1 and o_2 , with $V(o_1) > V(o_2)$, choice probabilities are given by the logit function:

$$\Pr(o_1) = \frac{1}{1 + \exp[-\mu_i (V(o_1) - V(o_2))]},$$

where $\mu_i \in \mathbb{R}_+$ is a choice consistency parameter. Higher values of μ_i imply more consistent choices.

Question selection is based on expected Kullback–Leibler divergence. Consider a finite set of possible parameter vectors θ_k , for $k = 1, \dots, K$, where $\theta_k = (\rho_k, \lambda_k, \delta_k, \mu_k)$, and let p_k be the posterior probability assigned to parameter vector θ_k . For a possible question Q_j , the expected Kullback–Leibler divergence is:

$$KL(Q_j) = \sum_{k=1}^K \sum_{a \in A} p_k l_k(a; Q_j) \log \left(\frac{l_k(a; Q_j)}{\sum_{m=1}^K p_m l_m(a; Q_j)} \right),$$

where $a \in A$ indexes the possible answers to the question and $l_k(a; Q_j)$ is the likelihood of answer a given θ_k . DOSE selects the question that maximizes $KL(Q_j)$, observes the participant's answer, updates the posterior over parameter vectors, and repeats this process for the next question.

The DOSE measures used in our analyses are posterior estimates of latent individual-level preference parameters rather than raw choice scores. The adaptive structure allows the elicitation procedure to tailor later questions to earlier answers, improving the precision of estimated risk and loss parameters. Figure A.1 represents the progress of one such DOSE task. We presented 33 tasks to participants, one for each of the 3 practice periods and 30 main market periods. Tasks are distinguished by varying the payouts presented to the participant, which in turn varied the percentages presented on screen in the pie charts.

A visual example of how the model generates ρ is shown in Figure A.1.

$$r^H = 18.25 \quad r^L = 1.75 \quad s^H = 9.5 \quad s^L = 9.0$$

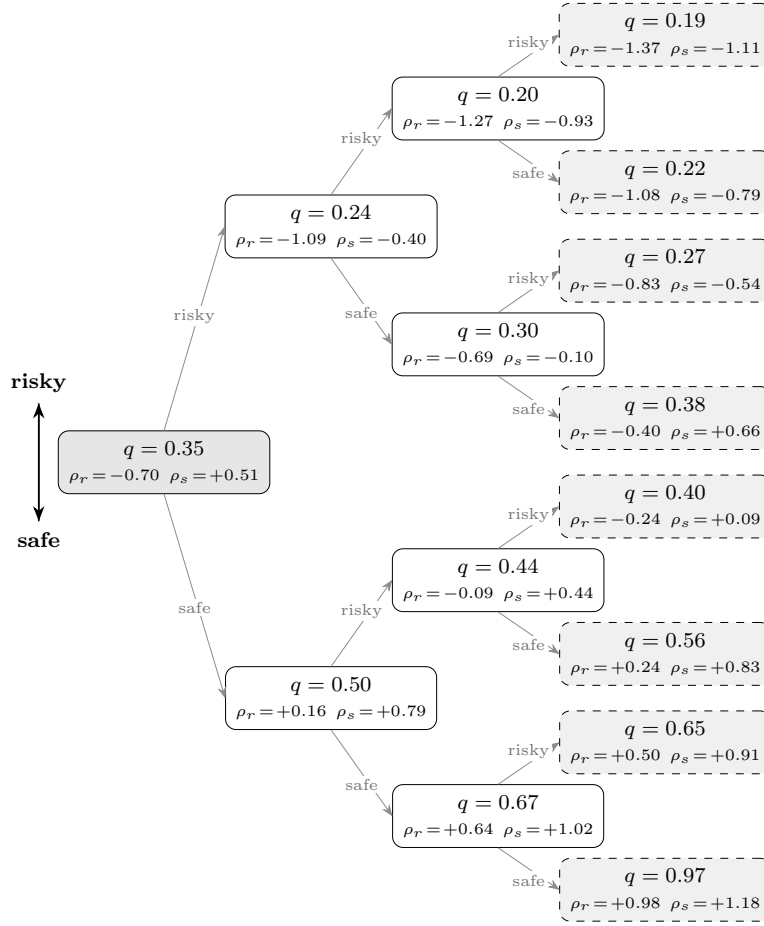


Figure A.1: DOSE Decision Tree (Round 14)

Branches upward are risky choices; branches downward are safe choices. Dashed nodes are terminal leaves. q = choice probability threshold; ρ_r = CRRA risk aversion parameter under the risky lottery; ρ_s = CRRA risk aversion parameter under the safe lottery. r^H & r^L are the high and low payouts of the risky lottery. s^H & s^L are the high and low payouts of the safe lottery.

A.3 Sample and Participants

A.3.1 Subject Differences In-Lab vs. Prolific

Laboratory participants are Caltech students, who tend to score highly on standardized academic measures, while online Prolific participants come from a broader population with more heterogeneous backgrounds. Because of this difference, we performed multiple analyses to quantify whether and by how much the trading behavior differed between these two groups. This allows us not only to understand our data better but also to gain some insight (albeit somewhat crudely) into whether differences in measurable analytical skills, rather

than education labels, are associated with systematic differences in trading behavior. We find that Caltech students statistically outperformed the Prolific students on the forecast bonus and quiz, and, within a session, on the market bonus.

Figure A.2 summarizes the differences between the two groups of in-lab and Prolific participants. Below is the demographic information for each of our sessions.

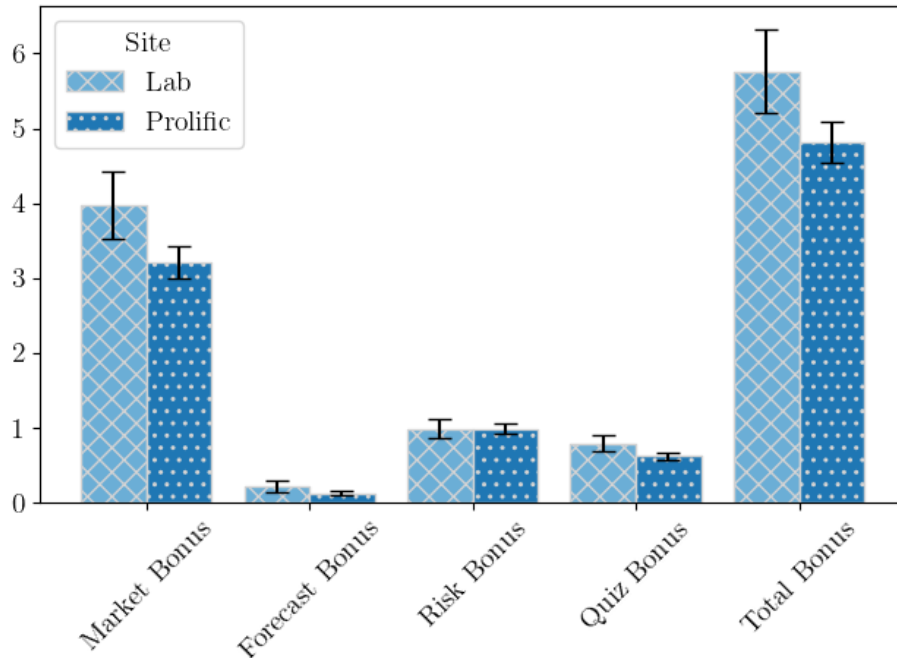


Figure A.2: **Mean Bonuses by Site**

Standard sessions only. 95 percent confidence intervals are shown.

Alt text: Bar chart of mean bonuses for in-lab versus online participants in Standard sessions with 95 percent confidence intervals; the in-lab bars are higher.

The market bonuses in a given session were all correlated since there is a predetermined amount of wealth in the session. Therefore, a high bonus for one participant or a group of participants must result in a low bonus for another participant or group. Because the two groups (in-lab vs. Prolific) are not i.i.d., a simple t -test will not suffice to compare the relative performances.

We instead employed a bootstrapping technique in which the market bonuses for the participants within each session were randomly shuffled, and the new group means were compared. This gave us a distribution against which we could compare our observed market bonus means. Figure A.3 shows the results of 20,000 iterations of this exercise. Assuming that the bootstrapped distribution of group mean differences is Student's t with 19,999 degrees of freedom, the observed difference (\$0.763) was significantly different from the

Table A.1: Summary Statistics by Site

Session	Site	General Risk				Financial Risk				Invest 100				Bonus				Trades		DOSE	
		Age	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Quiz	Mkt		FC	Risk	Att	Exe	ρ	μ		
100hm05f	L	30.33	5.33	5.00	3.00	2.33	66.67	58.33	4.00	3.45	0.02	1.55	50.33	12.00	-0.03	3.09					
	P	40.76	4.53	4.35	4.12	4.12	53.53	54.12	3.06	3.43	0.01	1.23	35.59	8.82	0.38	3.09					
19jfyhnh	L	27.00	3.80	4.50	2.80	3.75	57.00	60.00	4.00	4.44	0.05	0.95	33.80	12.60	0.34	3.06					
	P	34.50	4.80	5.00	3.80	4.67	43.00	54.44	2.17	2.65	0.01	1.14	30.45	7.45	0.81	3.17					
3olyo3ti	L	31.60	2.40	2.00	3.00	1.67	44.00	30.00	3.80	4.85	0.55	0.78	29.00	9.60	0.55	2.90					
	P	37.94	4.06	3.29	3.94	3.65	47.18	39.53	2.53	2.93	0.49	0.82	33.47	7.71	0.79	3.08					
898aphuw	L	30.00	2.67	3.00	2.67	3.00	40.00	47.50	3.75	3.53	0.04	0.75	27.25	10.50	0.80	3.37					
	P	31.25	3.42	3.58	3.75	3.50	49.58	45.83	3.17	3.35	0.04	1.00	33.25	12.25	0.76	3.28					
8h0lckuw	L	24.86	4.00	4.00	3.29	3.17	48.57	42.50	0.00	8.37	1.38	2.00	33.71	7.71	0.46	3.10					
	P	41.85	3.85	3.64	4.23	4.27	87.08	58.00	0.00	5.44	0.87	2.00	40.43	10.14	0.61	2.89					
b4opeq8h	L	28.50	4.00	2.50	2.00	2.00	63.50	65.00	3.00	3.95	0.06	0.95	35.50	11.00	0.88	3.14					
	P	36.94	3.89	4.27	3.22	3.40	46.39	48.93	2.17	3.34	0.04	0.83	39.89	10.61	0.56	3.15					
ea9wnj36	L	34.40	3.20	3.80	3.80	3.60	40.00	42.00	3.40	4.22	0.01	1.02	32.20	9.00	0.59	3.16					
	P	35.82	5.00	3.90	4.73	3.90	38.27	31.50	2.38	2.92	0.01	0.91	42.00	12.50	0.39	3.09					
j9yi8qx8	L	23.67	3.50	3.17	2.50	2.17	34.17	30.83	3.50	4.23	0.60	0.95	41.00	11.33	0.83	3.13					
	P	39.93	3.64	3.29	2.71	2.64	49.29	47.29	2.29	3.02	0.47	1.07	36.00	8.79	0.76	3.02					
jtf9f20p	L	22.40	3.00	3.00	3.40	3.60	68.00	83.00	4.40	3.74	0.02	1.01	28.20	6.00	-0.01	2.99					
	P	37.00	4.00	3.44	3.78	3.67	45.44	42.78	1.80	3.21	0.02	0.59	50.90	14.60	0.40	2.92					
lrx4o1bb	L	34.67	4.00	3.83	3.67	3.50	51.67	50.00	3.33	3.39	0.06	0.85	38.17	10.67	0.41	3.10					
	P	38.53	4.53	4.43	4.80	4.71	54.33	50.71	3.19	3.44	0.05	0.77	38.06	9.81	0.52	3.05					
m48xf7cp	L	38.25	4.00	3.50	3.25	3.00	37.50	40.00	3.00	2.14	0.02	0.69	42.50	12.50	0.29	2.82					
	P	31.39	4.22	3.75	4.06	3.44	38.89	44.38	2.79	2.89	0.01	0.75	33.21	11.00	0.70	3.12					
mhuck8jp	L	24.00	3.50	3.25	3.25	3.75	75.00	65.00	3.50	1.36	0.03	0.31	30.00	8.75	0.52	2.78					
	P	37.33	4.33	3.89	4.17	3.89	59.17	57.22	2.75	2.25	0.02	0.64	35.58	10.25	0.25	3.07					
nhreg4w	L	21.83	4.00	4.40	4.50	4.20	100.00	100.00	4.33	3.61	0.05	0.61	27.17	7.83	-0.36	2.90					
	P	33.27	4.36	3.80	3.64	3.40	47.73	42.50	2.92	3.22	0.02	1.02	32.82	9.09	0.73	3.27					
pmyan048	L	20.33	3.67	4.00	3.33	3.33	40.00	50.00	4.20	2.89	0.00	0.92	35.60	12.80	0.99	3.12					
	P	35.89	4.53	4.25	4.21	4.12	51.58	52.19	3.53	3.21	0.01	0.95	36.79	11.05	0.66	3.04					
rcnn52km	L	30.00	4.33	4.33	3.00	3.33	50.00	66.67	3.50	3.70	0.01	1.16	31.50	8.50	0.11	3.04					
	P	38.00	3.50	4.00	3.38	4.14	48.12	59.29	3.22	3.14	0.02	1.10	26.89	7.44	0.54	3.02					
txyy1leq	L	20.00	4.75	4.00	4.25	3.25	46.25	43.75	3.40	2.88	0.02	0.75	41.80	9.00	0.83	3.15					
	P	37.94	3.06	2.80	2.81	2.53	25.94	21.67	3.12	3.54	0.02	1.03	34.75	9.00	0.90	3.08					
wvjdm40	L	24.00	3.00	3.50	2.80	3.00	45.00	46.25	4.20	3.35	0.05	1.10	37.60	12.00	0.51	3.14					
	P	39.46	3.77	3.77	3.23	3.69	40.00	44.23	2.93	3.07	0.03	0.82	41.36	12.29	0.34	3.18					
yh2bjr4n	L								3.00	4.35	0.31	1.36	30.20	11.80	1.02	3.18					
	P								2.00	2.93	0.19	0.99	34.09	7.56	1.00	3.15					
z1vrkziq	L	23.50	4.00	3.50	2.50	2.00	66.25	53.75	4.00	4.28	0.02	0.59	19.50	8.25	0.97	3.04					
	P	35.50	4.06	3.81	4.00	3.81	57.25	55.69	2.12	2.86	0.02	1.06	47.12	14.88	0.31	3.10					

Notes: Bolded values indicate a significant difference at the 5% level. L = Lab, P = Prolific. *Survey measures:* General and Financial Risk are self-reported risk-taking, in general and in financial matters, on a 1–7 scale (higher is more risk-taking); Invest 100 is the dollar amount (of \$100) allocated to a risky asset in a hypothetical investment task, elicited before (Pre) and after (Post) the experiment. *Market measures:* Quiz is the initial comprehension-quiz score; Mkt/FC/Risk are the market, forecast, and DOSE-lottery bonuses; Att/Exe are attempted and executed trades; ρ and μ are the DOSE risk-aversion and choice-consistency parameters.

Table A.2: Mean Bonuses by Group

Bonus Type	Lab	Prolific	p -value	95% CI
Market	3.976	5.144	0.041	[-1.846, -0.489]
Forecast	0.218	0.125	0.004	[0.007, 0.179]
Risk Assessment	0.983	0.991	0.903	[-0.152, 0.135]
Quiz	0.787	0.682	0.035	[-0.006, 0.216]
Total	5.763	6.852	0.068	[-1.855, -0.322]

Notes: Mean bonuses by group with p -values and 95% confidence intervals. μ_{Lab} and μ_{Prolific} denote mean bonuses for in-lab and Prolific online participants, respectively. The p -value is from a two-sample Welch’s t -test of $H_0 : \mu_{\text{Lab}} = \mu_{\text{Prolific}}$, which does not assume equal variances. The 95% confidence interval is for the difference $\mu_{\text{Lab}} - \mu_{\text{Prolific}}$. The Total row reports each participant’s mean total bonus; it is slightly below the sum of the component means because the quiz bonus is missing for some in-lab participants ($n = 67$ of 90), so the columns use different denominators.

bootstrapped mean of \$0.099 ($p = 0.002$).

A.3.2 Summary Statistics

Table A.3 shows bubble metrics, number of participants, peak period, and price for all sessions.

A.3.3 Experienced Participants Selection Effects

To identify the impact of any selection effects, specifically, whether participants who returned for follow-up sessions differed from those who did not, we compared the various survey responses and market performance measures of an Experienced participant’s first session with those of participants who did not repeat the experience (see Table A.5). We found that the only statistically significant differences between the two groups (as determined by a t -test) were the number of attempted and executed trades.

The survey measures were: Age; whether they had participated in a similar experiment before (coded as “Similar” 0–1); a self-reported preference for general and financial risk appetite (exact measures can be found in Section E.3.3). Performance measures included market and forecast bonuses, their grade on the post-instruction quiz (Section E.2), the number of orders submitted, and the number of orders executed.

On average, Experienced participants attempted 32.93 trades compared to 39.03 trades for the general participant population ($p = 0.023$). We also found that on average future repeat participants executed fewer trades (8.47) than participants who did not go on to repeat sessions (11.20) ($p = 0.005$).

We also analyzed how repeat participant behavior changed over time. We find that compared to their first experiment, repeat participants seem to attempt and execute fewer

Table A.3: Session-Level Summary Statistics

Type	<i>N</i>	Float	RAD	Bias	Disp	PA	Dur	Pk. Price	Pk. Period
Standard	21	84	1.86	25.76	850	62	19	77	20
	20	80	0.77	13.94	460	32	19	47	23
	16	32	2.67	41.27	1362	87	23	102	24
	22	88	1.23	18.30	604	32	18	47	19
	22	88	0.51	7.12	235	14	3	28	17
	21	84	0.59	8.09	273	21	16	34	19
	17	68	0.95	14.33	473	34	13	48	14
	18	72	0.75	11.58	384	31	12	44	13
	16	64	0.56	7.76	260	13	6	26	11
	23	92	1.16	15.85	525	38	14	51	19
	24	96	2.12	31.79	1049	67	24	82	25
	19	76	0.65	9.61	317	26	9	40	19
	16	64	0.79	13.03	430	26	7	40	21
	18	72	1.18	15.73	519	40	18	55	19
	15	60	0.70	11.00	363	29	9	43	23
	20	80	1.94	26.88	887	66	16	80	20
	13	52	1.47	21.03	704	52	11	64	18
	20	80	1.18	15.76	520	27	22	42	24
	21	84	0.91	13.21	436	23	9	38	10
Experienced	15	60	1.15	14.15	467	38	9	52	10
	21	84	1.58	29.91	987	68	21	83	22
	13	52	0.51	5.61	197	17	4	29	12
	22	88	0.59	7.24	239	22	5	36	6
	22	88	1.16	18.45	615	54	10	67	11
High-Stakes	17	68	0.25	3.52	116	7	2	21	15
	12	48	1.23	18.06	596	33	10	48	21
	13	52	0.98	14.06	464	34	15	49	27
	17	68	0.41	6.21	209	12	6	25	25
<i>Averages by treatment</i>									
Standard	19.05	74.53	1.16	16.95	560.58	37.89	14.11	52.00	18.84
Experienced	18.60	74.40	1.00	15.07	501.00	39.80	9.80	53.40	12.20
High-Stakes	14.75	59.00	0.72	10.46	346.25	21.50	8.25	35.75	22.00

Notes: Each row is one session (by date), grouped by treatment, with treatment averages at the bottom. *N* is the number of participants and Float the total shares outstanding (four per participant). RAD (relative absolute deviation), Bias (average bias), Disp (total dispersion), PA (price amplitude), and Dur (duration) are the bubble metrics defined in Section B.2. Pk. Price and Pk. Period are the peak price and the period in which it occurred.

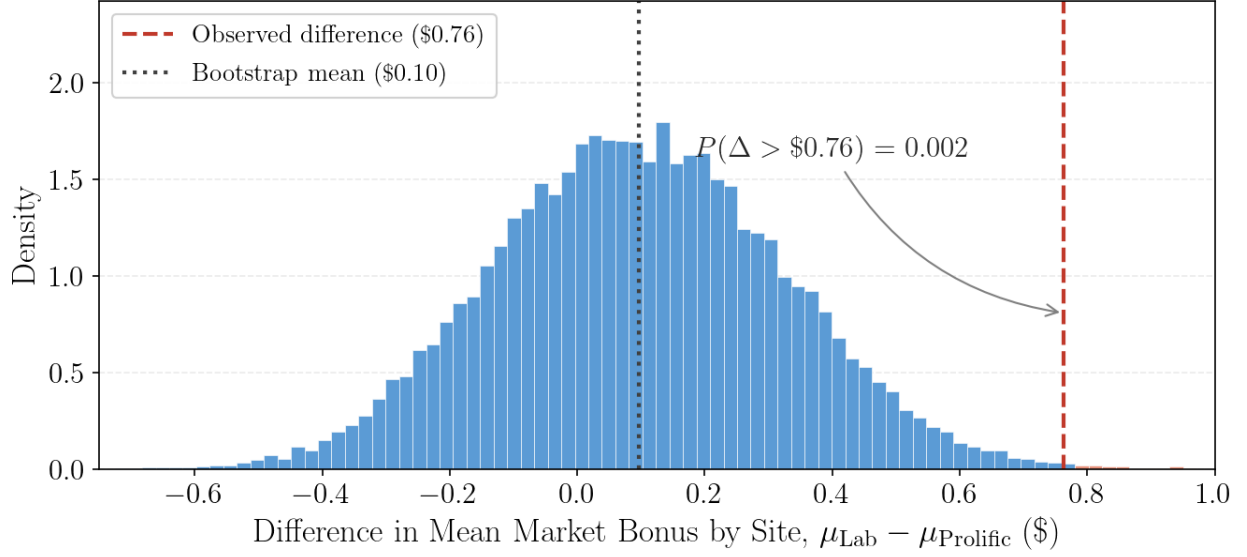


Figure A.3: Bootstrap Distribution of the Market-Bonus Difference by Site

Comparison of the mean market bonus of in-lab (μ_{Lab}) and Prolific (μ_{Prolific}) participants in the Standard sessions. Within each session, bonuses are shuffled and the difference in site means (Lab minus Prolific) is recomputed; repeating this 20,000 times generates the plotted distribution (mean \$0.099, SD \$0.226). The red dashed line marks the observed difference of \$0.763 and the dotted line the bootstrap mean. The shaded right tail is the one-sided bootstrap p -value—the proportion of resamples exceeding the observed difference ($p = 0.002$); a Student’s t approximation ($\text{df} = 19,999$) gives $p = 0.002$. The observed difference is highly significant.

Alt text: Histogram of 20,000 permutation differences in mean market bonus between in-lab and online participants, centered near zero, with a red dashed line at the observed difference of 0.763 dollars in the shaded right tail.

trades, though these differences are not statistically significant. A full table detailing how these Experienced participants traded in each repeat period is available below in Table A.6.

The mean market bonus for one-time Prolific participants ($\mu_o = \$3.09$) was compared to the first market bonus earned by repeat participants ($\mu_r = \$3.47$). The observed difference in means was $\mu_r - \mu_o = \$0.39$. Because market bonuses are correlated within a session, gains for some traders necessarily imply losses for others; a simple t -test would not suffice. Instead, we bootstrapped the means by shuffling the market bonuses within each session and recalculating the group means ($\mu_{ri} - \mu_{oi}$ where i is the i^{th} iteration of the bootstrapping process). This process was repeated $B = 10,000$ times to create a distribution of market bonus outcomes with mean $\mu_b = \$0.17$ and standard deviation $\sigma_b = \$0.26$ (Figure A.4). We then tested the null hypothesis ($H_0 : \mu_r - \mu_o = \mu_b$) by counting the number of instances in the bootstrapping process that generated a difference greater than the observed difference of \$0.39 and dividing by B . This generated a p -value of 0.208, so we cannot reject H_0 .

Table A.4: **Summary Statistics**

	All	Lab	Prolific	High-Stakes	Experienced
<i>N</i>	514	90	424	59	93
Age	35.77	27.05	37.53	40.64	37.31
Quiz Grade	2.86	3.42	2.74	2.69	3.31
Quiz 1	0.66	0.60	0.68	0.69	0.80
Quiz 2	0.53	0.53	0.53	0.53	0.57
Quiz 3	0.56	0.79	0.51	0.47	0.60
Quiz 4	0.49	0.76	0.43	0.39	0.58
Quiz 5	0.62	0.74	0.59	0.61	0.76
Market Bonus	4.94	3.98	5.14	16.84	3.37
Forecast Bonus	0.14	0.22	0.13	0.27	0.03
Risk Bonus	0.99	0.98	0.99	0.94	1.05
Total Bonus	6.66	5.76	6.85	18.73	5.28
Risk Gen (pre)	3.97	3.69	4.03	3.76	4.04
Risk Gen (post)	3.79	3.65	3.82	3.57	3.94
Invest 100 (pre)	49.58	54.27	48.63	48.55	47.80
Invest 100 (post)	48.55	53.80	47.48	48.77	48.59
<i>Race</i>					
American Indian or Alaska Native	1	1	0	0	0
Asian or Asian Indian	107	44	63	4	18
Black or African American	90	1	89	7	21
White Caucasian	266	29	237	46	51
Other	13	4	9	1	1
Prefer Not to Say	37	11	26	1	2
<i>Gender</i>					
Female	213	39	174	32	37
Male	259	39	220	25	54
Non-binary	5	1	4	1	0
Prefer Not to Say	37	11	26	1	2

Notes: Continuous variables are reported as means; Race and Gender are reported as participant counts.

We also calculated a Student's t distribution to compare the observed difference with the bootstrapped distribution with 9,999 degrees of freedom to get a p -value of 0.204.

Table A.5: Repeat Participants versus First-Time Participants

	Only Once	Repeated	<i>p</i> -value
<i>N</i>	189.00	74.00	
Age	36.25	37.44	0.431
Similar	0.11	0.05	0.191
Forecast Bonus	0.12	0.15	0.531
Market Bonus	3.09	3.47	0.204 †
Quiz Grade	2.43	2.77	0.075
Attempted Trades	39.03	32.93	0.023
Executed Trades	11.20	8.47	0.005
General Risk	4.17	3.81	0.117
Financial Risk	3.85	3.64	0.368

Notes: Bolded values indicate a significant difference at the 5% level. “Repeated” counts any participant who took part in more than one session ($N = 74$), two more than the 72 Experienced-session participants because two took part in a second Standard session. ‘Similar’ is the percentage of participants who answered “yes” to the survey question: “Have you participated in similar trading experiments before?” Market and forecast bonuses are the bonuses earned during the experiment. Quiz Grade is the score out of 5 on the comprehension quiz. Attempted Trades are the number of orders submitted, while Executed Trades are the orders that were actually filled. General Risk is a Likert response to the question: “How would you rate your risk-taking behavior in general, on a scale from 1–7,” where 1 is risk-averse, and 7 is risk-taking. Financial Risk is a Likert response to the question: “How would you rate your financial risk-taking behavior, on a scale from 1–7?” The dagger (†) on the market bonus indicates that the *p*-value was computed by bootstrapping differences due to within-session correlations (see Figure A.4).

Table A.6: Mean Trades by Experienced Sessions

Attempt	<i>N</i>	Att Trades	Exe Trades	Market Bonus	Forecast Bonus
1	72	33.24	8.62	3.47	0.14
2	72	31.18	8.59	3.22	0.03
3	24	25.50	5.75	3.80	0.05
4	4	35.25	8.75	3.40	0.03

Notes: Summary of mean trade activity and market performance based on experience level. A full breakdown is provided in the doctoral dissertation [Spoon \(2026\)](#), Table B.25.

Appendix B. Bubbles and Return Prediction

B.1 Individual Session and Forecast Plots

B.1.1 Comparison of Descriptive Statistics Across Experimental Treatments

B.1.2 Standard Session Plots

We ran 19 Standard sessions. Figure B.2 shows all 19 price paths (continued over two pages).

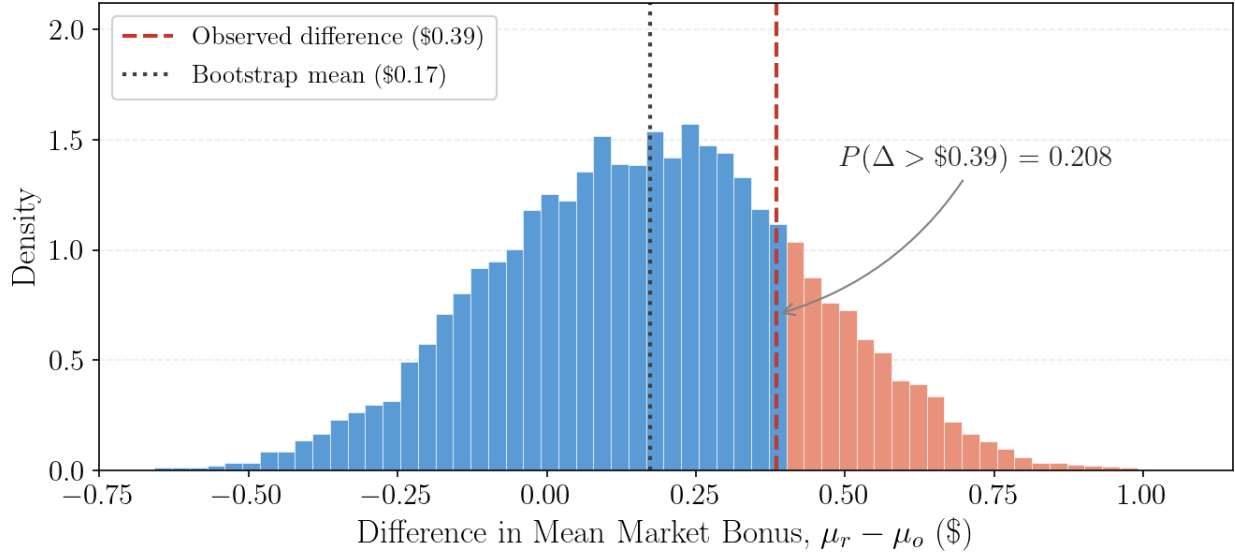


Figure A.4: **Bootstrap Distribution of the Market-Bonus Difference: One-Timers vs. Repeaters**

The mean market bonus for one-time Prolific participants ($\mu_o = \$3.09$) is compared to the first market bonus earned by repeat participants ($\mu_r = \$3.47$). Within each session, bonuses are shuffled and the difference in group means (repeaters minus one-timers) is recomputed; repeating this 10,000 times generates the plotted distribution (mean \$0.17, SD \$0.26). The red dashed line marks the observed difference of \$0.39 and the dotted line the bootstrap mean. The shaded right tail is the one-sided bootstrap p -value—the proportion of resamples exceeding the observed difference ($p = 0.208$); a Student’s t approximation ($df = 9,999$) gives $p = 0.204$. We cannot reject equality of the two means.

Alt text: Histogram of 10,000 permutation differences in mean market bonus between repeat and one-time online participants, centered near zero, with a red dashed line at the observed difference of 0.39 dollars.

B.1.3 High-Stakes Sessions

We ran four High-Stakes sessions. Each session had a payout five times greater than in the Standard sessions; we changed the final-payout conversion rate from \$1.00 per 200 CASH to \$1.00 per 40 CASH. None of these participants had taken part in a prior session.

There were bubbles and crashes in all the High-Stakes sessions, shown in Figure B.3a. High-Stakes sessions peaked about three periods later than Standard sessions (22.0, $p = 0.178$) and had smaller price peaks (35.8, $p = 0.256$). More details are in Figure B.1. With these small samples, there is too little power to detect genuine differences among the alternative market conditions. Their main purpose is to show robustness of the general bubble pattern to stakes and experience, an aim that has been important in the history of experimental economics.

We also find that participants in High-Stakes sessions were better forecasters. Table D.2

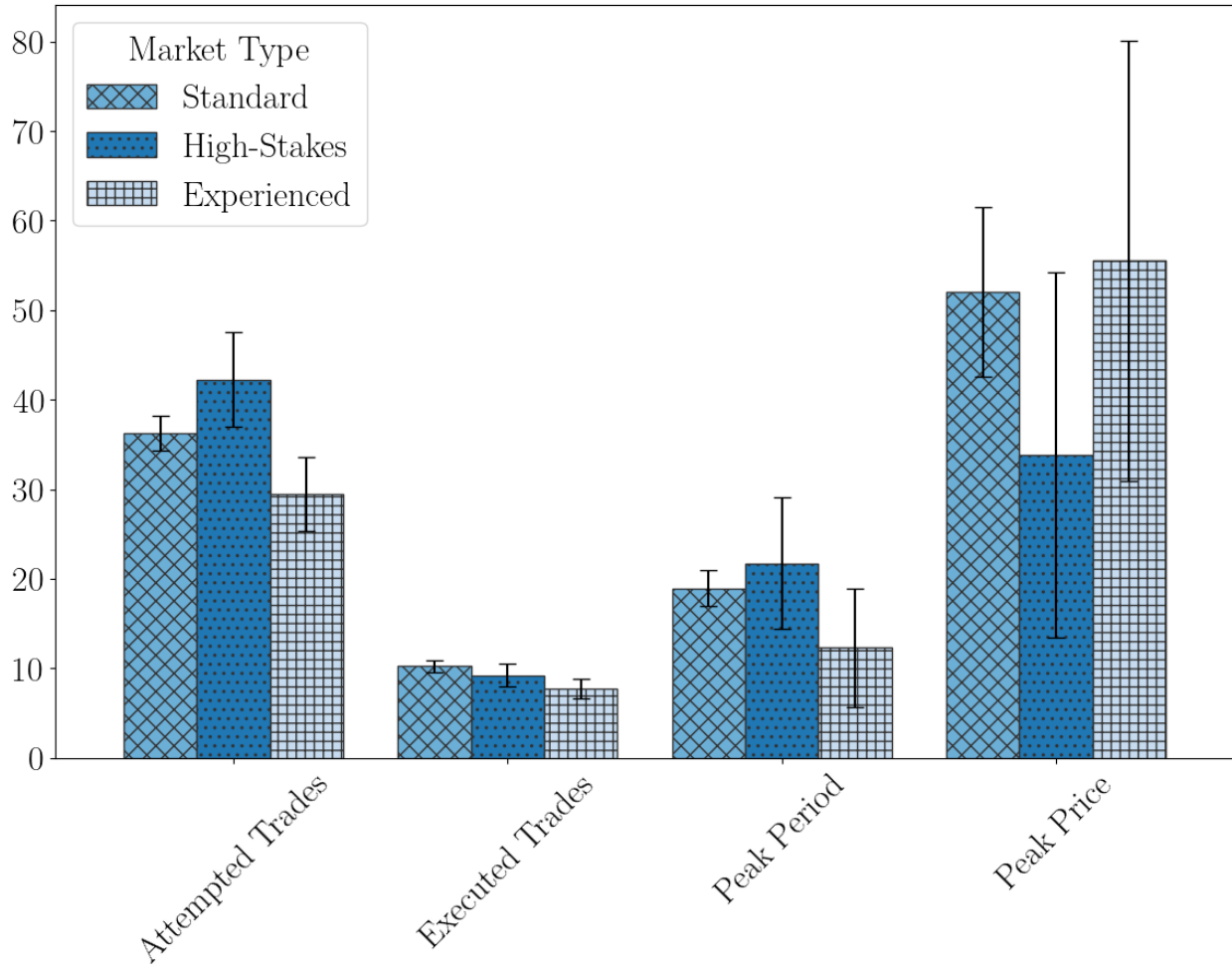
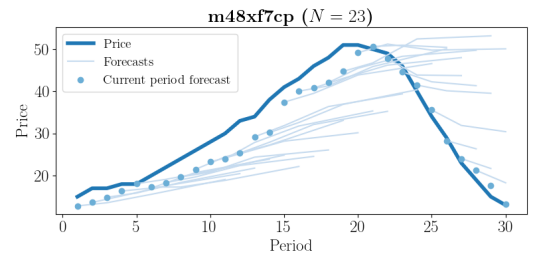
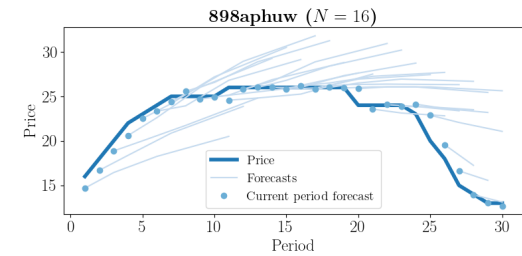
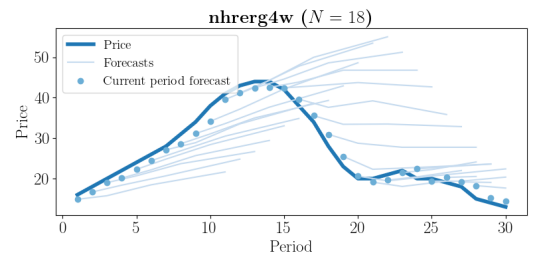
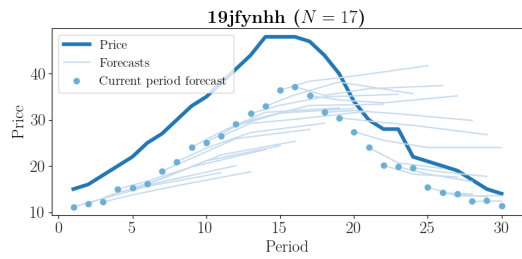
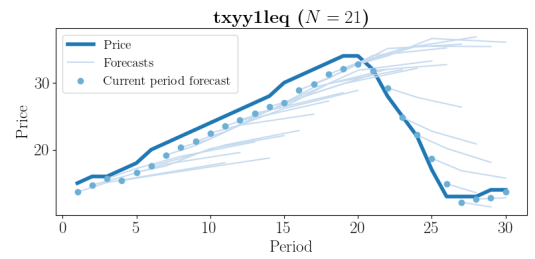
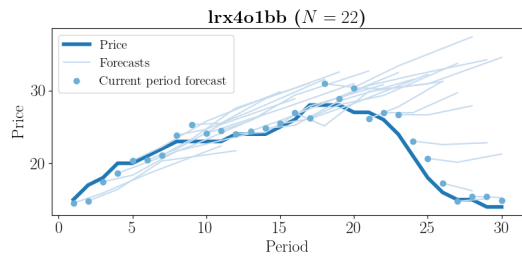
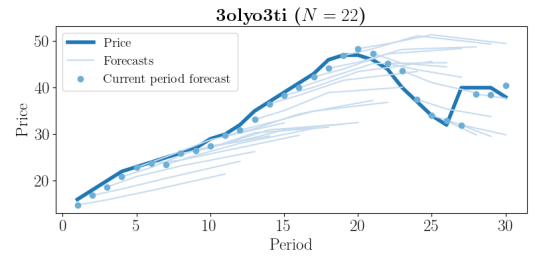
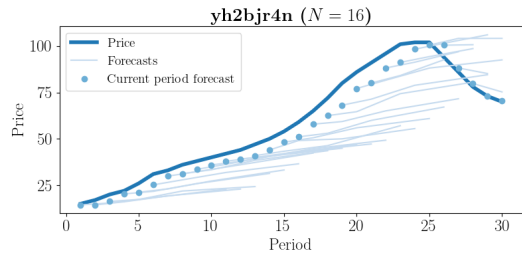
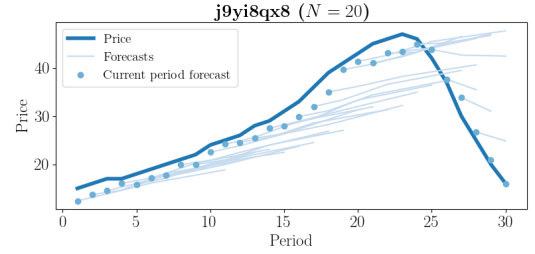
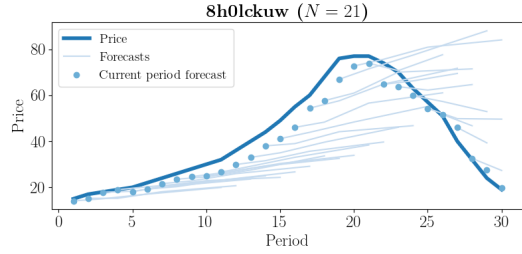


Figure B.1: Market Measures by Session Type

95 percent confidence intervals shown. Because there are only four High-Stakes sessions and five Experienced, the tests are too weakly powered to detect significant differences in most of these measures. Notice the standard error bars are larger for High-Stakes and Experienced statistics compared to the Standard sessions. Attempted (Executed) Trades refer to the average number of trades attempted (or executed) per participant in each session.

Alt text: Grouped bar chart of session-level market measures for Standard, High-Stakes, and Experienced sessions with 95 percent confidence intervals; the intervals for High-Stakes and Experienced sessions are much wider.



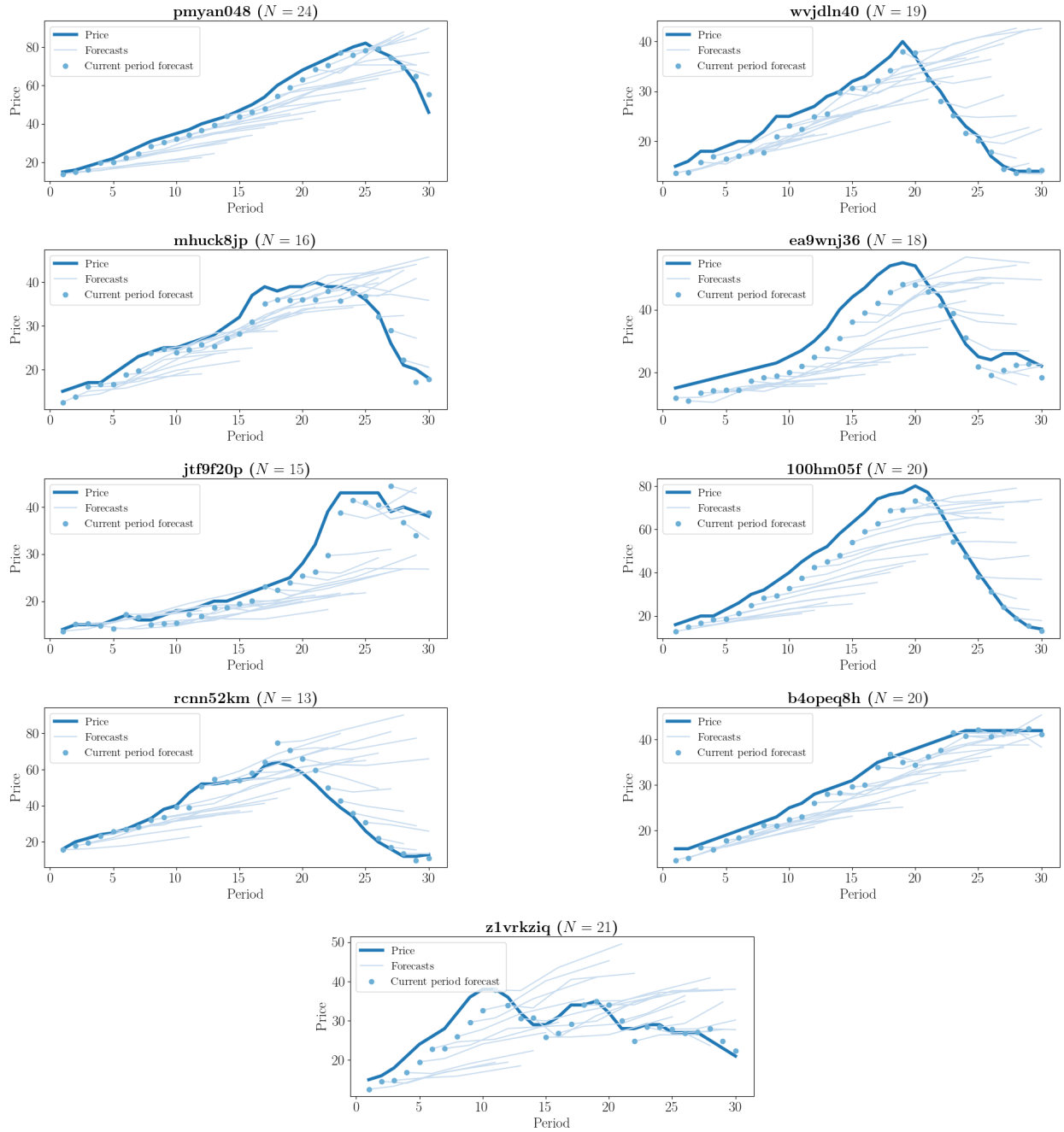
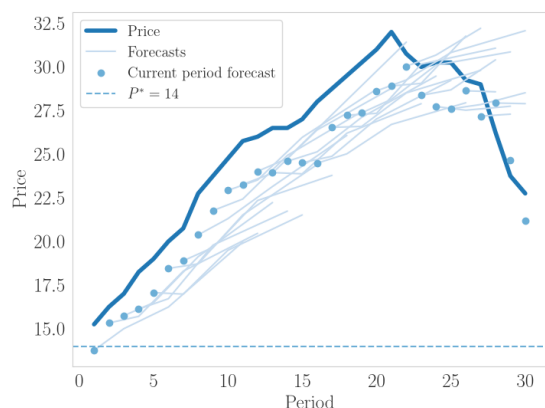


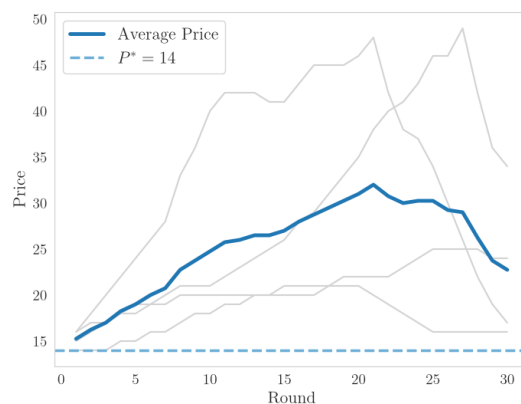
Figure B.2: All Standard Session Price Paths

The lines represent the dynamics of price levels for 30 periods. Each point represents the current period forecast, and subsequent lines represent average price forecasts for the 0-, 2-, 5-, and 10-period-ahead forecasts. Note: there was an error in session yh2bjr4n; participants were only endowed with an average of two shares of stock rather than a set number of four. The qualitative performance was similar to that of our other sessions, so we combined it into our analysis.

Alt text: Grid of 19 small line charts, one per Standard session, each showing the price path over 30 periods with current-period forecast points and forecast tongues.



(a) Average High-Stakes Sessions



(b) All High-Stakes Sessions

Figure B.3: **Prices and Forecasts in High-Stakes Sessions**

(a) Average prices and forecasts across four High-Stakes sessions. The thick line shows the average price path. Each circular point on the graph represents the current period forecast (just before the current price is announced) averaged across participants. The thin lines extending from the circular point are the “tongues” of future forecasts in period t for each of the 2-, 5-, and 10-period ahead prices. Session-specific graphs can be found in Figure B.4. (b) All High-Stakes sessions are shown here as thin light gray lines. The average High-Stakes price over time is shown as the bold line. Fundamental value is shown as a light dashed line at price 14.

Alt text: Two line charts for the four High-Stakes sessions: panel (a) shows the average price and forecasts by period; panel (b) shows all four session price paths in gray, the average in bold, and the fundamental value dashed at 14.

shows that log mean squared error ($\log(\text{MSE})$) is about half as high in High-Stakes sessions compared to Standard sessions.

Individual session plots are shown below.

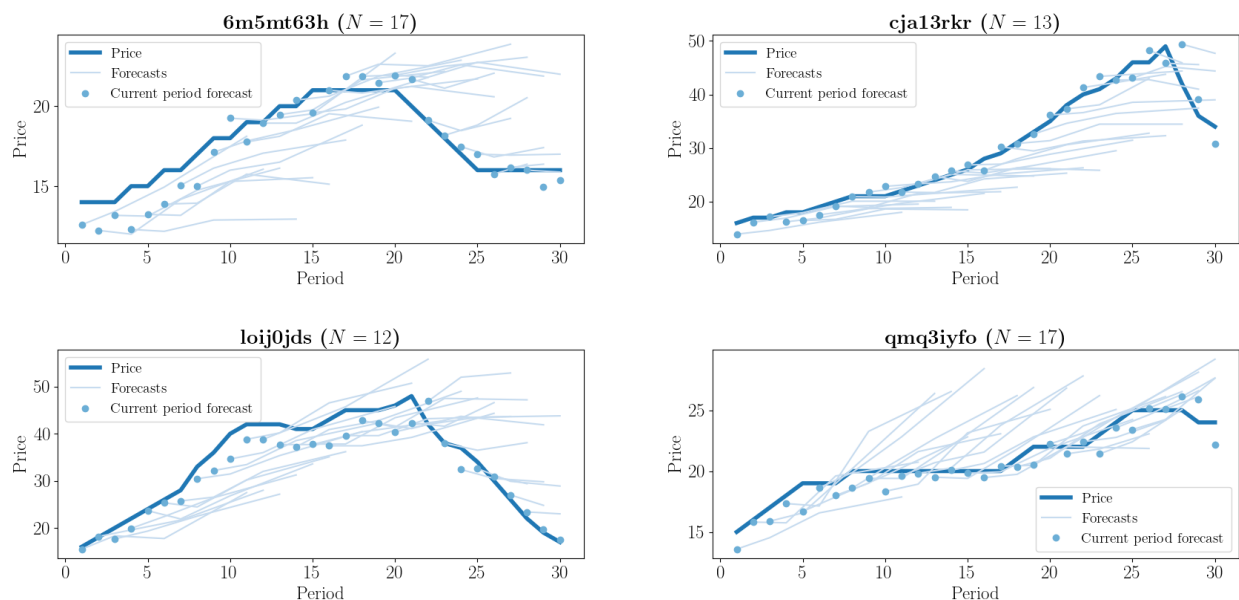


Figure B.4: **All High-Stakes Session Price Paths**

The lines represent the dynamics of price levels for 30 periods. Each point represents the current period forecast, and subsequent lines represent average price forecasts for the 0-, 2-, 5-, and 10-period-ahead forecasts.

Alt text: Grid of four line charts, one per High-Stakes session, showing price paths over 30 periods with forecast points and forecast tongues.

B.1.4 Experienced Sessions

We ran five Experienced sessions using Prolific participants who had previously traded in one or more Standard sessions.

Prior experience in the same design sometimes reduces SSW bubbles. The closest CFV design to ours, [Kopányi-Peuker and Weber \(2021\)](#), differs mainly in that its terminal period is a range rather than precisely known;³² there, three consecutive market experiences within a single session reduced price variation but not price levels.

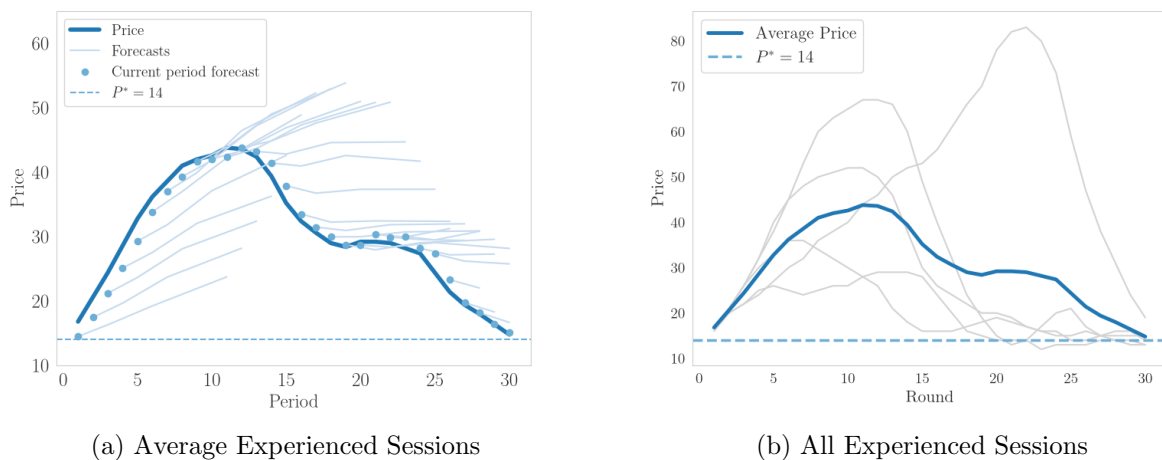


Figure B.5: **Prices and Forecasts in Experienced Sessions**

(a) Average prices and forecasts across five Experienced sessions. The thick line shows the average price path. Each circular point on the graph represents the current period forecast (just before the current price is announced) averaged across participants. The thin lines extending from the circular point are the “tongues” of future forecasts in period t for each of the 2-, 5-, and 10-period ahead prices. Session-specific graphs can be found in Figure B.6. (b) All Experienced sessions are shown here as thin light gray lines. The average Experienced price over time is shown as the bold line. Fundamental value is shown as a light dashed line at price 14.

Alt text: Two line charts for the five Experienced sessions: panel (a) shows the average price and forecasts by period; panel (b) shows all five session price paths in gray, the average in bold, and the fundamental value dashed at 14.

There were bubbles and crashes in all the Experienced sessions, shown in Figure B.5a. Peak prices were similar to Standard sessions (53.4 compared to 52.0, $p = 0.831$), but the Experienced sessions peaked significantly earlier (period 12.2 compared to 18.8, $p = 0.038$). More details are in Figure B.1. With only five Experienced sessions this comparison is

³²They have only six traders per session, interest cash cannot be used to buy assets, and subjects are told the terminal period is sometime between 25 and 40 periods.

exploratory, but it is consistent with replicated evidence that trader experience reduces bubbles (Huber et al., 2024).³³

Unlike the High-Stakes participants, Experienced participants showed a significant reduction in forecast error only for the current period, $h = 0$ (Table D.2). Perhaps the early market peaks of these sessions caused experienced traders to misjudge the timing of the crash, which threw off their longer-horizon forecasts.

We attribute these differences to experience rather than to self-selection. In their first session, the returners who populate the Experienced sessions were statistically indistinguishable from non-returners on demographics, earned bonuses, risk attitudes, and quiz performance, differing only in that they attempted and executed somewhat fewer trades.³⁴ A participant’s market earnings in their first and second Experienced sessions are essentially uncorrelated ($r = -0.11$, $p = 0.64$), consistent with superior market performance being more luck than skill.

Individual session plots are shown below.

B.2 Bubble Metrics

We use several common measures of market bubbles to compare price paths of our treatments. Most of the measures are intended for use in continuous auctions in which trades are executed instantly and the market price can fluctuate during a given period. With the call market setup that we employ, there is a single price per period, which simplifies measures that involve the mean or median price for a period. Further, several of these measures are designed around a changing fundamental value. Since our fundamental value is constant ($P^* = 14$) for all $T = 30$ periods, we can further simplify these measures.

The measures are outlined briefly below.

- Relative Absolute Deviation (RAD): A measure of the average price deviation from the fundamental value within a given session.
- Average Bias (AB): A measure of how far the market price is, on average, from the fundamental value.
- Total Dispersion (TD): A measure of the absolute difference between the market price and the fundamental value.

³³Huber et al. (2024) conduct high-powered, pre-registered replications of 17 claims from prior experimental asset-market studies. They replicate prior evidence that experience reduces but does not eliminate bubbles.

³⁴A detailed breakdown of selection effects for these participants is in Table A.5; Table A.6 reports their trading activity by number of prior sessions.

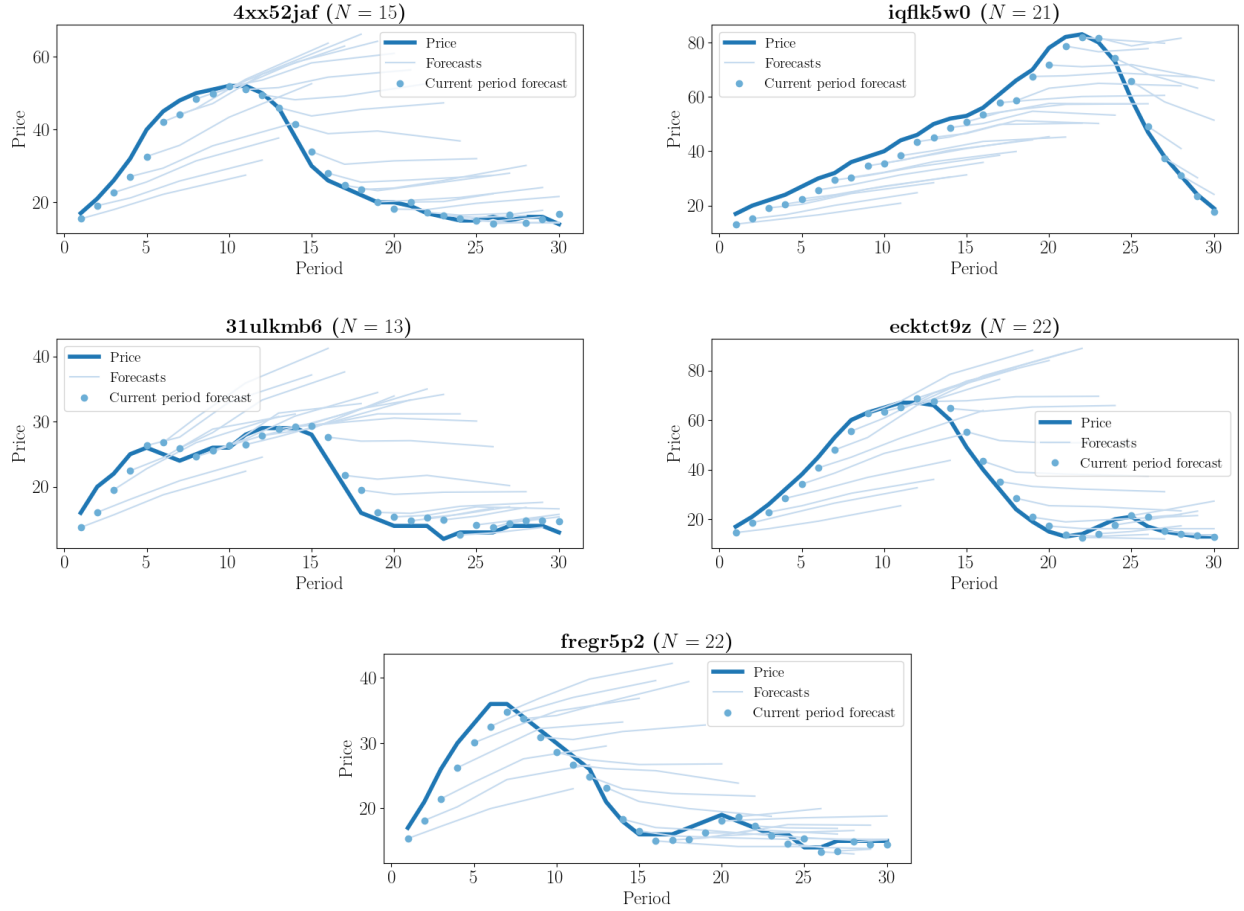


Figure B.6: **All Experienced Session Price Paths**

The lines represent the dynamics of price levels for 30 periods. Each point represents the current period forecast, and subsequent lines represent average price forecasts for the 0-, 2-, 5-, and 10-period-ahead forecasts.

Alt text: Grid of five line charts, one per Experienced session, showing price paths over 30 periods with forecast points and forecast tongues.

- Price Amplitude (PA): A measure of the largest difference in price deviations from the fundamental value.
- Duration (DUR): The greatest number of consecutive periods over which the price deviation from the fundamental value increases.

B.2.1 Relative Absolute Deviation (RAD)

Relative Absolute Deviation was introduced by [Stöckl, Huber, and Kirchler \(2010\)](#) in an attempt to develop a measure that can be compared across market configurations. It is

calculated by:

$$\text{RAD} = \frac{\sum_{t=1}^T v_t \left(\frac{|P_t - 14|}{14} \right)}{\sum_{t=1}^T v_t}$$

where P_t is the market price in period t and v_t is the number of shares traded in period t .

B.2.2 Average Bias (AB)

Average Bias ([Haruvy and Noussair, 2006](#)) measures how far the market price is, on average, from the fundamental value. As given by Haruvy, the expression is:

$$\text{AB} = \frac{1}{T} \sum_{t=1}^T \text{Median}(P_t) - 14$$

where $\text{Median}(P_t)$ is the median price realized during period t and 14 is the fundamental value for period t . Our market design allows us to simplify those terms to constants, and the expression becomes a simple average of the differences between price and fundamental value.

$$\text{AB} = \frac{1}{T} \sum_{t=1}^T P_t - 14$$

B.2.3 Total Dispersion (TD)

Haruvy and Noussair also introduced Total Dispersion, which is very similar to the AB measure except that it is a measure of the absolute difference between the market price and the fundamental value (14).

$$\text{TD} = \sum_{t=1}^T |P_t - 14|$$

B.2.4 Price Amplitude (PA)

Price Amplitude ([Porter and Smith, 1995](#)) is a measure of the largest difference in price deviations from the fundamental value (14).

$$\text{PA} = \max_{t \in T} (P_t - 14) - \min_{t \in T} (P_t - 14)$$

Because our constant fundamental value cancels out, this measure reduces to the difference between the maximum price and the minimum price achieved in a session.

$$\text{PA} = \max_{t \in T} (P_t) - \min_{t \in T} (P_t)$$

B.2.5 Duration (DUR)

Duration ([Porter and Smith, 1995](#)) is the greatest number of consecutive periods over which the price deviation from the fundamental value (14) increases.

$$\text{DUR} = \max(D : P_t - 14 < P_{t+1} - 14 < \dots < P_{t+D-1} - 14)$$

Since the fundamental value is constant, all the (14)'s cancel out, and the duration is simply the longest streak of increasing prices.

$$\text{DUR}_s = \max(D : P_t < P_{t+1} < \dots < P_{t+D-1})$$

B.3 Predicting Market Phenomena

B.3.1 Pre-Registered Simple OLS

B.3.2 Model Features

We suppress the session notation s for clarity; every variable below is calculated in a given session s . All explanatory variables are constructed using information available at or before the end of period $t - 1$, unless otherwise noted.

The specific explanatory variables are listed in [Table B.2](#).

B.3.2.1 Feature selection These variables serve as inputs to our predictive models (machine learning and LASSO), with a few adjustments relative to [Table B.2](#). Lagged returns are excluded, so that predictive power is measured net of return autocorrelation. The backward-looking forecasts enter at four horizons ($f_t^t, f_{t-2}^t, f_{t-5}^t, f_{t-10}^t$); the DOSE task contributes both the risk-aversion parameter ρ_t and the choice-consistency parameter μ_t ; and tonic EDA enters not only as its period mean but also through its cross-sectional standard deviation and skewness. Lagged volume V_{t-1} and remaining periods $30 - t$ enter directly.

B.3.3 Standard Session Regressions

We performed a couple of robustness checks of the pre-registered model. First, we looked at the difference between backward- and forward-looking forecasts with no major changes in results (shown in the Pre-registration analysis section in [Table B.1](#)).³⁵ We report both backward- and forward-looking forecast specifications to distinguish between the predictive content of beliefs formed prior to trading and beliefs stated contemporaneously about future prices. Next, we suspected multicollinearity among the forecast variables, and [Figure B.8](#)

³⁵Regression notation and additional details can be found in [Section B.3.7](#).

Table B.1: Predicting Next-Period Log Returns Across Session Types

	Standard		High-Stakes		Experienced	
	Bwd	Fwd	Bwd	Fwd	Bwd	Fwd
Intercept	-2.392 (2.979)	-0.933 (2.540)	5.938 (3.175)	9.441 (8.590)	-6.036 (11.702)	-15.059 (7.711)
R_{t-1}	0.839*** (0.039)	0.822*** (0.038)	0.488* (0.169)	0.572** (0.170)	0.716*** (0.076)	0.722*** (0.060)
ρ	0.020 (1.504)	-2.836** (1.117)	-1.502 (1.494)	-0.617 (2.875)	10.908** (3.535)	13.766 (6.670)
$30 - t$	0.015 (0.084)	0.105 (0.090)	-0.070 (0.101)	-0.212 (0.242)	-0.066 (0.465)	0.129 (0.377)
Ω_t	0.015 (0.029)	-0.021 (0.029)	-0.052 (0.055)	0.079* (0.025)	0.008 (0.061)	-0.043 (0.061)
Υ_t	0.008 (0.008)	0.002 (0.007)	-0.013 (0.024)	-0.022 (0.043)	-0.032 (0.034)	-0.003 (0.030)
V_{t-1}	-0.036 (0.062)	-0.055 (0.053)	-0.220 (0.226)	-0.454 (0.224)	0.210 (0.351)	0.057 (0.371)
EDA_t^{ton}	2.786 (2.004)	-0.152 (1.901)				
f_t^t	0.322 (0.244)	0.282 (0.171)	-1.457** (0.326)	0.745 (0.598)	1.262 (0.967)	0.367 (1.154)
f_{t-2}^t	-0.541** (0.204)		0.894* (0.289)		-1.873 (1.089)	
f_{t-5}^t	0.202 (0.153)		0.494 (0.220)		0.549*** (0.092)	
f_{t-10}^t	-0.012 (0.141)		0.021 (0.188)		0.070 (0.126)	
f_t^{t+2}		-0.215 (0.199)		-1.043 (0.843)		0.119 (1.440)
f_t^{t+5}		0.162 (0.229)		-0.357 (0.659)		-0.501 (0.784)
f_t^{t+10}		-0.164 (0.146)		0.503 (0.269)		0.151 (0.462)
N	355	336	75	71	93	88
R^2	0.697	0.689	0.492	0.479	0.695	0.688
Adjusted R^2	0.687	0.679	0.413	0.392	0.658	0.647

Notes: The dependent variable is the future log return R_t (multiplied by 100). Each session type is reported with a backward-looking (f_{t-h}^t , “Bwd”) and a forward-looking (f_t^{t+h} , “Fwd”) forecast specification; the pre-registered specification is the backward-looking one. EDA^{ton} is recorded only in Standard sessions. Standard errors (in parentheses) are heteroskedasticity- and cluster-robust (clustered by session). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: **Explanatory Variables Used in the Return-Prediction Models**

Symbol	Name	Description
R_{t-1}	Lagged return	Log return over the previous period.
ρ_t	Risk aversion	DOSE risk-aversion parameter.
f_{t-h}^t	Backward-looking	Forecast of the period- t price, made h periods earlier.
V_{t-1}	Lagged volume	Trading volume in the previous period.
$30 - t$	Periods remaining	Periods left in the session.
Ω_t	Order imbalance	$\Omega_t = \frac{\sum_{p_b < P_{t-1}} v_b p_a - \sum_{p_a > P_{t-1}} v_a p_b}{\text{TOV}_{t-1}}$; TOV_{t-1} is total order volume, the best ask p_a pairs with outstanding bid volume v_b , and the best bid p_b with outstanding ask volume v_a .
Υ_t	Asset allocation	CASH-to-stock value ratio.
$\text{EDA}_t^{\text{ton}}$	Tonic EDA	$\text{EDA}_t^{\text{ton}} = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{EDA}_{i,t}^{\text{ton}}$, over the N_t in-lab participants.
	Buying ratio	Net order-flow imbalance, $\frac{B_{t-1} - S_{t-1}}{B_{t-1} + S_{t-1} + \epsilon}$; B and S are total buy and sell orders and $\epsilon = 10^{-5}$; ranges from -1 (net selling) to $+1$ (net buying).

Notes: Risk aversion, the forecast, and asset allocation are reported as cross-sectional means across participants. All variables are constructed from information available at or before the end of period $t - 1$ unless otherwise noted; the session subscript s is suppressed. Forecast notation matches Table 1.

shows a high degree of correlation among them (f_t^t , f_{t-2}^t , f_{t-5}^t). Furthermore, a PCA of those variables shows that the first principal component explains over 91% of the variation. We reran the main regression substituting the forecast variables with their first principal component (Table B.3). The results are not much different from the main result with no major changes in significance.

B.3.3.1 Predicting returns in High-Stakes and Experienced sessions Table B.4 repeats the same OLS regressions predicting returns as for the Standard sessions, except that there are no EDA recordings in these sessions. The OLS model explains roughly 58% of the variance in next-period returns ($R^2 = 0.587$) for the High-Stakes sessions and about 63% for the Experienced sessions ($R^2 = 0.629$). As seen in the Standard sessions, the lagged return R_{t-1} is highly predictive, but even when R_{t-1} is omitted, the forecasts have similar predictive power.³⁶ It is notable that predictability of returns is similar for all three types of sessions.

³⁶This pre-registered specification is reported for all three session types in Table B.1; EDA enters only in the Standard columns, as it is not collected from online-only subjects.

Table B.3: **Predicting Next-Period Log Returns with the First Principal Component of Forecasts**

	(1)	(2)	(3)	(4)	(5)
Intercept	-0.621*** (0.115)	-5.032* (2.660)	-4.240** (1.719)	-19.007*** (4.152)	-4.340** (1.954)
R_{t-1}	0.827*** (0.034)		0.698*** (0.043)		0.698*** (0.043)
$\text{EDA}_t^{\text{ton}}\text{MA}_5$		9.783* (5.420)		-0.620 (4.366)	0.208 (1.557)
ρ			0.654 (0.879)	-1.078 (1.499)	0.671 (0.906)
V_{t-1}			-0.115* (0.061)	-0.137 (0.082)	-0.114* (0.063)
$30 - t$			0.199** (0.090)	0.916*** (0.186)	0.198** (0.091)
Ω_t			0.034 (0.029)	0.181** (0.064)	0.034 (0.029)
Υ_t			0.002 (0.006)	0.017 (0.011)	0.002 (0.006)
Forecast PCA 1			-0.899* (0.494)	-1.982* (0.999)	-0.899* (0.494)
N	456	456	456	456	456
R^2	0.668	0.012	0.699	0.470	0.699
Adjusted R^2	0.668	0.010	0.694	0.461	0.693

Notes: The dependent variable is the future log return R_t (multiplied by 100). Standard errors (in parentheses) are heteroskedasticity- and cluster-robust (clustered by session). This model shows the first principal component of the three forecast measures as a check for multicollinearity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3.4 Support Vector Regression (SVR)

Support Vector Regressions (SVRs) estimate a function $f(\mathbf{x})$ that approximates the data within a tolerance ϵ , treating deviations smaller than ϵ as negligible. This allows the model to capture the underlying trend while ignoring minor fluctuations. Simultaneously, SVR enforces flatness of the function by minimizing the norm of the weight vector, thereby controlling model complexity. The trade-off between approximation accuracy and simplicity is governed by the regularization parameter C , while kernels (e.g., linear, RBF) enable the method to handle both linear and nonlinear relationships.

Below are the details of our SVR setup and corresponding hyperparameters.

- **Hyperparameters:**

- C : Regularization parameter controlling the trade-off between model complexity and the amount up to which deviations larger than ϵ are tolerated.
- **kernel**: The kernel function used to transform data into higher-dimensional space (e.g., `linear` or `rbf`).

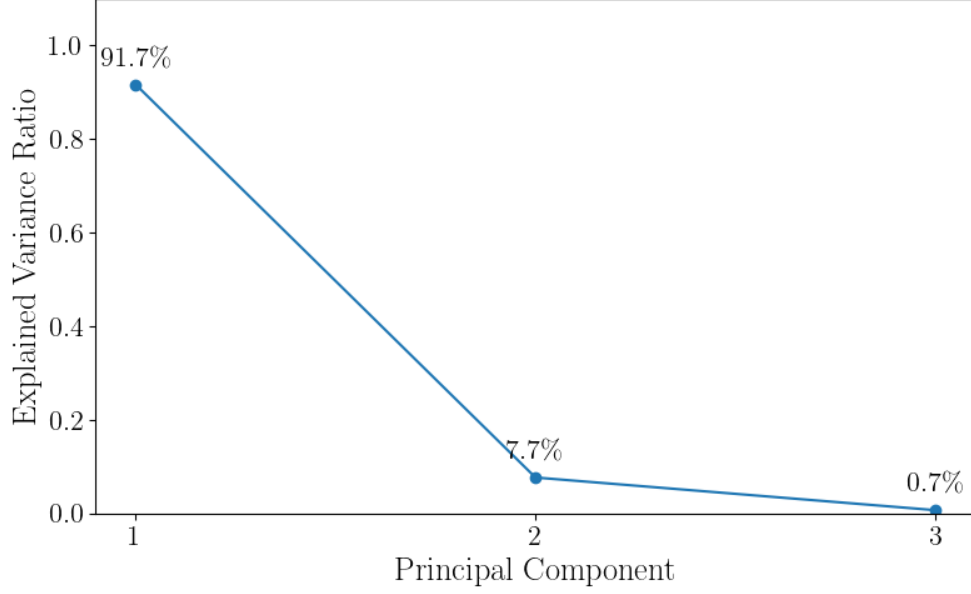


Figure B.7: **Scree Plot of Forecast-Measure Principal Components**

Scree plot showing the explained variation of each principal component of the three forecast measures used in the main regression (Table 2). Forecast variables are f_t^t , f_{t-2}^t , f_{t-5}^t . Exact explained percentage of each component can be found beside the data point.

Alt text: Scree plot of the variance explained by each of the three principal components of the forecast measures; the first component explains most of the variance.

- ϵ : The width of the ϵ -insensitive tube within which no penalty is associated in the training loss.

B.3.5 SVR Feature Pre-Processing

1. **Imputation:** Missing feature values are imputed using the median of each feature.
2. **Scaling:** The features are standardized using Z-score normalization:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}, \quad (16)$$

where μ and σ are the mean and standard deviation of the feature in the training set.

B.3.5.1 Mathematical formulation Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^m$ be our training data, where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$. The goal of ϵ -SVR is to learn a function $f(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x} \rangle + b$ (in the linear case) or

$$f(\mathbf{x}) = \sum_{i=1}^m (\alpha_i - \alpha_i^*) K(\mathbf{x}_i, \mathbf{x}) + b \quad (\text{in the kernelized case}),$$

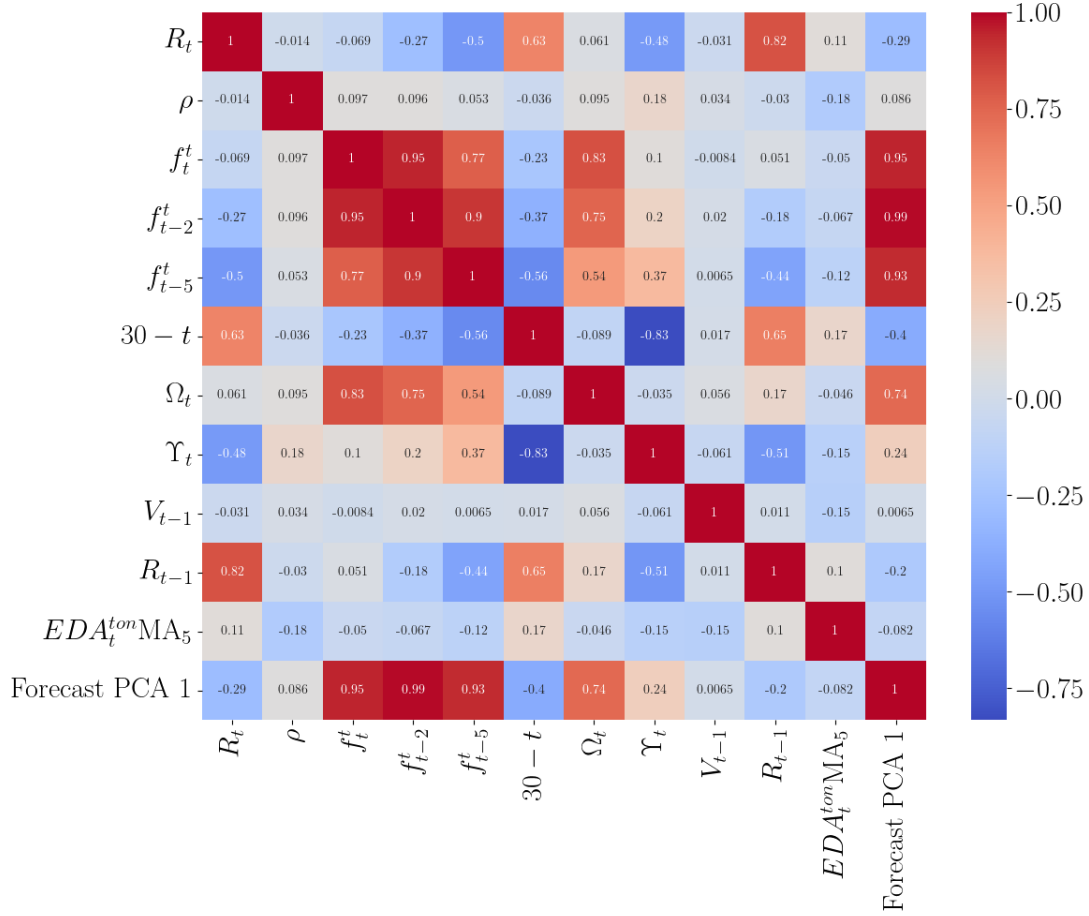


Figure B.8: **Correlation Matrix of OLS Regressors**

Correlation matrix of the regressors in the future-return prediction OLS (Table 2). The image shows a high level of correlation among the three forecast measures, indicating the possibility of multicollinearity. The first principal component of the three forecasts is included to avoid this in the regressions provided in the main text.

Alt text: Heatmap of pairwise correlations among the OLS regressors, with darker cells indicating stronger correlations, especially among the forecast measures.

where $K(\cdot, \cdot)$ is the kernel function.

Primal Problem. SVR solves the following optimization problem:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m (\xi_i + \xi_i^*)$$

subject to

$$\begin{aligned} y_i - \langle \mathbf{w}, \mathbf{x}_i \rangle - b &\leq \epsilon + \xi_i, \\ \langle \mathbf{w}, \mathbf{x}_i \rangle + b - y_i &\leq \epsilon + \xi_i^*, \\ \xi_i, \xi_i^* &\geq 0, \quad i = 1, \dots, m, \end{aligned}$$

Table B.4: Predicting Future Returns in High-Stakes and Experienced Sessions

	High-Stakes			Experienced		
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	-0.158 (0.395)	4.269 (4.672)	3.429 (2.510)	-1.483*** (0.269)	2.622 (4.806)	4.092 (4.974)
R_{t-1}	0.716*** (0.089)		0.479** (0.083)	0.744*** (0.085)		0.532*** (0.081)
ρ		-3.359 (1.616)	-1.925 (1.056)		1.563 (6.801)	-2.350 (4.730)
V_{t-1}		-0.022 (0.215)	-0.128 (0.140)		-0.153 (0.317)	-0.077 (0.288)
$30 - t$		0.163 (0.193)	0.088 (0.095)		0.107 (0.195)	0.087 (0.101)
Ω_t		0.012 (0.090)	-0.061 (0.051)		-0.023 (0.079)	0.005 (0.064)
Υ_t		-0.006 (0.021)	-0.003 (0.014)		-0.006 (0.018)	-0.004 (0.011)
f_t^t		0.532* (0.180)	0.332** (0.096)		0.904** (0.226)	0.303 (0.258)
f_{t-2}^t		-0.225 (0.153)	-0.168 (0.085)		-0.889*** (0.181)	-0.401 (0.210)
f_{t-5}^t		-0.442*** (0.074)	-0.194 (0.115)		-0.158 (0.076)	-0.033 (0.042)
N	96	96	96	120	120	120
R^2	0.503	0.482	0.587	0.572	0.521	0.629
Adjusted R^2	0.497	0.435	0.544	0.569	0.487	0.599

Notes: The dependent variable is the future log return R_t (multiplied by 100). Standard errors (in parentheses) are heteroskedasticity- and cluster-robust (clustered by session). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

where ξ_i and ξ_i^* are positive slack variables penalizing predictions outside the ϵ -insensitive tube.

Dual Problem. By introducing Lagrange multipliers α_i and α_i^* , one obtains the dual form, and the resulting decision function can be written as:

$$f(\mathbf{x}) = \sum_{i=1}^m (\alpha_i - \alpha_i^*) K(\mathbf{x}_i, \mathbf{x}) + b.$$

Only a subset of the training points (the *support vectors*) will have non-zero $(\alpha_i - \alpha_i^*)$.

B.3.5.2 Loss function In ϵ -SVR, the loss function is known as the ϵ -insensitive loss. It penalizes deviations only when the prediction is outside the ϵ -insensitive region:

$$L_{\epsilon}(y, \hat{y}) = \begin{cases} 0, & \text{if } |y - \hat{y}| \leq \epsilon, \\ |y - \hat{y}| - \epsilon, & \text{otherwise.} \end{cases}$$

However, our models, when performing parameter selection (e.g., via cross-validation), often use a standard regression metric such as the Mean Squared Error (MSE) or Mean Absolute Error (MAE) to compare models. In our case, we use MSE:

$$\mathcal{L} = \frac{1}{m} \sum_{i=1}^m (y_i - f(\mathbf{x}_i))^2.$$

B.3.5.3 Optimization algorithm The SVR model is typically trained via an iterative method that solves the underlying convex Quadratic Programming problem. In scikit-learn, this is handled internally by the `SVR` class (e.g., using `libsvm` or `liblinear` backends). We use `GridSearchCV` with 5-fold cross-validation to select the optimal hyperparameters. Our chosen grid for `C` and `kernel` is given by:

Parameter	Values
<code>C</code>	{0.1, 1, 10}
<code>kernel</code>	{ <code>linear</code> , <code>rbf</code> }

We train the model to maximize `neg_mean_squared_error` (i.e., maximize $- \text{MSE}$), and then select the parameter combination yielding the best cross-validation performance.

Final SVR Model. After the grid search completes, we use the best combination of (`C`, `kernel`) to re-train the model on the full training set. The final prediction function is:

$$\hat{y} = f(\mathbf{x}) = \sum_{i=1}^m (\alpha_i - \alpha_i^*) K(\mathbf{x}_i, \mathbf{x}) + b,$$

where the learned weights and support vectors reflect the best-fit regression surface.

B.3.6 Random Split SVR

We selected a random train/test split to represent the 100 random splits tested against the no-EDA data, as is customary in machine learning. The split held out sessions with indexes 1, 2, 6, and 12. Here, we report much of the SVR analysis conducted on the session cross-validation reported in the main body of the paper.

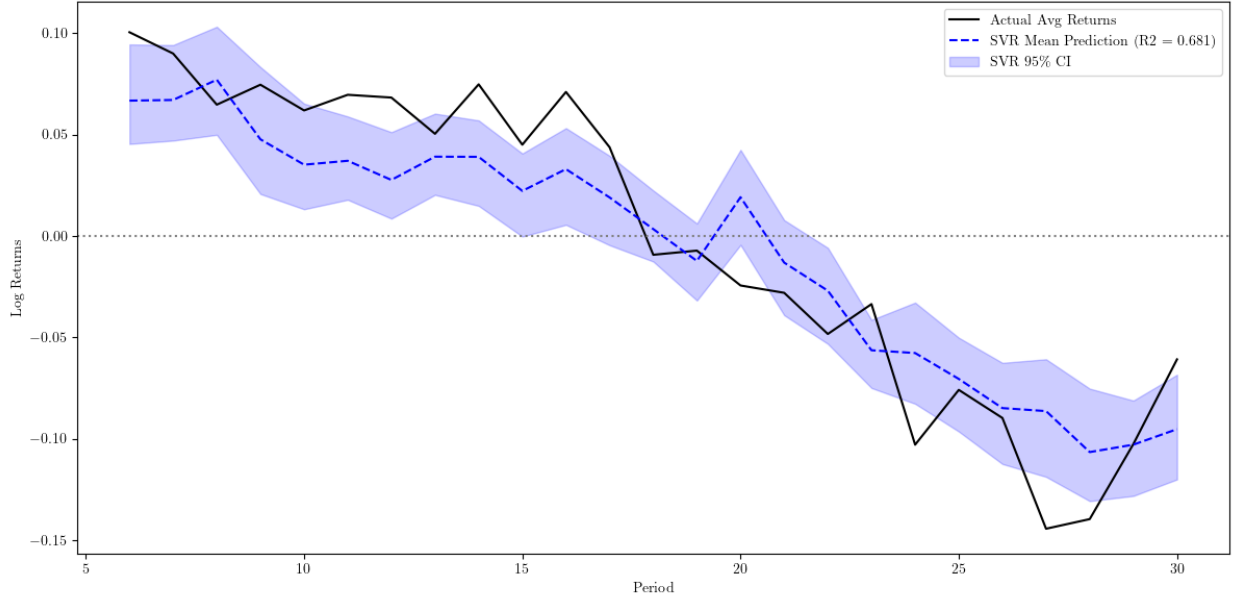


Figure B.9: **SVR Predictions vs. Actual Returns (with Tonic EDA)**

The SVR mean prediction curve represents the model's out-of-sample predictions averaged across all (four; [market: 1, 2, 6, 12]) held-out test sessions for each trading period. Each SVR was trained on the remaining sessions (15), cross-validated on training folds for tuning, and then evaluated once on the four held-out sessions. Confidence intervals were generated using a nonparametric bootstrap applied to the training data: for each of 1000 iterations, the training data were resampled with replacement, a new SVR was fit to the bootstrapped sample, and its performance was evaluated on the fixed held-out test sessions.

Alt text: Line chart by period comparing realized returns averaged over four held-out sessions with the mean SVR prediction, with shaded bootstrap confidence intervals.

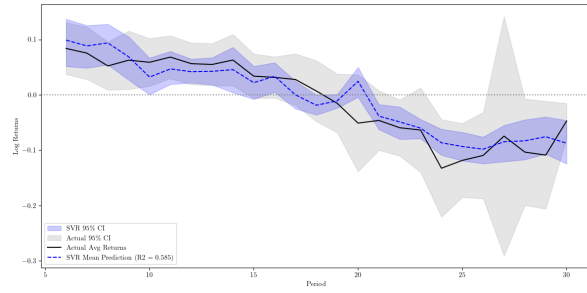
The random-split evaluation (Panel A) tells the same story: the non-EDA model attains MAE mean = 0.041 and R^2 mean = 0.671, and adding EDA^{ton} data does not improve performance. Details of feature weights can be found in Table 3 and Table B.6. Notation is also explained further in Section B.3.2.1.

B.3.6.1 Regressions with real-valued confidence intervals Figure B.10 shows the out-of-sample SVR return predictions with held-out confidence bands for each group of held-out sessions.

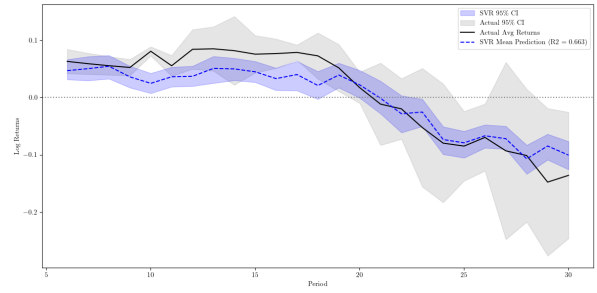
B.3.7 Regression Details

Below is the outline of our regression model, which is used to predict returns and examine the relationship between our biometrics and asset prices.

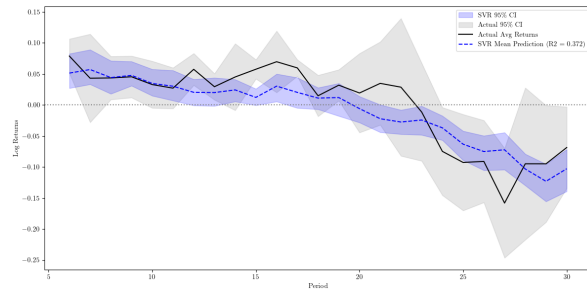
Let $P_{s,t}$ be the price of the risky asset in session s , period t , and let:



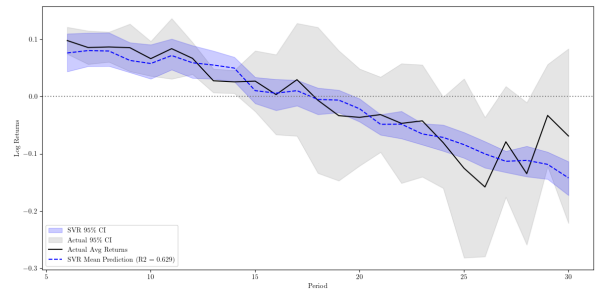
(a) Holdout: Sessions 1–4



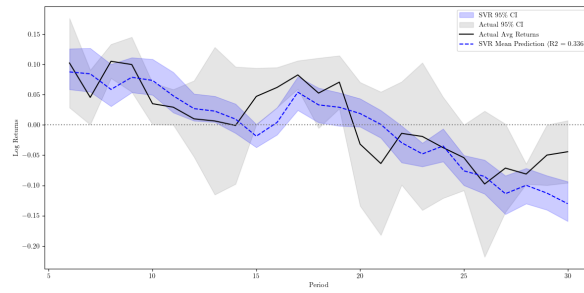
(b) Holdout: Sessions 5–8



(c) Holdout: Sessions 9–12



(d) Holdout: Sessions 13–16



(e) Holdout: Sessions 17–19

Figure B.10: Mean Out-of-Sample SVR Return Predictions with Held-Out Confidence Bands

Shaded purple regions show 95% confidence intervals from 1,000 bootstrap resamples of the training data, evaluated on a fixed held-out test set. Shaded gray regions show 95% confidence intervals from the standard deviation of the held-out test data. The majority of realized returns lie within the bootstrap confidence bands across all held-out session groups, suggesting the model captures common market dynamics and provides well-calibrated out-of-sample uncertainty.

Alt text: Five panels, one per held-out session group, plotting realized returns and SVR predictions with two sets of shaded 95 percent bands: bootstrap bands from the training data and bands from the held-out data.

Table B.5: **SVR Return-Prediction Performance With and Without Tonic EDA**

Model	Split	MAE			R^2		
		Value	Low	High	Value	Low	High
<i>Panel A: Random Train/Test Split</i>							
SVR (w/ EDA ^{ton})	Train	0.042	0.038	0.051	0.631	0.477	0.660
	Test	0.043	0.039	0.051	0.645	0.525	0.693
SVR (w/o EDA ^{ton})	Train	0.041	0.029	0.048	0.628	0.483	0.776
	Test	0.041	0.040	0.050	0.671	0.548	0.695
<i>Panel B: Held-Out Session Cross-Validation</i>							
SVR (w/ EDA ^{ton})	Train	0.044	0.043	0.044	0.627	0.609	0.646
	Test	0.048	0.044	0.052	0.517	0.336	0.663
SVR (w/o EDA ^{ton})	Train	0.043	0.042	0.045	0.626	0.610	0.651
	Test	0.046	0.042	0.056	0.544	0.261	0.680

Notes: Mean absolute error (MAE) and out-of-sample R^2 for the linear SVR return-prediction model, with and without tonic EDA (EDA^{ton}) features, on the training and test splits. Panel A uses a single random train/test split (held-out sessions [1,2,6,12]), as is standard in machine learning: “Value” is the split estimate and Low/High are 95% bootstrap confidence intervals (see Figure B.9). Panel B uses held-out session cross-validation (Figure 3): “Value” is the mean across held-out session groups and Low/High are the minimum and maximum across those groups. Including EDA^{ton} does not improve performance under either scheme.

$$R_{s,t} = \log \left(\frac{P_{s,t}}{P_{s,t-1}} \right)$$

be the log price ratio between periods t and $t - 1$ in session s . Then, return dynamics can be captured by the following equation:

$$R_{s,t} = \alpha_s + \sum_{v=1}^K \beta_v X_{v,s,t} + \epsilon_{s,t} \quad (17)$$

where:

- α_s is the session-specific intercept for session s ,
- β_v represents the coefficient for the v -th variable in the vector of variables X_v ,
- K is the number of variables,
- $X_{v,s,t}$ represents the v -th explanatory variable in session s , period t ,
- $\epsilon_{s,t} \sim N(0, \sigma^2)$ is the error term, with $\mathbb{E}(R_{s,t} \mid X_v)$ representing the conditional expectation.

Table B.6: **SVR Feature Importance With and Without Tonic EDA (Random Split)**

Feature	Normalized Importance	
	w/ EDA	w/o EDA
f_t^t	0.372	0.422
f_{t-2}^t	0.292	0.353
$30 - t$	0.130	0.137
f_{t-5}^t	0.059	0.037
standard deviation EDA^{ton}	0.054	—
skewness $\text{EDA}_t^{\text{ton}}$	0.029	—
DOSE μ	0.026	0.021
DOSE ρ	0.017	0.008
f_{t-10}^t	0.011	0.021
$\text{EDA}_t^{\text{ton}}$	0.008	—
V_{t-1}	0.002	0.000

Notes: Normalized absolute coefficients from the linear SVR model (each column sums to one); higher values indicate a stronger influence on predicted returns. The two columns are the model estimated with and without tonic EDA (EDA^{ton}) features; the EDA-based features are absent (—) from the without-EDA model. Both use the same random split as Panel A of Table B.5 (held-out sessions: [1,2,6,12]).

B.3.7.1 LASSO regression In empirical settings such as ours—where predictors can be highly correlated, and the number of potentially predictive variables is large—traditional Ordinary Least Squares (OLS) regression can yield highly variable estimates. This instability arises from multicollinearity, which makes it difficult to distinguish the contributions of correlated predictors. Moreover, including too many weakly informative variables can reduce out-of-sample predictive performance and lead to overfitting.

To address these issues, we estimated a Least Absolute Shrinkage and Selection Operator (LASSO) model using the same pre-registered set of explanatory variables as in our OLS specification. LASSO adds an ℓ_1 -penalty to the regression loss function, shrinking small or redundant coefficients toward zero and effectively performing variable selection:

$$\hat{\boldsymbol{\beta}}^{\text{LASSO}} = \arg \min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}. \quad (18)$$

The regularization parameter λ was selected via a grid-search minimizing the Bayesian Information Criterion (BIC) over $\lambda \in [10^{-3}, 1]$. The optimal value was $\lambda = 0.091$. At this level of penalization, the LASSO retained only five non-zero predictors: lagged return, periods remaining, near-term forecast (f_t^t), 10-period forecast (f_{t-10}^t), and lagged volume. These variables were subsequently re-estimated using OLS with cluster-robust standard errors to

obtain unbiased coefficient estimates and valid inference.

Table B.7: **Predicting Next-Period Returns Using LASSO**

	Coefficient	SE	t	p -value
Intercept	-0.755	1.209	-0.624	0.540
f_t^t	-0.068*	0.034	-2.014	0.059
f_{t-10}^t	0.046***	0.014	3.249	0.004
$30 - t$	0.141***	0.047	3.013	0.007
V_{t-1}	-0.088	0.051	-1.718	0.103
R_{t-1}	0.689***	0.036	19.017	< 0.001
N		532		
R^2		0.700		
Adjusted R^2		0.697		

Notes: Coefficients estimated by post-LASSO OLS with cluster-robust standard errors (clustered by session); SE denotes standard error. The dependent variable is the next-period log return R_t (multiplied by 100). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The resulting post-LASSO model achieves a similar R^2 to the standard OLS regressions, an $R^2 = 0.700$, indicating that the retained predictors explain 70% of the variation in next-period returns. Lagged returns are the dominant driver, but backward-looking forecasts and session-level variables (periods remaining, lagged volume) also contribute significantly to short-horizon return dynamics. LASSO helps control the multicollinearity issue present in our standard OLS measures.

We explored other machine learning models, the details of which can be found in the doctoral dissertation (Spoon, 2026).

Appendix C. Arousal and Earner Behavior

C.1 Physiological Arousal (EDA) and Trading Behavior

C.1.1 Peak EDA (EDA^{peak}) of High vs. Low Earners

We also wanted to understand whether there were different arousal patterns between the highest earners and the lowest earners. Are our higher earners calmer than our lower earners? To investigate, we ran a mixed linear model comparing the period-by-period EDA^{peak} levels of individuals in the highest-earning tercile with those in the lowest-earning tercile. We found a trend ($p < 0.06$) showing that higher earners, contrary to our expectation, actually operated closer to their EDA^{peak} throughout the experiment than lower earners.³⁷ The data suggest that higher earners operate closer to their maximum arousal state than lower earners, although this relationship appears to be reversed in the early periods of the experiment.

³⁷This measure is standardized by participant, so it is possible that this effect is because higher earners never reach the EDA spike seen in low earners.

C.1.2 Tonic EDA (EDA^{ton}) Activity by Earnings Tercile

For much of the individual-level analysis, we divided participants by earning terciles. This tercile split is done at the session level, so there is some overlap in earnings. That is, a medium-level earner in one session might actually earn more than a high earner in another. Table C.1 shows the earnings ranges by tercile.

Table C.1: **Ranges of Market Earnings by Tercile**

Earnings Tercile	Min	Max
High	3.48	15.43
Medium	1.38	3.80
Low	0.00	3.21

Notes: The terciles are determined at the session level, which explains any inter-tercile overlaps.

We also investigated whether there were differences in EDA^{ton} activity among high earners compared to other earners. To do so, we sorted participants into terciles based on their individual market bonuses in each session. For clarity, the main text only shows the top and bottom terciles. Here is the same plot showing all three terciles (Figure C.1a).

We also performed the same analysis with a median split. The Levene test for variance by median is similarly significant ($\sigma_{hi} = 0.025, \sigma_{lo} = 0.049, p = 0.009$).

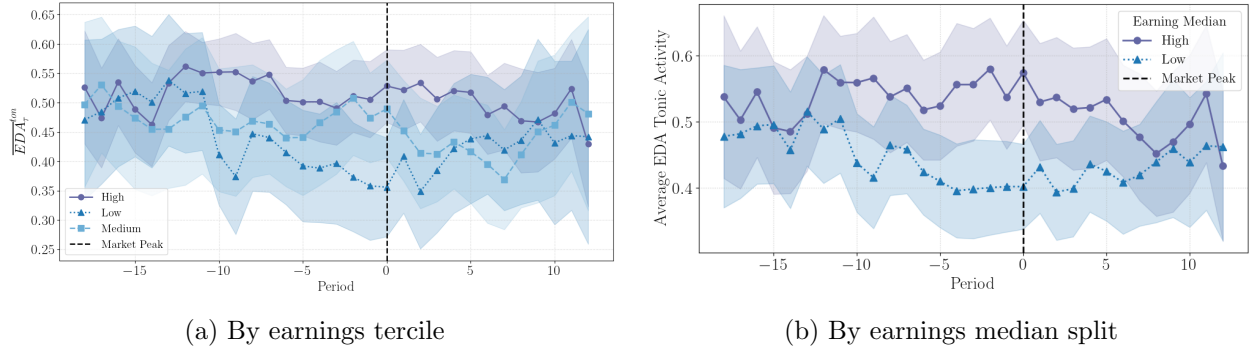


Figure C.1: **Average Tonic EDA by Earnings Group**

Average EDA^{ton} over the session for participants grouped by market-bonus earnings. Panel (a) shows all three terciles (90% confidence intervals); panel (b) an above- versus below-median split (95% confidence intervals).

Alt text: Two line charts of average tonic EDA over the 30 periods, panel (a) by earnings tercile and panel (b) by earnings median split, with confidence bands; high earners are more stable and low earners rise late in the session.

C.1.2.1 Forward returns prediction from smoothed tonic EDA While direct averages of EDA_t^{ton} did not predict subsequent returns R_t , we did an exploratory analysis

Table C.2: Quadratic Regressions of Tonic EDA Around the Market Peak

	High Tercile	Low Tercile
Intercept α	52.878*** (0.876)	39.749*** (1.716)
Distance From Peak τ (β)	-0.211*** (0.064)	0.062 (0.290)
Squared Distance τ^2 (γ)	-0.029*** (0.007)	0.050** (0.023)
N	45	42
R^2	0.545	0.388
Adj. R^2	0.523	0.356

Notes: The dependent variable is mean tonic EDA ($\overline{\text{EDA}}_{\tau}^{\text{ton}}$) multiplied by 100 for readability. Regressions are estimated separately for High- and Low-earning terciles on binned means by distance from the market peak, $\tau = t - t_p$. Thus, observations denote the number of observed τ bins for each tercile, not participant-period observations. Heteroskedasticity-robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

of whether future returns might be predicted by smoothed EDA^{ton} . We find that when smoothed, there is a significant relationship between EDA^{ton} and forward returns.

The justification for this analysis is that emotion researchers distinguish between transitory emotions (often associated with events, and which are short-lived—minutes or hours) and moods, which are longer-lived (days or weeks) (Russell, 2003; Beedie, Terry, and Lane, 2005).

Neuroscientists distinguish between fast ‘phasic’ responses and slower time-averaged ‘tonic’ responses (Dawson, Schell, and Filion, 2007; Boucsein, 2012). Maybe our EDA^{ton} measures are associated with ‘market moods’ (several periods) rather than ‘market emotions’ (one period ahead)?

To investigate this relationship, we first computed the average EDA^{ton} signal across participants for each period t within a given session. We then constructed a 5-period backward-looking moving average,

$$\text{EDA}_t^{\text{ton}}\text{MA}_5 = \frac{1}{5} \sum_{i=0}^4 \text{EDA}_{t-i}^{\text{ton}},$$

computed separately within each session.³⁸ This procedure yields a session-specific, smoothed measure of session-level tonic arousal for each period.

Within each session, we then split observations at the median of $\text{EDA}_t^{\text{ton}}\text{MA}_5$ into a “Top”

³⁸The first four periods of each session are excluded because a full 5-period lookback is unavailable.

and “Bottom” half, ensuring that each session contributes an equal number of observations to both groups. For each group, we computed the mean subsequent 5-period forward return,

$$\frac{p_{t+5} - p_t}{p_t}.$$

Figure C.2 shows a clear and statistically significant difference ($p < 0.001$)³⁹ between the returns in future periods for EDA below the EDA^{ton} median and the returns in future periods above the EDA^{ton} median. Higher emotional arousal measured by EDA^{ton} is associated with higher future returns.

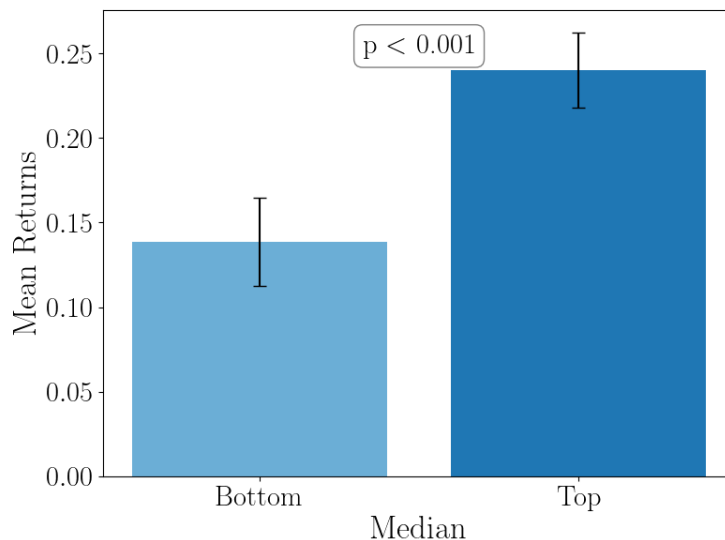


Figure C.2: Mean Returns by Tonic-EDA Median Split

EDA^{ton} activity predicts future returns of the risky asset. Periods within each session were split into high and low activity based on a 5-period moving average of EDA^{ton} activity. The bars show that the mean 5-period forward return is higher for above-median EDA^{ton} activity.

Alt text: Bar chart of the mean 5-period forward return for periods with above-median versus below-median smoothed tonic EDA; the above-median bar is higher.

This is a rather strong and interesting exploratory finding. It could easily have been the case that a stronger market ‘mood’ is associated with rapid crashes. It is consistent with the hypothesis that the association between EDA^{ton} and future returns is slower (and hence mood-like)⁴⁰, while other kinds of emotion-returns association are either too noisy or too fast to be detected by our EDA^{ton} measures.

³⁹The p -value is from a permutation test for equality of distributions.

⁴⁰Note in emotion literature “mood” is often associated with much longer timescales than we discuss in this work.

C.1.3 De-Risking and Physiological Arousal

We previously documented a reduced-form relationship between physiological arousal and de-risking behavior. Here, we decompose this effect into distinct trading actions to identify whether arousal predicts intentions to sell, successful execution, or net selling within a period.

Those who had heightened physiological arousal (measured by in-period EDA peak EDA^{peak}) placed more aggressive sell orders. To uncover this, we conducted an exploratory analysis where we compared de-risking on different levels, each represented by a binary variable that takes a value of 1 if the given behavior was observed. Logit regressions were then run on these variables with EDA^{peak} as the independent variable. Participants who attempted to sell more shares than they bought are referred to as Net Sellers. We find that EDA^{peak} is not predictive of Net Selling, though the relationship is also positive. Likewise, EDA^{peak} does not significantly predict an attempt to sell ($\beta = -0.0678$; $p = 0.663$). However, EDA^{peak} 's effect on participants executing a sell is significant ($\beta = 0.602$; $p = 0.005$). This dissociation suggests that the decision to attempt a sale may be driven by multiple factors — such as hedging or arbitraging expected returns against the fixed return of cash — whereas a strong motivation to de-risk may lead participants to price the asset more aggressively, thereby increasing the likelihood of completing the trade.

Table C.3: **EDA De-Risk Logistic Regressions**

	Executed Sell	Attempted Sell	Net Sell
EDA Peak	0.602*** (0.2120) [0.005]	-0.068 (0.1556) [0.663]	0.050 (0.1694) [0.767]
N	2280	2280	2280
Pseudo R^2	0.004	0.000	0.000

Notes: All are logit regressions with robust standard errors, clustered by subject (shown in parentheses). p -values are shown in square brackets. Executed Sell takes a value of 1 when a participant manages to sell shares. Attempted Sell is a boolean that takes a value of 1 when a participant places at least one sell order (regardless of quantity and price). Net Sell takes a value of 1 when a participant attempts to sell more shares than they bought. Attempted Trade takes a value of 1 if a participant in a given period places any order at all (buy or sell) and zero otherwise. Across the 2,280 period-level observations, the share of zeros is 82.98% for Executed Sell, 60.22% for Attempted Sell, and 71.36% for Net Sell. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table C.4: Tests for Differences in Quadratic EDA Profiles by Earnings Tercile

Test	F -statistic	p -value
Classical Chow	16.917	1.27×10^{-8}
Robust Wald Interaction	28.569	1.06×10^{-12}

Notes: Tests compare the High and Low earning-tercile quadratic EDA profiles estimated from $\overline{\text{EDA}}_{\tau}^{\text{ton}} = \alpha + \beta\tau + \gamma\tau^2 + \epsilon$, where τ denotes periods relative to the market peak. The test is implemented by estimating a pooled interaction model with Low-tercile indicators interacted with τ and τ^2 . The null hypothesis is equality of the intercept, linear, and quadratic terms across High and Low terciles. The robust Wald test uses heteroskedasticity-robust standard errors. Regressions are estimated on tercile-by- τ binned means.

C.1.4 Arousal and Time-Varying Risk Premia

Our experimental design also enables us to investigate the widely held belief that bubbles are a rational consequence of changing individual risk premia over time, rather than some form of behavioral irrationality. To investigate this, we incorporated a separate task in which participants chose between multiple lotteries that utilized the Dynamically Optimized Sequential Experimentation (DOSE) methodology developed by [Chapman et al. \(2024\)](#) to measure individual subject risk premia at each market period. The task is encompassed in the fourth window in the sequence of windows given to the participants (see Section A.2.5 for details). This measurement enabled us to obtain an exogenous measure of risk at each time period and directly compare it to price movements to determine if there is any relationship.

Figure C.3a shows the plots of the average log return for our risk measure ρ (risk-aversion) and Figure C.3b shows the plots of the average log return for our risk measure μ (choice consistency), obtained from DOSE. We then also fit a regression, given the quadratic shape of our return curve, in order to see if the lack of relationship between ρ and average returns may be due to a misspecification.

$$R_t = \beta_0 + \beta_1\rho_t + \beta_2\rho_t^2 + \epsilon_t$$

In our experimental context, despite the findings of [Cohn et al. \(2015\)](#); [Guiso, Sapienza, and Zingales \(2018\)](#), we find little evidence supporting the idea that there is a significant relationship between risk premia and the formation of bubbles. Despite some partial correlations with next-period returns in specific specifications, the effects are small and do not appear in the specifications we describe above.

We find no evidence that time-varying risk premia explain bubble formation. The quadratic regression yielded an R^2 of 0.000, an adjusted R^2 of -0.003, and an F-statistic of 0.902, indicating no explanatory power. The linear and quadratic term estimates were

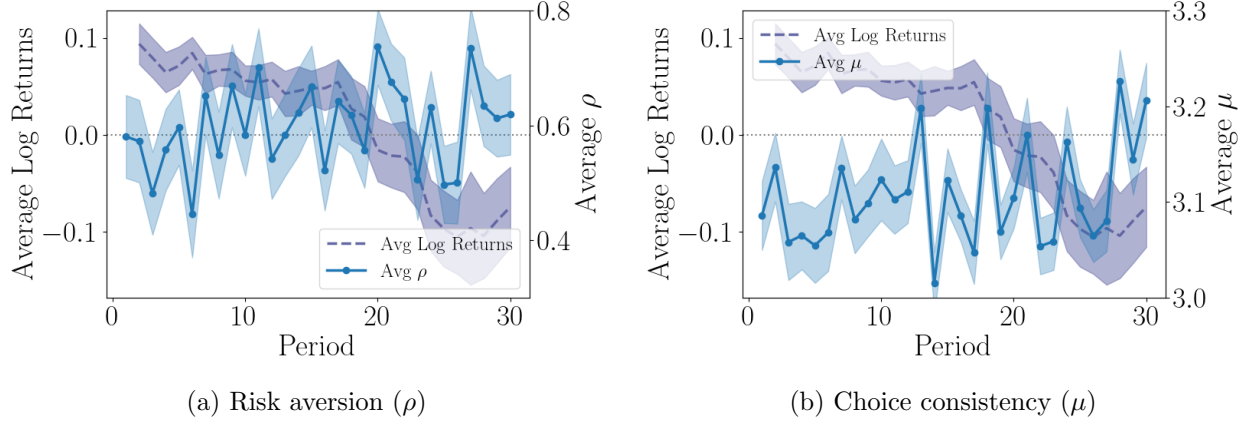


Figure C.3: Log Returns vs. DOSE Risk Parameters Across Periods

Each parameter is estimated for each subject from the risk-elicitation task (Chapman et al., 2024). Panel (a) plots the risk-aversion coefficient ρ (higher values indicate greater risk aversion) and panel (b) the choice-consistency coefficient μ (higher values indicate greater consistency). Neither shows a statistically significant relationship with R_t , suggesting that time-varying risk premia are not the drivers of bubble-like returns.

Alt text: Two panels plotting log returns against the DOSE risk-aversion parameter in panel (a) and the choice-consistency parameter in panel (b) across periods, with confidence intervals and no visible relationship.

close to zero ($\beta_{linear} = -0.004$, $p = 0.767$; $\beta_{quadratic} = 0.004$, $p = 0.876$). These findings provide evidence against the time-varying risk premia hypothesis of bubble formation.

C.2 High vs. Low Earner Behavior in Experimental Markets

There is substantial evidence that asset-market outcomes often reflect systematic differences between traders with more information or stronger task-relevant skills and those with less expertise (Brunnermeier and Nagel, 2004).

Our hybrid lab–online design provides new leverage on this issue. In our study, we analyze the differences in behavior and biometrics that exist between our highest-earning participants and our lowest-performing participants. The results are from exploratory analyses. Analysis of shareholding patterns revealed three clusters of behavior: high earners bought shares early and sold later, medium earners did not make many transactions, and low earners sold early and bought late. Because of this, we split our participant analysis into these three groups. These earning terciles are determined within each session, so it is possible that a medium earning participant in one session might actually earn a larger market bonus than a high earner in another. Indeed, we do observe some overlap in earnings by tercile (Table C.1). We also analyze the strategic differences between our high and low earner participants using NLP techniques in Section C.3.

C.2.1 Trading Behavior of High vs. Low Earners

Our goal in these analyses is to understand the behavioral differences that exist between high and low earners. Figure C.4 shows that the top market-earnings tercile tended to accumulate shares early in the experiment, and then steadily reduce their holdings in later market periods, which suggests that most of the trading was done between the high and low earners. This unloading behavior intensified just before the market peak. In contrast, the low-earning participants showed the opposite pattern: they gradually increased their share holdings throughout the market’s rise and continued to do so even through the later market periods. This behavior suggests that high-earning participants may anticipate the market’s run-up and crash more effectively than low-earning participants. We analyzed the strategy differences outlined by the participants in the different terciles; high-earning participants are much more likely to adopt a strategy of “buy early, sell later” (Figure C.7).

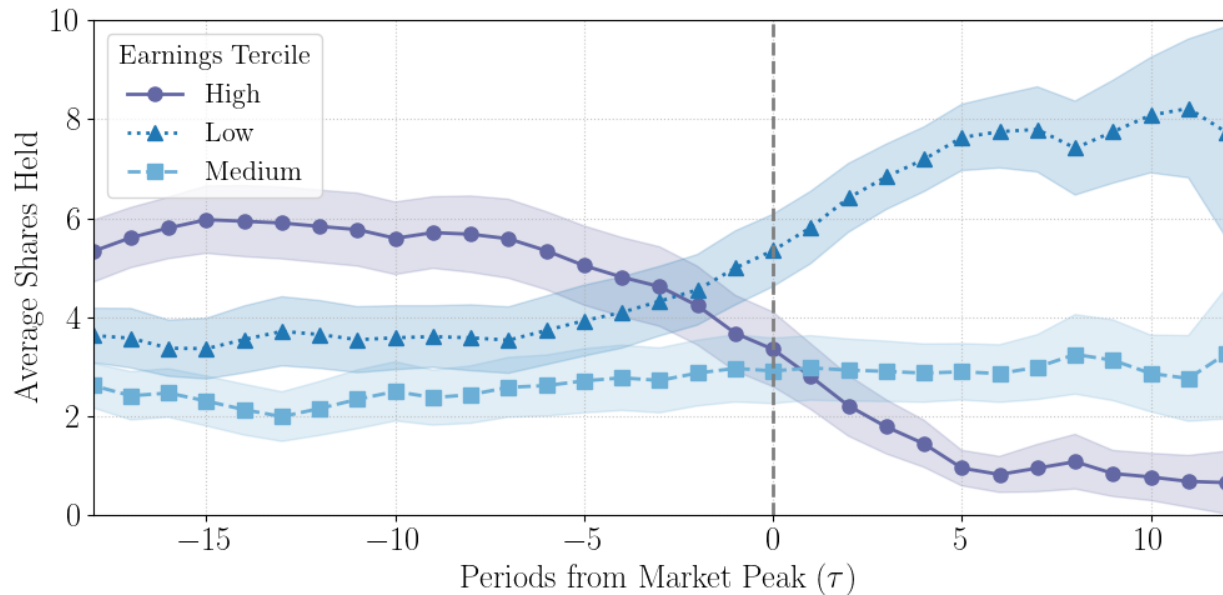


Figure C.4: **Average Shares Held by Earnings Tercile**

The high-earning group accumulated shares early in the session and sold them off later. The lowest-earning group increased their holdings late in the market, particularly after the average peak period.

Alt text: Line chart of average shares held by period for high, medium, and low earnings terciles with confidence bands; high earners accumulate shares early and sell before the peak, while low earners hold more later.

C.2.2 Forecast Behavior of High and Low Earners

The idea that the highest-earning participants were better able to anticipate market patterns is substantiated in our results, which show that high-earning participants are significantly more accurate forecasters at all time horizons, as shown in Figure C.5 and Table C.5.

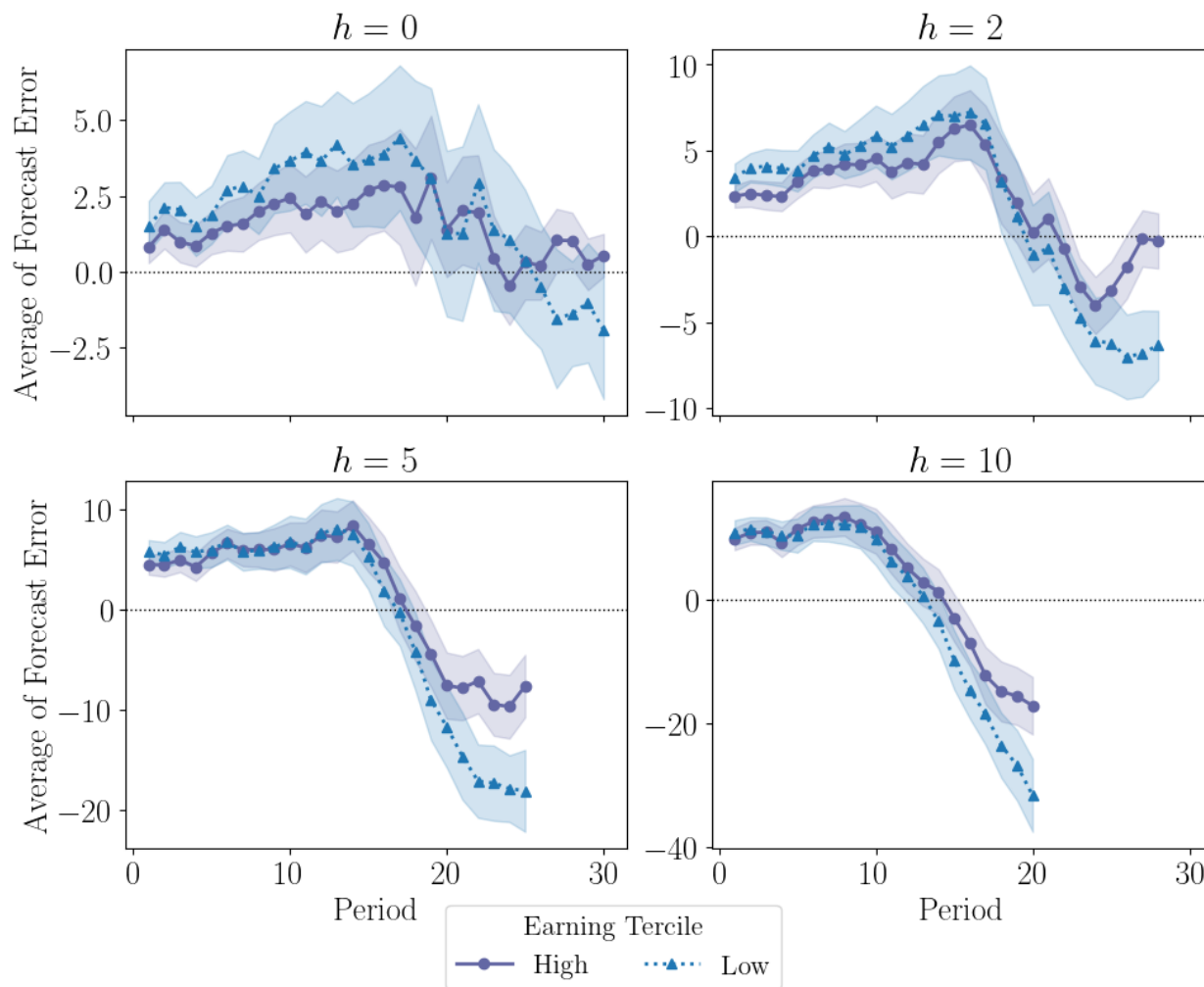


Figure C.5: **Mean Forecast Error by Earnings Tercile**

Shaded regions represent 95 percent confidence intervals. h denotes the forecast horizon.

Alt text: Line charts of mean forecast error by period for high and low earnings terciles at each horizon, with 95 percent bands; low earners' errors are larger after the peak.

Together, these findings suggest that higher earnings are not merely the result of luck or random variation but reflect a systematic forecasting advantage. This advantage may in turn enable them to strategically accumulate and offload shares in anticipation of market peaks and crashes.

Table C.5: **Forecast Error Comparison**

h	High	Low	p -value
0	4.304	7.604	0.001
2	7.828	11.010	0.001
5	13.363	17.359	0.001
10	19.841	23.756	0.012

Notes: Root mean squared error was calculated for each participant and each horizon (h). Participants were then grouped by earnings tercile, and t -tests were conducted to compare RMSE between the High and Low Earners.

C.3 Debriefing and Language Modeling

We sought to gain an understanding of the types of strategies our participants used. To do so, we included a free-response question in the post-experiment survey in which we asked the participants to describe the strategy they used during the session. The exact text of the question can be found in the *Post-experimental questionnaire* section of the supplementary material. For this analysis, we excluded the High-Stakes and Experienced sessions to maintain consistency of participants’ risk attitudes and experience. Of the 362 participants recruited, we received 302 responses.

Figure C.6 presents a word cloud visualizing the raw frequency of words used in the responses, with a word’s size corresponding to its frequency of use. The most frequently mentioned words are “sell” (140 occurrences), “buy” (133), “stock” (116), “price” (94), “market” (56), “low” (54), and “high” (48). A cursory glance shows that participants are keenly focused on buying and selling, and the appearance of “low” and “high” suggests that the classic “buy low / sell high” strategy is often employed. Indeed, it is a prominent category of strategy groups discussed below.

We then sought to group participants into strategy clusters. To do so, we computed the similarity of strategies using sentence embeddings and then performed a k-means clustering algorithm, after dimensionality reduction via principal component analysis (PCA).⁴¹ Next, we stemmed⁴² the words with `PorterStemmer` from Python’s Natural Language Toolkit (`nltk`). This step combines similar words, increasing their impact on the subsequent analysis. We then vectorized the individual strategies using sentence embeddings and ran a k-means cluster analysis on the results to classify the strategies into five groups. Figure C.8 presents the clustering results, categorizing the reported strategies into five distinct groups. The first two principal components explained 20.8 percent of the variance.

⁴¹We preprocessed strategies by removing all stop words, except for the word “no”. Thirteen participants responded with a single-word answer, “no”, which we still wanted to classify.

⁴²Removed the prefixes and suffixes of the words.

throughout the market. In contrast, the “Buy Early” participants typically made most (or all) of their purchases in the early periods, seemingly anticipating the boom-and-bust nature of these sessions. This group and their strategy were likely the drivers of the early surge in the average share path of high-earning median participants, as seen in Figure C.4. The participants most often assigned to these two strategy groups were the high-earning median participants, while the lowest-earning median participants appeared in them the least.

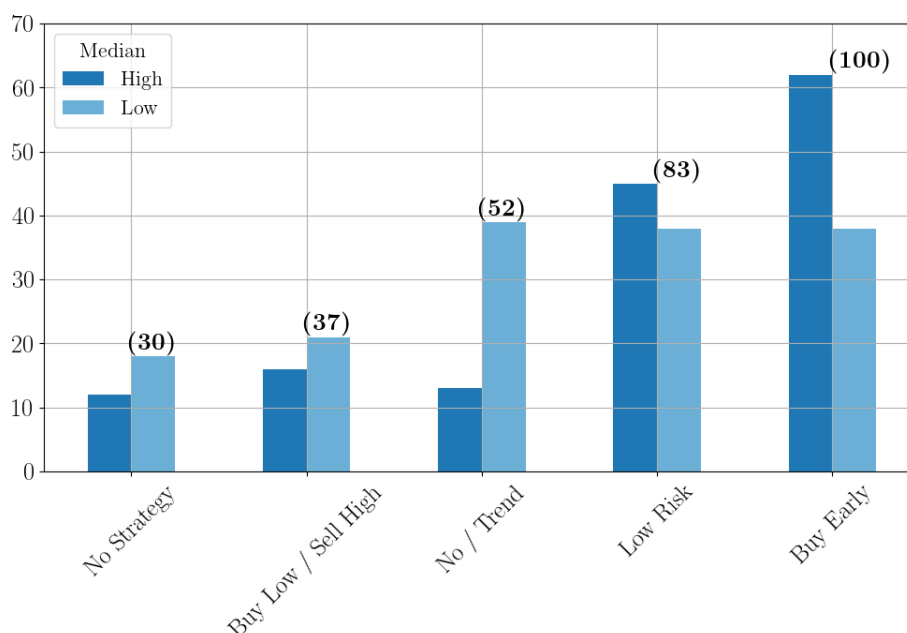


Figure C.7: **Strategy Group Assignments by Earnings Median**

Counts of strategy group assignment by earnings median. The count of all strategies assigned to a group is shown above the bar clusters. The strategy group **No Strategy** consists of those who responded with “No”, “None”, or the like. **No / Trend** mostly stated that they did not follow a strategy, but also indicated that they followed the market trend or guessed. **Buy Low / Sell High** are those that express that classic strategy. **Low-Risk** strategies focused on accruing interest and selling shares as soon as possible. **Buy Early** strategies involve acquiring the stock at an early stage with the intention of selling it later.

Alt text: Grouped bar chart of counts of strategy-cluster assignments for participants above and below median earnings, with cluster totals printed above each group.

Appendix D. Forecasting Behavior

D.1 Background literature about forecasting models

How people form expectations about the future is one of the oldest questions in economics, and one that has resurfaced repeatedly as new data and theories emerge. This Appendix section traces four connected strands of that literature:

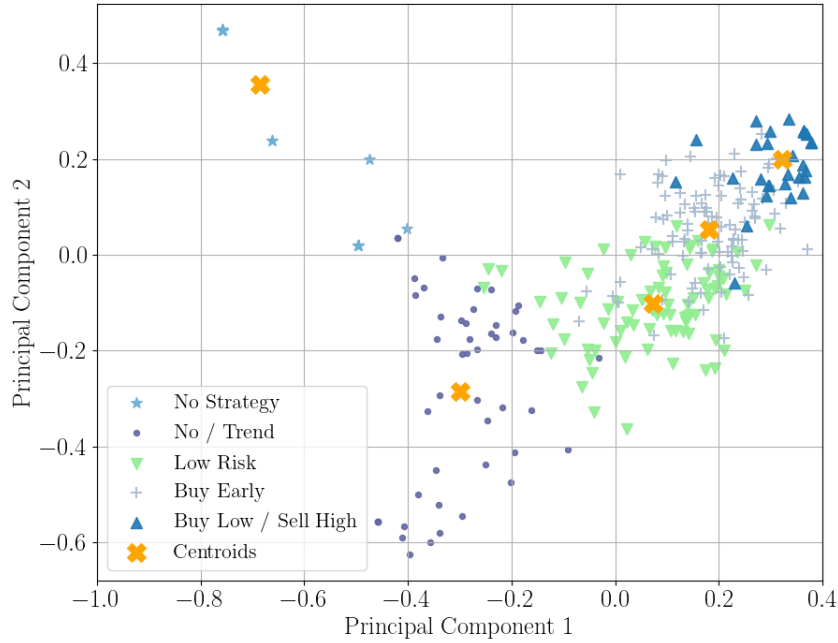


Figure C.8: **PCA of Sentence Embeddings with K-Means Clustering**

After vectorizing all strategies, we ran a k-means clustering algorithm and used principal component analysis (PCA) to reduce the vector space to two dimensions. The first principal component accounts for 13.7 percent of the variance, and the second component accounts for 7.1 percent.

Alt text: Scatter plot of the first two principal components of sentence embeddings of strategy descriptions, with points colored by k-means cluster.

- The earliest debate between rational and adaptive models of expectations formation and the survey-based tests that debate spawned;
- A body of experimental psychology from the 1970s and early 1980s showing that ordinary people extrapolate simple numerical time series in systematically biased ways;
- The resurgence of interest, beginning around 2010, driven by survey data on macroeconomic variables and investor expectations of stock returns;
- And the recent resurgence of interest which uses controlled laboratory forecasting tasks reminiscent of the 1970s psychology experiments, to test whether there is a unified bias — such as overreaction — across settings ranging from individual investors to professional macroeconomic forecasters (e.g., [Afrouzi et al., 2023b](#)).

D.1.1 Foundations: Rational and Adaptive Expectations

The modern literature began in the 1950s and 1960s with a split between two ways of modeling how agents use information about the past to predict the future. [Cagan \(1956b\)](#) modeled expectations of inflation during hyperinflations as an adaptive, backward-looking process: agents revise their forecast by some fraction of their most recent forecast error, so that expectations are a geometrically declining weighted average of past realizations. [Nerlove \(1958\)](#) generalized this error-learning logic to agricultural supply decisions in his treatment of cobweb phenomena, giving adaptive expectations a foothold in applied macroeconomics and industrial organization through the 1960s. Both models make expectations a function only of past data.

[Muth \(1961b\)](#) introduced the idea of “rational expectations” which challenged the adaptive expectations view. Muth argued that if adaptive expectations were systematically wrong in a way that agents could detect and correct, then at some point, adjustment and profit-seeking behavior would eliminate such biases. More aggressively, Muth proposed that a model of the economy should treat expectations as numerically equal to the mathematical expectation implied by the model itself (conditional on agents’ information sets). This rational expectations hypothesis was initially a modest econometric proposal but was subsequently generalized and popularized by Lucas, Sargent, and Prescott and became an accepted part of macroeconomics. Rational expectations made policy models immune to the Lucas critique, and also made general equilibrium models tractable because the model and expectations about it were the same (up to error).

A weaker version of expectations rationality is that agents should make efficient use of information. This hypothesis is easily testable if there are data on expectations. This gap motivated an enduring empirical tradition of eliciting and testing expectations directly through surveys. [Carlson and Parkin \(1975\)](#) proposed an influential method for converting qualitative survey responses — the fraction of respondents who say prices will rise, fall, or stay the same — into a quantitative measure of expected inflation. This procedure, and many variants, allowed researchers to test the rational expectations hypothesis against long-lived samples of survey data (the Livingston Survey of professional forecasters, the Michigan Survey of Consumers, and others) well before administrative or experimental data on beliefs became widely available. The resulting empirical literature repeatedly found deviations from strict rationality — serial correlation in forecast errors, sluggish responses to news, and heterogeneity across respondents that a representative-agent rational expectations model could not easily accommodate — without offering a settled alternative model of how those errors arose. [Lovell \(1986\)](#) synthesized this body of survey-based tests directly against Muth’s hypothesis, and concluded that the cumulative weight of the evidence was strong enough to

warrant suspending the maintained assumption of rational expectations in applied work.

D.1.2 Some Cognitive Foundations: Experimental Psychology and Time-Series Extrapolation

In parallel with the econometric literature, but not in conversation with it, cognitive psychologists in the late 1960s and 1970s were also documenting systematic and predictable errors in how people reason about probability, sequences, and time series growth. Most famous and impactful were studies about heuristics which could be identified by the biases from statistical logic and Bayesian rationality they created ([Tversky and Kahneman, 1971](#); [Kahneman and Tversky, 1972](#); [Tversky and Kahneman, 1974](#)).

A more specialized psychological experimental literature asked subjects to do something very close to what a forecaster does: look at a short numerical or graphical time series and predict its next value, or extrapolate its value to a future date. [Wagenaar and Sagaria \(1975\)](#) presented subjects with data or graphs following exponential growth and asked them to extrapolate; subjects consistently and substantially underestimated future values, a bias that a brief statistics lecture reduced only modestly. [Timmers and Wagenaar \(1977\)](#) extended this to an “inverse” version of the task — asking how long a quantity would take to reach a target level — and found the same qualitative underestimation of exponential growth. [Wagenaar and Timmers \(1978\)](#) showed that the bias was not remedied by giving subjects more data points to extrapolate from. [Wagenaar and Timmers \(1979\)](#), in the well-known “pond-and-duckweed” experiments, showed that the same misperception of exponential growth persisted whether the series was presented numerically, graphically, or as an animated simulation of duckweed spreading across a pond. Taken together, this cluster of studies established that misperception of nonlinear growth in a simple time series is a robust perceptual and judgmental phenomenon, largely insensitive to the amount of information provided or the format in which it is displayed.

A parallel experiment in economics was conducted by [Schmalensee \(1976b\)](#), using a repeated time-series forecasting task incentivizing both forecasts and confidence (implicit dispersion). The estimated speed of adjustment fell sharply around turning points in the series, and self-reported confidence tracked recent forecasting performance.

[Eggleton \(1982\)](#) brought this experimental paradigm directly into an economically relevant domain, asking subjects — including some with accounting and finance training — to extrapolate time series resembling firm-level accounting and financial data. Eggleton’s subjects displayed systematic anchoring on recent values and “trend-damping”: forecasts for series with an upward trend were too low, and forecasts for series with a downward trend were too high, relative to what a correct linear or exponential extrapolation would imply.

(Note that this damping is exactly what is seen in our own bubble forecasting data.) This trend-damping finding proved durable; later work modeled it explicitly as an anchoring-and-adjustment process in which forecasters anchor on the recent average of the series and adjust insufficiently for its trend. The economic significance of this line of research is that it isolates a purely perceptual and cognitive source of forecast bias, stripped of the strategic, informational, and incentive confounds present in field data. The value of isolating the psychology is also what motivated the controlled laboratory approach revived four decades later in behavioral economics by [Afrouzi et al. \(2023b\)](#).

D.1.3 The Post-2010 Resurgence: Investor Expectations Surveys and Extrapolative Beliefs

Interest in directly measured expectations experienced a marked revival beginning around 2010, driven substantially by work that combined longitudinal investor survey data with formal behavioral models of belief formation. [Lakonishok, Shleifer, and Vishny \(1994\)](#) had already shown, using realized return data rather than surveys, that “glamour” stocks with strong recent growth were priced as though investors extrapolated that growth indefinitely, while contrarian strategies betting against such extrapolation earned superior returns. [Greenwood and Shleifer \(2014b\)](#) made the underlying belief mechanism directly observable, assembling six independent survey series of investor return expectations spanning 1963 to 2011. They documented that these six measures are highly correlated with one another and with recent past stock returns, but strongly *negatively* correlated with model-implied expected returns of the kind that rational, representative-investor asset-pricing models would predict.

[Barberis et al. \(2015b\)](#) gave this extrapolative behavior a tractable theoretical home in the “X-CAPM,” in which a subset of investors form beliefs about future returns by extrapolating a weighted average of past returns, generating both the excess volatility and the return predictability patterns documented in the extrapolation literature. [Bordalo et al. \(2019b\)](#) took a different, more explicitly psychological route, formalizing [Kahneman and Tversky](#)’s representativeness heuristic as “diagnostic expectations”: forecasters over-weight the states of the world that have become more likely in light of recent news, generating excessive optimism after good news and excessive pessimism after bad news, followed by predictable reversals. Diagnostic expectations were shown to fit the joint dynamics of earnings expectations and stock returns, and the same framework was subsequently applied to credit markets in [Bordalo, Gennaioli, and Shleifer \(2018\)](#), where diagnostic beliefs about default risk generate boom-bust credit cycles.

A parallel empirical literature asked whether the same overreaction pattern found in

investor and analyst expectations also characterizes expectations about macroeconomic aggregates, where the earlier survey-testing tradition of [Carlson and Parkin \(1975\)](#) had generally found the *opposite* sign of bias — sluggish, underreacting forecast revisions consistent with costly information acquisition or “rational inattention.” [Coibion and Gorodnichenko \(2012, 2015\)](#) formalized this information-rigidity framework and showed, using Survey of Professional Forecasters data, that consensus forecast revisions underreact to news in a way consistent with sticky or noisy information. [Bordalo et al. \(2020b\)](#) revisited the same class of surveys — the SPF along with the Livingston Survey and the Blue Chip survey of professional forecasters — and showed that at the level of *individual* forecasters, rather than the cross-sectional consensus, expectations of inflation, unemployment, and output growth display significant overreaction to news, not underreaction; aggregation across heterogeneous individually overreacting forecasters can mechanically produce underreaction in the consensus, reconciling the two literatures. [Bordalo, Gennaioli, and Shleifer \(2022\)](#) later surveyed this overreaction and diagnostic-expectations research program for a general audience.

Complementary survey evidence accumulated on other kinds of economic beliefs during this period: [Amromin and Sharpe \(2014b\)](#) used the UBS/Gallup survey of individual investors to show that expected returns and expected risk move together with recent macroeconomic and market conditions in ways not well captured by rational updating. [Bouchaud et al. \(2019a\)](#) showed that sell-side analysts’ earnings forecasts are excessively sticky relative to news, generating a profitability-based return anomaly. [Kumar et al. \(2015\)](#) found, using a firm-level survey in New Zealand, that even a long-standing, credible inflation-targeting regime does not fully anchor firms’ inflation expectations, providing some of the survey groundwork on which the later individual-level overreaction results would build.

D.1.4 The Afrouzi et al. (2023, *QJE*) Synthesis

The literature above left two loose ends: field survey data on expectations, however rich, always confounds forecasters’ cognitive biases with their private information sets and their incentives, making it hard to isolate a ‘pure’ bias; and the overreaction findings in finance and the underreaction findings in macroeconomics needed—ideally—to be reconciled within a single framework rather than treated as evidence for competing models. [Afrouzi et al. \(2023b\)](#) address both problems by returning to the controlled experimental design used in psychology, at a larger experimental scale with online participants. Participants were asked to forecast the next value of stable autoregressive, AR(1), stochastic processes whose persistence the experimenters varied directly, so that the true, model-implied optimal forecast was known exactly and any deviation from it could be attributed unambiguously to a bias in belief formation rather than to private information or strategic behavior.

The paper reports three central facts. Forecasts overreact to recent values; overreaction is stronger for more mean-reverting processes; and overreaction is stronger for longer forecast horizons. They show that other models cannot explain these data but a recency bias can. This single mechanism reproduces the paper’s laboratory findings and, the authors argue, is consistent with overreaction patterns previously documented separately in analyst forecasts, individual investor return expectations, and (once aggregation effects are accounted for) individual-level macroeconomic forecasts. This evidence has already led to some new theory: [Angeletos, Huo, and Sastry \(2021\)](#) develop a general imperfect-information framework nesting both over- and under-reaction as special cases depending on the information structure.

More recently, the approach to behavioral economics called “cognitive economics” has incorporated formal, biologically grounded ideas about memory and similarity to microfound a wide range of economic decisions. Many of these revolve around forecasting (see [Bordalo et al., 2020b, 2023](#); [Gennaioli et al., 2024](#); [Bordalo et al., 2025](#)).

D.2 Evaluating Individual Forecast-Trade Consistency

While forecasting accuracy is a natural benchmark for assessing beliefs, accurate or biased forecasts alone do not determine market outcomes unless they are acted upon. A separate, conceptually distinct question is whether participants trade in a manner consistent with their stated expectations. [Carlé et al. \(2019\)](#) is directly relevant because it shows that heterogeneous beliefs predict individual purchases, holdings, bids, and asks in experimental asset markets. Even if forecasts are biased or noisy, a trader who systematically conditions their trades on their beliefs may still behave coherently, whereas a trader who disregards their own forecasts exhibits a disconnect between beliefs and actions. Assessing this internal consistency, therefore, provides insight into whether forecasts meaningfully guide trading decisions and whether heterogeneity in performance reflects differences in how beliefs are operationalized, rather than differences in beliefs themselves.

In our experiment, participants provide forecasts prior to the revelation of actual prices. Specifically, at each period t , participants submit forecasts for the current period as well as for the next two, five, and ten periods. Let $f_{i,t}^{t+h}$ denote the forecast made by participant i at period t for the price in period $t + h$, where $h \in \{0, 2, 5, 10\}$. Thus, $f_{i,t}^t$ specifically refers to the forecast for the current period’s price P_t made by participant i . For example, before the price for period 8 is revealed, each participant provides the forecasts $f_{i,8}^8$, $f_{i,8}^{10}$, $f_{i,8}^{13}$, and $f_{i,8}^{18}$. Throughout this paper, session identifiers are omitted from the forecast notation for simplicity.

To analyze the link between expectations and trading decisions, we regressed each participant’s net trading on an indicator for whether their forecasted next-period price increase exceeded the 5% risk-free rate, estimating the regression separately for each individual.

If a participant forecasts that the price will rise by more than the risk-free rate of 5%, a consistent response (in the absence of constraints) would be to increase net buying. Therefore, our indicator regression provides a proxy for internal consistency: a significant, positive relationship would indicate that participants trade consistently with their own beliefs.

$$(\hat{\beta}_{0,i}, \hat{\beta}_{1,i}) = \arg \min_{\beta_0, \beta_1} \sum_t \left(Y_{i,t} - \beta_0 - \beta_1 \mathbf{1} \left\{ \frac{f_{i,t}^t - P_{t-1}}{P_{t-1}} + \frac{0.7}{f_{i,t}^t} > 0.05 \right\} \right)^2. \quad (19)$$

The estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ are stored for each participant. To ensure accurate estimation, the following preprocessing steps were performed.

1. Buy orders were assigned positive values (+ q), while sell orders were assigned negative values ($-q$).
2. Net trading orders were computed as the sum of buy and sell orders for each participant in each period t :

$$Y_{i,t} = \sum_j q_{i,t}, \quad (20)$$

where $q_{i,t}$ represents an order’s quantity in period t .

Figure D.1 shows the distribution of participant-level coefficients. A positive coefficient indicates increased net buying when a participant forecasts returns above the sum of the risk-free rate and the dividend yield. More than half of the participants exhibit coefficients that are positive, and the mean coefficient is 0.89. That is, on average participants buy 0.89 more units when they expect $\geq 5\%$ returns compared to when they do not.

The positive consistency of forecasts and trades is consistent with experimental findings of Grosshans and Zeisberger (2025). However, the relation is not very strong and is often negative. This weak relation is also evident in equity markets, where there is often a “beliefs-portfolio” gap. This was first noted by Vissing-Jørgensen (2003) in data from wealthy investors. The gap was documented further by Giglio et al. (2021) and corroborated in a representative US sample (Choi and Robertson, 2020) and a high-wealth sample (Bender et al., 2022).

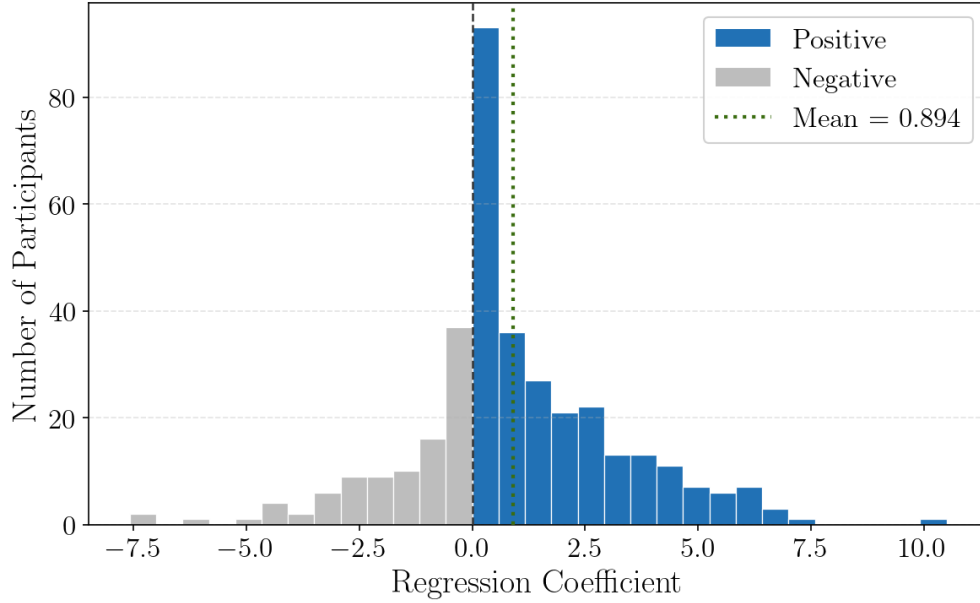


Figure D.1: Net Orders vs. Forecasted Excess Returns

Distribution across participants (Standard sessions) of the coefficient from a participant-level regression of net orders on an indicator equal to one when the participant’s forecasted return exceeds the 5% risk-free rate. The coefficient measures how much net buying increases (or decreases) in high-expected-return states relative to low- or sub-risk-free-return states; more positive coefficients indicate a higher degree of internal consistency between beliefs and trading behavior. The dashed line marks zero and the dotted line the mean. The mean coefficient is significantly positive (0.894; one-sample t -test versus zero, $p < 0.001$), indicating that participants tend to place more net buy orders when they expect returns above the risk-free rate.

Alt text: Histogram across participants of the coefficient from regressing net orders on an indicator for forecasted returns above the risk-free rate, with a dashed line at zero; most coefficients are positive.

D.3 Forecast Rationality and Expectation Models

Figure D.2 shows histograms of current-period and 2-period forecast errors. Means are slightly positive for both and about the same. The distribution for current-period errors is more leptokurtic than that for the 2-period errors. This makes sense because current-period prices are easier to predict.

Figure D.3 repeats the main-text mean forecast-error plot (Figure 5) using the *median* forecast across participants, which is less sensitive to extreme individual forecasts. The pattern is the same: errors grow with the horizon and become increasingly negative through the peak and crash.

The first column of Figure D.4 displays the distribution of the estimated δ parameters. Most agents have values close to 1, indicating they place substantial weight on the most recent

Table D.1: **Normality and Rationality Acceptance Rates (%) by Forecast Horizon**

Statistic	Test	h			
		0	2	5	10
$E_{i,t}^h = 0$	Normality	42.5%	45.9%	44.7%	46.5%
	Unbiasedness	62.0%	62.4%	57.1%	43.7%
$\beta_1 = 0$ (Equation 5)	No Autocorrelation	42.2%	—	—	—
$\beta_1 = 0$ (Equation 6)	No Err-Fcast Correlation	62.2%	35.1%	14.9%	11.8%

Notes: “Normality” refers to results from the Shapiro–Wilk test, with normality rejected at $p < 0.05$; the percentages for each horizon $h \in \{0, 2, 5, 10\}$ indicate the proportion of agents whose forecast errors are normal. “Unbiasedness” denotes that the mean forecast error is not significantly different from zero ($p > 0.05$), and the reported percentages reflect the proportion of unbiased agents at each horizon. “No Autocorrelation” and “No Err-Fcast Correlation” indicate agents whose forecast errors exhibit no significant autocorrelation or no significant correlation with forecasts at the 0.05 level, respectively, with the percentages representing the proportion of such agents for each horizon. Scatter plot of β ’s shown in Figure 6.

Table D.2: **OLS Regressions of $\log(\text{MSE})$ on Distance from Price Peak**

	$h = 0$	$h = 2$	$h = 5$	$h = 10$
Intercept	4.392*** (0.101)	4.975*** (0.089)	5.786*** (0.112)	6.383*** (0.132)
τ	−0.030*** (0.004)	−0.039*** (0.005)	−0.049*** (0.007)	−0.083*** (0.014)
τ^2	−0.003*** (0.000)	−0.004*** (0.000)	−0.005*** (0.000)	−0.005*** (0.001)
High-Stakes	−0.626*** (0.105)	−0.988*** (0.111)	−1.013*** (0.128)	−1.303*** (0.139)
Experienced	−0.467*** (0.128)	−0.033 (0.119)	0.081 (0.149)	0.183 (0.167)
R^2	0.582	0.704	0.626	0.641
Adj. R^2	0.569	0.694	0.613	0.626
N	133	127	118	103
Vertex (τ)	−5.20	−4.88	−5.14	−7.90

Notes: The dependent variable is the $\log(\text{MSE})$ of grouped mean forecast errors by periods from the price peak (τ). The Standard session type is the omitted category. Columns represent the forecast horizon $h \in \{0, 2, 5, 10\}$. HC3 heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

price when forming expectations. The mean estimate of 0.917 (Table 4) further supports this finding. The same pattern emerges in the High-Stakes and Experienced sessions (middle and bottom rows), where participants consistently weight the most recent price heavily. The second column of Figure D.4 shows that most agents have ϕ_i estimates between 0 and 1. The mean estimate of 0.199 suggests a moderate degree of extrapolation in the Standard sessions.

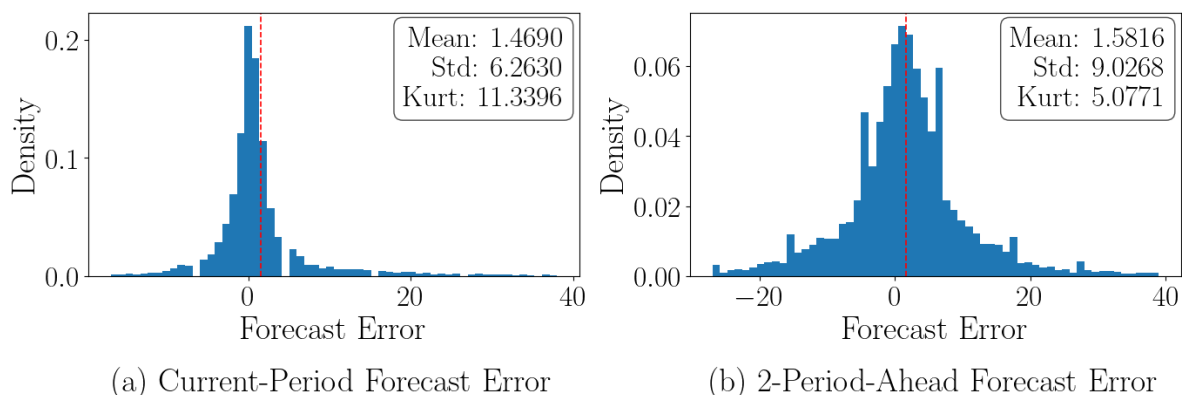


Figure D.2: **Histograms of Forecast Errors for the Current Period and 2 Periods Ahead**

Each panel reports the mean, standard deviation, and kurtosis, with the dashed line marking the mean. Tails are trimmed at the 1st and 99th percentiles.

Alt text: Two histograms of forecast errors, for the current period and two periods ahead, each annotated with mean, standard deviation, and kurtosis and a dashed line at the mean; both are peaked with heavy tails.

Table D.3: **Summary Statistics of Forecasts and Forecast Errors**

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Price	31.31	16.42	12	19.00	26.00	40.00	102
f_t^t	29.65	17.80	1	18.00	25.00	38.00	147
f_t^{t+2}	30.69	18.18	1	19.00	26.00	39.00	137
f_t^{t+5}	33.80	20.02	1	20.00	29.00	42.00	180
f_t^{t+10}	35.56	21.05	1	21.00	30.00	45.00	190
E_t^t	1.67	8.70	-84	-1.00	0.00	2.00	92
E_t^{t+2}	1.27	9.36	-68	-2.00	0.00	3.00	95
E_t^{t+5}	-1.23	12.01	-107	-6.00	-2.00	3.00	93
E_t^{t+10}	-3.63	15.14	-132	-10.00	-3.00	3.00	77

Table D.4: **Joint Significance of Forecast-Rationality Coefficients by Treatment**

	Standard	High-Stakes	Experienced	All
Both significant	86 (24.0%)	13 (22.0%)	19 (20.7%)	118 (23.1%)
One significant	164 (45.7%)	35 (59.3%)	53 (57.6%)	252 (49.4%)
Neither significant	109 (30.4%)	11 (18.6%)	20 (21.7%)	140 (27.5%)
N	359	59	92	510

Notes: Each participant is classified by how many of the two forecast-rationality coefficients—zero auto-correlation (Equation 5) and zero error-forecast correlation (Equation 6)—are significant at the 5% level. Percentages are column shares within each treatment and match the significance-group legends in Figures 6 and D.5.

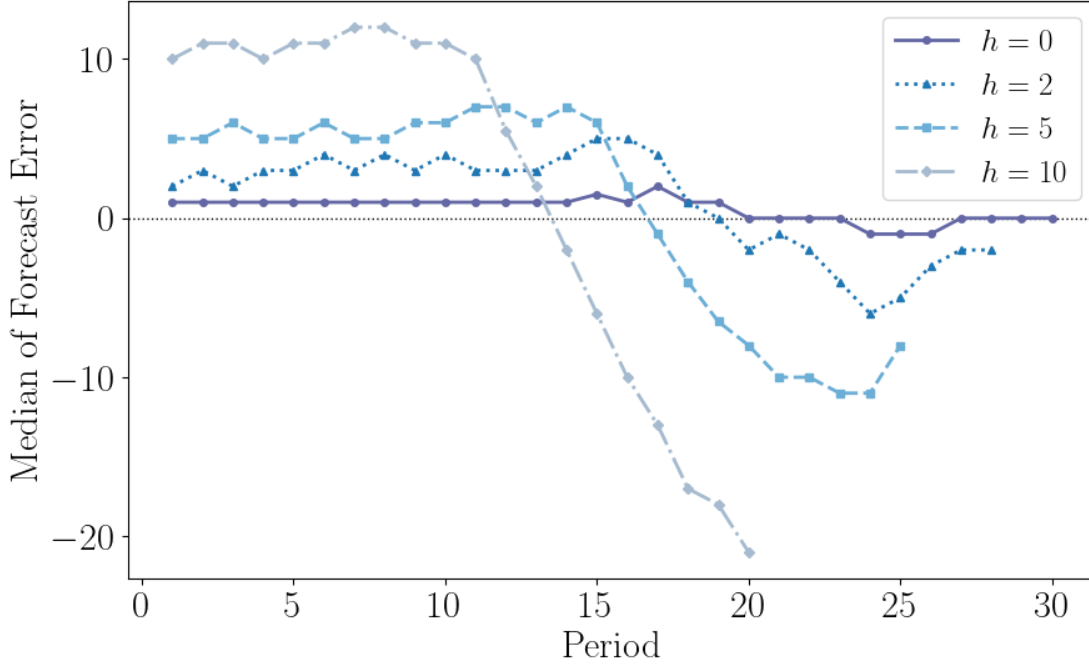


Figure D.3: **Median Forecast Error by Period and Horizon**

Median forecast error across trading periods for each horizon h , with 95% confidence intervals shaded. This is the median counterpart to the mean forecast error in Figure 5; the two display the same pattern.

Alt text: Line chart of median forecast error by period for horizons 0, 2, 5, and 10 with 95 percent bands, showing the same positive-then-negative pattern as the mean forecast error.

This extrapolative tendency strengthens in the High-Stakes sessions (mean $\phi = 0.273$) and the Experienced sessions (mean $\phi = 0.389$).

The fourth and fifth columns of Figure D.4 display the parameter distributions for the adaptive level and adaptive trend, respectively. Most estimates are positive and bounded between 0 and 1. The mean adaptive level parameter is 0.648, and the mean adaptive trend parameter is 0.572 (Table 4).

D.3.1 Tests of Competing Models of Forecasts

As a more demanding joint test, we regress both experimental bonus measures on all five expectation-formation parameters simultaneously (Table D.8). In the OLS specifications, these parameters explain a modest share of the variation in bonuses ($R^2 = 0.084$ for market bonus; $R^2 = 0.059$ for forecast bonus). Including session fixed effects increases explanatory power, slightly for the market bonus ($R^2 = 0.094$) and substantially for the forecast bonus ($R^2 = 0.219$). These figures are for the Standard sessions (Panel A); the High-Stakes and

Table D.5: **Summary Statistics of Parameter Estimates and Fit by Forecasting Model Across Session Types**

Model	Mean	SD	Mean R^2	N
<i>Panel A: High-Stakes Sessions</i>				
Adaptive (δ)	0.746	0.407	0.463	58
Extrapolation (ϕ)	0.273	2.520	0.162	58
Overreaction (β)	-0.214	0.337	0.175	59
DAE: Level Error (η)	0.608	0.369	0.572	58
DAE: Trend Error (γ)	0.644	0.569		
<i>Panel B: Experienced Sessions</i>				
Adaptive (δ)	1.036	0.446	0.548	91
Extrapolation (ϕ)	0.389	0.447	0.386	90
Overreaction (β)	0.121	0.395	0.162	92
DAE: Level Error (η)	0.772	0.364	0.754	92
DAE: Trend Error (γ)	0.483	0.330		

Notes: Panel A reports the High-Stakes sessions and Panel B the Experienced sessions. Mean and SD denote the mean and standard deviation of the model parameter estimate across participants for each model, respectively. Mean R^2 reports the average fit across participants, and N indicates the number of participants with non-missing estimates. The slight variation in sample size N across models is due to differences in model specifications (e.g., lag structures), which prevented estimation for a small subset of participants with insufficient observations. For the Double-Adaptive Expectations model, results are reported separately for the level and trend forecast error parameters.

Experienced sessions are reported in Panels B and C of the same table.

The adaptive parameter positively predicts the forecast bonus in both OLS and fixed-effects models, but lacks precision for the market bonus. Anchoring heavily on recent prices improves short-term prediction accuracy, yet it does not reliably translate into profitable trading decisions. Extrapolation positively correlates with both bonuses under OLS but becomes small and statistically insignificant once session fixed effects are included. This indicates that the apparent payoff to trend-chasing is driven by overall market conditions rather than individual skill.

The overreaction parameter (β) strongly predicts the market bonus across all specifications. A higher β corresponds to more cautious belief revision, which likely protects participants from suboptimal trades like buying at market peaks. Interestingly, its relationship with the forecast bonus flips from positive in OLS to significantly negative in the fixed-effects model. This reversal highlights a clear trade-off: while underreacting safeguards trading profits, it causes agents to lag behind in fast-moving markets, thereby hurting short-term forecast accuracy. The two DAE terms lack general precision, though the forecast-error component is significant for the market bonus under OLS, likely because the combined responses to price levels and trends offset each other in determining overall payoffs.

Table D.6: **Market & Forecast Bonus Explained by Forecast Parameters and Controls**

Dependent Variable	(1) MB	(2) FB	(3) MB w/ EDA	(4) FB w/ EDA
Adaptive (δ)	0.030 [†] (0.055)	0.063* (0.037)	0.152 [†] (0.117)	-0.060 [†] (0.121)
Extrapolation (ϕ)	0.082** (0.032)	0.029** (0.013)	0.483* (0.253)	0.118 [†] (0.164)
Overreaction (β)	0.081** (0.034)	0.012 [†] (0.022)	-0.026 [†] (0.185)	0.168 [†] (0.126)
DAE: Level Error (η)	0.020 [†] (0.058)	-0.017 [†] (0.024)	-0.153 [†] (0.194)	0.065 [†] (0.143)
DAE: Trend Error (γ)	0.036 [†] (0.073)	0.020 [†] (0.042)	-0.258 [†] (0.238)	0.219 [†] (0.189)
DOSE ρ	-	0.044** (0.019)	-	-
DOSE μ	-0.200* (0.119)	-	-	-
Net Orders	-	-	-0.101* (0.060)	-
Attempted Trades	-0.004*** (0.001)	-	-	-
Executed Trades	-0.003 (0.004)	-0.002 (0.001)	-0.017** (0.007)	-
Quiz Grade	-	-0.064*** (0.012)	-0.050 (0.071)	-0.045 (0.048)
General Risk (Pre)	-0.029** (0.014)	-	-	-
Own Stock	-	-	-0.108 (0.074)	-0.083* (0.046)
Self Place	-	-	-0.003* (0.002)	-
Invest 100 (Post)	-	-	-	-0.001 (0.001)
In-Lab	0.080* (0.046)	0.074** (0.032)	N/A	N/A
Fcast Normal	-	-0.060** (0.029)	-	-
Fcast Unbiased	0.010*** (0.003)	-	0.029** (0.012)	-
Fcast Auto Corr	-0.100** (0.047)	-	-	-
Fcast Err Corr	0.041 (0.058)	-	-	-
N	274	274	55	55
R^2	0.290	0.259	0.566	0.318

Notes: LASSO with five-fold cross-validation was used to select control variables. The five forecast-parameter variables are retained in all post-selection OLS specifications regardless of LASSO selection because they are the main variables of interest. Coefficients are from post-selection OLS with heteroskedasticity-robust standard errors in parentheses. [†] indicates forecast-parameter variables that LASSO would have set to zero but that were retained by design. Dependent variables are Market Bonus (MB) and Forecast Bonus (FB) transformed by $\log(1 + \text{bonus})$. Adaptive expectations, extrapolation, overreaction, and double-adaptive expectation parameters are individually estimated from their respective forecasting models and then used as regressors. Columns (1) and (2) show results from all participants. Columns (3) and (4) show only in-lab participants for whom we have biometric data. The “In-Lab” indicator equals one for participants who completed the study in the laboratory and zero for online (Prolific) participants; it is constant for models (3) and (4) and was therefore excluded (indicated with N/A). These two columns also include tonic EDA mean and tonic EDA standard deviation which were not selected by the LASSO regressions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

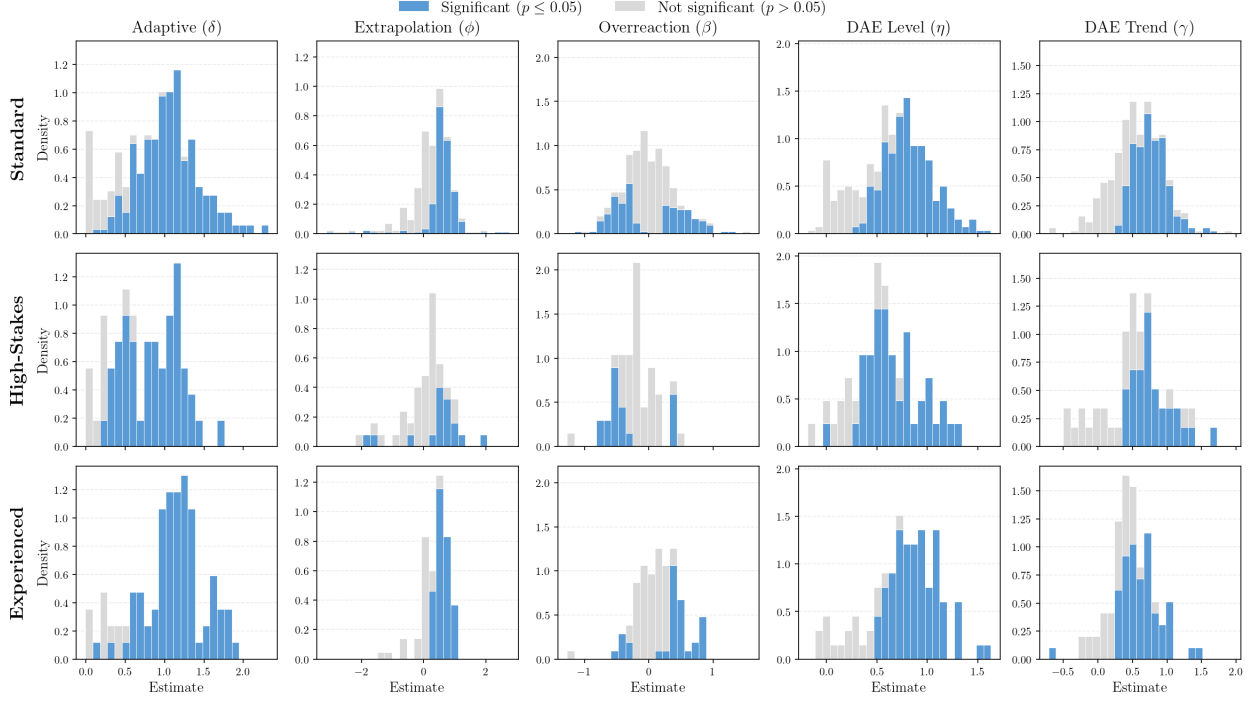
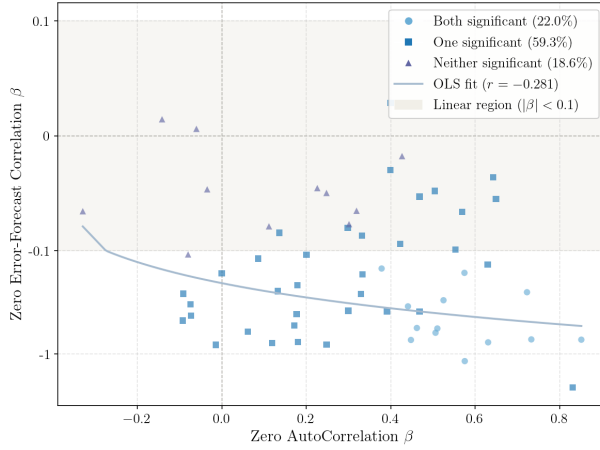
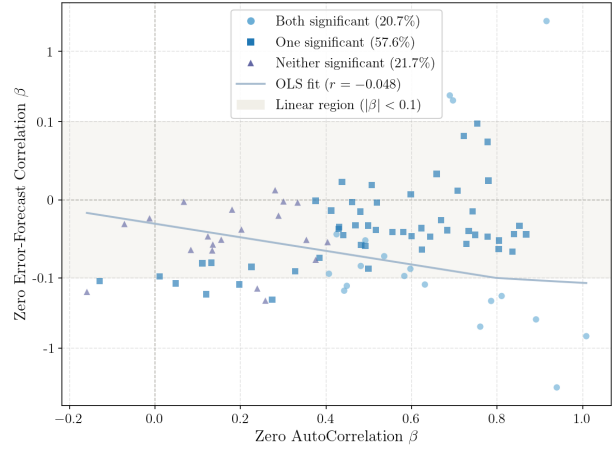


Figure D.4: Histograms of Parameter Estimates for Different Expectation Models Each column shows the cross-participant distribution of one expectation-model parameter—Adaptive (δ), Extrapolation (ϕ), Overreaction (β), and the double-adaptive level (η) and trend (γ) errors—and each row a session type: Standard (top), High-Stakes (middle), and Experienced (bottom). Models are estimated per participant by OLS with heteroskedasticity-robust (HC1) standard errors. Within each bar, light blue denotes estimates significantly different from zero ($p \leq 0.05$) and light gray those that are not; estimates more than three standard deviations from the parameter mean are omitted.

Alt text: Grid of histograms with one column per expectation-model parameter (adaptive, extrapolation, overreaction, double-adaptive level and trend) and one row per session type (Standard, High-Stakes, Experienced); light blue bars mark significant estimates and gray bars insignificant ones.



(a) High-Stakes sessions

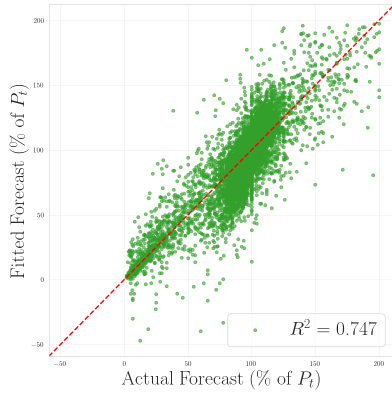


(b) Experienced sessions

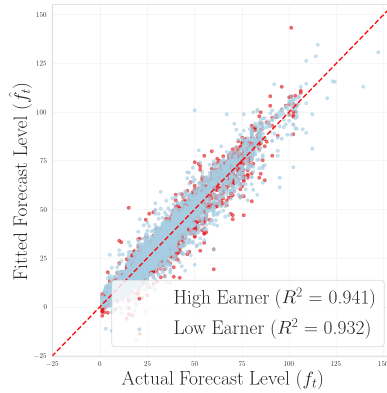
Figure D.5: Forecast-Rationality Coefficients in High-Stakes and Experienced Sessions

Individual-level autocorrelation coefficients (Equation 5, x -axis) against error-forecast correlation coefficients (Equation 6, y -axis). This is the appendix companion to Figure 6; the axes, the `symlog` y -scale, and the significance coloring are identical. Panel (a) shows the High-Stakes sessions and panel (b) the Experienced sessions.

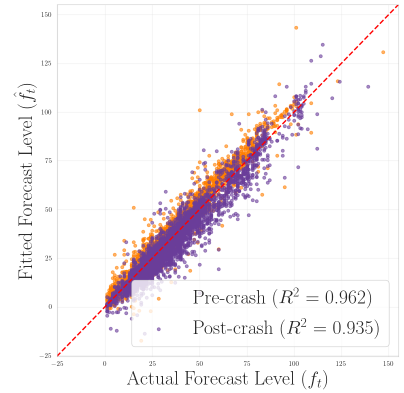
Alt text: Two scatter plots of error-forecast correlation coefficients against autocorrelation coefficients, for High-Stakes sessions in panel (a) and Experienced sessions in panel (b), with markers indicating joint significance.



(a) % of lagged price



(b) High vs. low earners



(c) Pre- vs. post-crash

Figure D.6: Double-Adaptive Model Fit Across Subgroups

Additional specifications of the double-adaptive-model fit shown in Figure 7, split three ways: panel (a) expresses forecasts as a percentage of the lagged price; panel (b) splits participants into high versus low earners; and panel (c) into pre- versus post-crash periods.

Alt text: Three scatter plots of fitted versus actual forecasts from the double-adaptive model, expressed as a percentage of lagged price, split by high versus low earners, and split into pre- and post-crash periods, each with a dashed 45-degree line.

Table D.7: **Expectation Model Parameters Across Sessions**

Model	Param.	Std.	HS	Exp.	Std.–HS	Std.–Exp.	HS.–Exp.
Adaptive	δ	0.917	0.746	1.036	0.005	0.027	< 0.001
Extrapolation	ϕ	0.199	0.273	0.389	0.826	0.009	0.729
Overreaction	β	0.009	−0.214	0.121	< 0.001	0.033	< 0.001
DAE: Level Error	η	0.648	0.608	0.772	0.441	0.004	0.009
DAE: Trend Error	γ	0.572	0.644	0.483	0.370	0.037	0.055

Notes: Reported values are cross-participant means from individual-level OLS estimates with heteroskedasticity-robust standard errors. The final three columns report pairwise Welch two-sample p -values comparing participant-level parameter estimates across market conditions. Session abbreviations are Std. = Standard, HS = High-Stakes, and Exp. = Experienced. Parameter interpretations are as follows: δ is the weight on the most recent price; ϕ is sensitivity to recent price trends; $\beta < 0$ indicates overreaction and $\beta > 0$ indicates underreaction; η adjusts to level forecast errors; and γ adjusts to trend forecast errors. A value of $\delta > 1$ implies overweighting the most recent price relative to prior expectations.

Table D.8: **OLS and Fixed-Effects Regressions of Bonuses on Expectation-Model Parameters, by Session Type**

	Market Bonus		Forecast Bonus	
	OLS	FE	OLS	FE
<i>Panel A: Standard Markets</i>				
Adaptive (δ)	0.003 (0.049)	0.033 (0.039)	0.076** (0.031)	0.069** (0.033)
Extrapolation (ϕ)	0.051* (0.028)	0.045 (0.035)	0.023** (0.009)	0.004 (0.006)
Overreaction (β)	0.109*** (0.032)	0.098*** (0.033)	0.027* (0.015)	-0.032** (0.013)
DAE: Level Error (η)	0.136** (0.068)	0.131 (0.089)	-0.003 (0.032)	0.016 (0.029)
DAE: Trend Error (γ)	0.072 (0.055)	0.108 (0.072)	-0.023 (0.017)	0.027 (0.020)
Constant	1.303*** (0.071)	1.260*** (0.094)	0.059** (0.029)	0.029 (0.054)
N	341	341	341	341
R^2	0.084	0.094	0.059	0.219
<i>Panel B: High-Stakes Markets</i>				
Adaptive (δ)	0.024 (0.117)	0.052 (0.159)	-0.006 (0.075)	0.014 (0.023)
Extrapolation (ϕ)	0.005 (0.006)	0.004 (0.007)	-0.014*** (0.005)	-0.010* (0.003)
Overreaction (β)	-0.139 (0.116)	-0.164 (0.163)	-0.066 (0.088)	0.027 (0.073)
DAE: Level Error (η)	-0.157 (0.135)	-0.180 (0.099)	-0.022 (0.089)	0.025 (0.047)
DAE: Trend Error (γ)	-0.015 (0.056)	-0.010 (0.036)	0.053 (0.035)	0.056 (0.024)
Constant	2.926*** (0.071)	2.910*** (0.097)	0.205*** (0.058)	0.177** (0.032)
N	57	57	57	57
R^2	0.040	0.041	0.091	0.089
<i>Panel C: Experienced Markets</i>				
Adaptive (δ)	0.216 (0.143)	0.226 (0.185)	0.036*** (0.010)	0.022 (0.011)
Extrapolation (ϕ)	0.188 (0.138)	0.242** (0.058)	0.011 (0.008)	0.018 (0.011)
Overreaction (β)	-0.047 (0.123)	-0.037 (0.119)	-0.030** (0.014)	-0.012** (0.003)
DAE: Level Error (η)	-0.526** (0.228)	-0.570** (0.131)	-0.049*** (0.018)	-0.035 (0.017)
DAE: Trend Error (γ)	-0.171 (0.158)	-0.151* (0.061)	-0.043*** (0.015)	-0.025 (0.014)
Constant	1.606*** (0.177)	1.598*** (0.136)	0.052*** (0.016)	0.044*** (0.007)
N	90	90	90	90
R^2	0.084	0.103	0.204	0.131

Notes: Dependent variables are $\log(1 + \text{bonus})$. Each panel regresses the market and forecast bonuses for one session type on the five expectation-model parameters (adaptive δ , extrapolation ϕ , overreaction β , and the double-adaptive level η and trend γ errors), which are individually estimated from their respective forecasting models. The OLS columns report robust standard errors in parentheses and the overall R^2 ; the FE columns include session fixed effects, the within R^2 (the share of variation explained after removing session-level means), and standard errors clustered by session in parentheses. Panels A, B, and C report the Standard, High-Stakes, and Experienced sessions, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix E. Experimental Materials

E.1 Instructions

Instructions were given by a video which can be seen here: https://youtu.be/232inD_HChc. Below is the transcript of those instructions.

Welcome.

This experiment is about economic decision-making in an experimental stock market. The experiment will consist of a series of 30 trading periods in which you will have the opportunity to buy or sell shares of an asset that can yield payments in the future. Understanding these instructions may help you earn money, and if you make good decisions you might earn a considerable amount of money which will be paid to you in cash at the end of the experiment.

There are two assets in this experiment cash and stock you begin with 100 units of cash and four shares of stock. Stock is traded in a session each period among all the experimental subjects in units of cash. When you buy stock the price you agreed to pay is deducted from your amount of cash. When you sell stock the price you sold at is added to your amount of cash. The reward from holding stock is called dividend. Each period, every unit of stock earns a low or high dividend of either 0.4 cash per unit or 1.00 cash per unit with equal probability. These dividend payments are the same for everyone in all periods the dividend in each period does not depend on whether the previous dividend was low or high. The reward from holding cash is given by a fixed interest rate of 5% each period.

Price conversion at the end of the 30 periods of trading each unit of stock is automatically converted to 14 cash. For example if the market price for round 30 is 20 and you have three stocks, you'll receive 42 cash not 60 cash. Then your experimental cash units are converted to US dollars at a rate of 200 cash per 1 US dollar to determine how much you will be paid at the end of the experiment. If you buy shares for more than 14 as you get near Round 30 it is possible those shares will be terminated at a value of 14 if you cannot sell them.

Let's take an example. Suppose one period you have 120 units of cash and 5 units of stock. The dividend is 0.4 per unit of stock. Then your new cash amount is going to be the sum of the initial 120 cash, a 5% increase in cash, and total dividends of 5 times 0.4. Your total cash would therefore be $120 + 6 + 2$ resulting in 128 cash. Notice that keeping cash will earn a return 5% per period and using cash to buy units of stock will also yield dividend earnings.

Each trading period will have four main screens that you will see one after the other. Order submission, forecast prices, round results, and choosing a lottery. This sequence is repeated 30 times. Let's look at each screen in a bit more detail.

On the order submission page in the upper right corner you will see the time left for this

page and the trading period. In this example the timer was set on purpose for 5 minutes but in the actual trading task you'll only have 20 seconds. For this page also, the next button will not be displayed in the actual trading task so you cannot click it.

Below that you will see the order submission rectangle where you can buy or sell stocks. The price is displayed on the y-axis with the Bold horizontal line being the previous market price. The quantity of shares you can buy is displayed on the x-axis. The vertical bold line represents the threshold between selling and buying. You can select the price level at which you want to buy or sell using the mouse or cursor. For example, I can enter a buy order at 15 and submit two sell orders at 16. You can see the orders you entered in the lower right corner. You can cancel orders by clicking on the X next to them. Orders are not carried from one period to the next. If I want to buy at 16 and sell at 15 I will get an error and the sell order will not go through similarly. If I place a sell order at 15 and then I want to submit a buy order at 16 the buy order will not go through. You can only buy stocks if you have enough cash, and you can only sell stocks you own. For example, if I want to sell five stocks I will get an error. Orders are limit orders. A limit order to buy one unit at a price of 15 means you would like to buy one unit at the price of 15 and at any lower price. Similarly a limit order to sell two units at 16 means you would like to sell two units at the price of 16 and at any higher price.

A single market price is determined by matching the number of people who would like to buy at that price with the same number of people who would like to sell. The lower left part of the screen displays your personal stats such as the number of shares you own, your current cash, and the total stock value. For the market stats you can see the current market price, the interest rate, the dividends value, and the buyback value of the stock at the end of the 30 trading periods; which is 14. The upper left corner displays the graph of the stock market from the beginning until the current trading period. We can also see the traded volume for each period.

In the Forecast Prices screen the upper right part of the screen will change. On this page, you will submit your best estimate of what you think the market price will be for this period, two periods in advance, five periods in advance, and 10 periods in advance. As you can see the slider can be dragged from left to right in this example I am setting the Price Forecast for this current trading period to 14. For two periods ahead I will set it to 14 as well. You can set the price forecast as you want. During the trading task, if the forecast is within 2.5 units from the actual realized price for each of the forecasted periods, then you will receive 5 units of cash as reward for each correct forecast. For example, if the actual price for period 1 is going to be 15 then you will be rewarded for that forecast. If the actual price for period 3 is 18 then you will not receive the reward for that forecast. On the round

results screen you will see what the market price was for this period, how many shares were traded, as well as the number of shares you traded. You will also see updates on the new current cash and stock value and the dividend you earned. For period one, as you had four shares with a dividend of 0.4 for that period you won \$1.60 from holding four stocks and \$5 from holding 100 cash.

On the last page you will see two pie charts with different monetary rewards and associated probabilities. You must select one of the pie charts. For example on the left side you have a lottery of winning either \$7.25 with a probability of 74% or \$9.00 with a probability of 26%. On the right side you have a lottery of winning either \$19.00 with 26% probability or \$11.50 with 74% probability. After you select one of the lotteries a new set of lotteries will appear. The next button will not appear in the actual trading task so you cannot click on it. There will be four iterations of different lotteries with different monetary rewards and probabilities. You will have 5 seconds for each pair of lotteries, so you must be quick. At the end, one Lottery will be selected at random from those that you chose throughout the experiment, and you will receive the outcome of the lottery divided by 10. It's in your interest to actually choose accordingly. Here are the key points. You will trade One stock for 30 trading periods using cash. You start with 100 units of cash and four stocks. Each period stock provides a dividend of either 0.4 or 1.00 while interest provides 5% reward each trading period. The experiment consists of four pages, order submission, forecasting prices, round results, and choosing a lottery. After the last trading round all of your Shares are converted to 14 cash units. If you buy shares for more than 14, as you get near Round 30 it is possible those shares will be terminated at a value of 14 if you cannot sell them. You will also take a five question quiz. After this, pay attention to the answers you give. You will earn a reward for each question you answer correctly on the first try. If you fail to answer at least one question correctly then you cannot continue.

The experiment will begin shortly.

E.2 Quiz

This quiz was administered at the end of the instructions. Participants answered the questions and submitted their responses. Feedback was given along with an explanation of any errors. Participants were asked to submit answers until they got all the questions correct. The quiz bonus was assessed based on their first submission.

Correct answers appear in **bold**

1. During order entry, you want to input a BUY order at a price of 8.52 and a SELL order at a price of 7.89. What will happen to your orders for that period?

- They'll be entered as usual
 - **The SELL order will not go through**
2. During order entry, the highest price at which you submit a BUY order is 16.78, and the lowest price at which others SELL is 16.22. The market price is 16.56 for that period. You will:
- BUY one unit at 16.78
 - **BUY one unit at 16.56**
 - SELL one unit at 16.56
 - SELL one unit at 17.22
 - Not Trade
3. You have 200 units of CASH at the start of a trading period and no STOCK. You do not trade that period. How much CASH do you have at the beginning of the next period?
- 200
 - **210**
 - 205
 - 190
4. Your account has 5 STOCK and 100 CASH at the start of a trading period, and you do not BUY or SELL during that period. The dividend for that round is 1.00. How much CASH do you have at the start of the next round?
- 105
 - **110**
 - 100
 - 114
5. After the final trading period, you have one remaining unit of STOCK. The market price in the final period is 29. How many units of experiment CASH do you receive in exchange for your STOCK?
- 1
 - 29

- 14
- 30

E.3 Surveys

E.3.1 Pre-experiment survey 1

Thank you for deciding to participate in the experiment. The answers you will provide herein will solely be used in academic research. The following is a pre-experimental consent form.

If you give your consent, you can participate in the experiment. If you do not give your consent, then please return the study on Prolific.

- Given the exclusion criteria, please confirm that you qualify to participate in the experiment:
 - Yes
 - No
- Print Prolific ID (as substitute for Name) *[text entry]*
- Please initial one option below to indicate whether or not you consent to us contacting you in the future:
 - I consent to having Caltech researchers contact me about other studies at Caltech that are funded by non-profit organizations, government agencies, universities, and other non-commercial entities.
 - I consent to having Caltech researchers contact me about other studies at Caltech that are funded by for-profit companies and other commercial entities.
 - I consent to having Caltech researchers contact me about any other studies at Caltech, regardless of who is funding the research.
 - I do not wish to be contacted about participating in future research studies.
- Print Participant's Initials (as substitute for signature) *[text entry]*
- Please indicate your consent
 - I consent
 - I do not consent

E.3.2 Pre-experiment survey 2

The following is a pre-experimental questionnaire.

The answers you will provide herein will solely be used in academic research.

After filling it in, you will perform the task in the actual experiment and then fill in the post-experimental questionnaire.

It is very important to be as honest as possible and to treat each question rigorously, as your participation will directly impact the subsequent research study.

The answers to this questionnaire are strictly confidential. Most of the questions are mandatory - you can distinguish them through the red asterisk (*) sign.

- Please enter your Prolific ID *[text entry]*
- Your age (in years) *[text entry]*
- Your gender
 1. Male
 2. Female
 3. Non-binary
 4. I prefer not to say
- Are you Hispanic, Latino/a, or of Spanish origin?
 - Yes
 - No
- What is your race?
 1. White/Caucasian
 2. Black or African American
 3. American Indian or Alaska Native
 4. Asian or Asian Indian
 5. Native Hawaiian or Other Pacific Islander
 6. What race do you consider yourself? Write here: *[text entry]*
- Studies background
 - Social sciences: economics, psychology

- IT and software
- STEM
- Humanities and arts
- I don't have higher education studies
- Other... *[text entry]*
- Are you a native English speaker (i.e., is English your mother tongue?)
 - Yes
 - No
- On a scale from 0 to 5, how good is your knowledge of English? (If YES to previous question, please mark five here) *[0 through 5]*
- Are you currently under medication treatment for depression, anxiety, or other psychiatric or neurological conditions?
 - Yes
 - No
- How would you rate your risk-taking behavior in general, on a scale from 1–7?
The scale goes from 1 = “Very risk-averse” to 7 = “Very risk-taking”
- How would you rate your financial risk-taking behavior, on a scale from 1–7?
The scale goes from 1 = “Very risk-averse” to 7 = “Very risk-taking”
- You have been given \$100. You can invest any part of this money in a risky asset. If the investment is successful, the risky asset returns 2.5 times the amount invested with a probability of one-half and nothing with a probability of one-half. You can also choose not to invest at all, in which case you will keep the money you have. How much would you like to invest? (write amount in dollars)

E.3.3 Post-experimental questionnaire

Thank you for deciding to fill in this questionnaire. It takes approximately 10 minutes to fill in. The answers you will provide herein will solely be used in academic research.

It is very important to be as honest as possible and to treat each question rigorously, as your participation will directly impact the subsequent research study.

The answers to this questionnaire are strictly confidential. Most of the questions are mandatory - you can distinguish them through the red asterisk (*) sign.

- Prolific ID *[text entry]*
- Please enter the 3-digit ID. If you do not have it, enter the Prolific ID. *[text entry]*
- Have you participated in similar trading experiments before?
 - Yes, multiple times
 - Yes, just once
 - No
- On a scale from 1–10, how well did you understand the task and were able to trade as required?
- Did you understand exactly what you had to do from the beginning?
 - Yes, entirely
 - Yes, but partially, I understood more as I continued trading
 - No, not at all
- Did you follow any strategy when trading? Please describe, be as detailed as possible *[text entry]*
- Do you have any suggestions to improve the experiment? i.e., the software, the trading instructions, time to enter orders, time to enter forecast, time to do the risk elicitation tasks, etc. *[text entry]*
- Do you own/trade stocks?
 - YES
 - NO
- Imagine you performed the tasks with another 99 persons. In terms of performance, in what place do you think you placed? How would you rate your outcome from 1–100? *[text entry]*
- How would you rate your risk-taking behavior in general, on a scale from 1–7?

The scale goes from 1 = “Very risk-averse” to 7 = “Very risk-taking”

- How would you rate your financial risk-taking behavior, on a scale from 1–7?

The scale goes from 1 = “Very risk-averse” to 7 = “Very risk-taking”

- You have been given \$100. You can invest any part of this money in a risky asset. If the investment is successful, the risky asset returns 2.5 times the amount invested with a probability of one-half and nothing with a probability of one-half. You can also choose not to invest at all, in which case you will keep the money you have. How much would you like to invest? (write amount in dollars)

[text entry]

- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$10
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$12
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$14
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$16
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$18
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$20
- Choose what you prefer:

- 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$22
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$24
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$26
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$28
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$30
- Choose what you prefer:
 - 100% chance of winning \$10
 - 50% chance of winning \$0, 50% chance of winning \$32